SingleRAN

# EPC-based NSA Performance Enhancement Feature Parameter Description

**Issue** Draft B

**Date** 2026-01-30

HUAWEI TECHNOLOGIES CO., LTD.

### Trademarks and Permissions



### Notice



## Huawei Technologies Co., Ltd.

| | |
|---|---|
| Address: | Huawei Industrial Base<br>Bantian, Longgang<br>Shenzhen 518129<br>People's Republic of China |
| Website: | https://www.huawei.com |
| Email: | support@huawei.com |

# Security Declaration

## Vulnerability

Huawei's regulations on product vulnerability management are subject to the *Vul. Response Process*. For details about this process, visit the following web page:
https://www.huawei.com/en/psirt/vul-response-process
For vulnerability information, enterprise customers can visit the following web page:
https://securitybulletin.huawei.com/enterprise/en/security-advisory

# Contents

# 1 Change History

The following describes the specific changes in different issues.

## SRAN22.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes

  None

- Editorial changes
  - Added the description of slot assignments for NR TDD cells supported by network-coordinated dynamic UE power sharing. For details, see **4.1.3.2 Network-Coordinated Dynamic UE Power Sharing**.
  - Modified the descriptions of PSCell addition after the target eNodeB receives a handover request message during an LTE handover of an NSA UE. For details, see **5.1.3 Blind PSCell Addition**.

## SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 05 (2025-08-30).

- Technical changes

| Change Description | Parameter Change | Base Station Model |
|---|---|---|
| Added incompatibility of BTS5929R with EN-DC performance enhancement. For details, see **4.3.3 Hardware**. | None | 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R |
| Added incompatibility of BTS5929R with EN-DC optimal carrier selection. For details, see **5.3.3 Hardware**. | None | 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R |

| Change Description | Parameter Change | Base Station Model |
|---|---|---|
| Added blacklist control over NSA PCC anchoring for RRC_CONNECTED UEs. For details, see:<br>• **5.1.1 NSA PCC Anchoring**<br>• **5.4.1.1 Data Preparation**<br>• **5.4.1.2 Using MML Commands** | Modified parameter on the LTE side: Added the **NSA_PCC_ANCHORING_FORBID_SW** option to the **UeCompat**.***BlacklistControlExtSwitch2*** parameter. | • 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R<br>• DBS3900 LampSite and DBS5900 LampSite |
| Added the impact relationship between SN-terminated MCG bearer transmission and LNR CRS muting. For details, see **4.3.2.3 Function Impacts**. | None | 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R |

- Editorial changes
  - Revised the descriptions of function impacts between UL and DL Decoupling and TDM power control in **4.3.2.3 Function Impacts**.
  - Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

# 2 About This Document

## 2.1 General Statements

### Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

**NOTE**

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

### Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

## 2.2 Applicable RAT

This document applies to LTE FDD, LTE TDD, and NR.

For definitions of base stations described in this document, see section "Base Station Products" in *SRAN Networking and Evolution Overview*.

## 2.3 Features in This Document

This document describes the following features.

| RAT | Feature ID | Feature Name | Chapter/Section |
| --- | --- | --- | --- |
| LTE FDD | MRFD-151223 | EN-DC Performance Enhancement (LTE FDD) | • 4.1.1.1 Blind PSCell Addition for EPS Fallback UEs<br>• 4.1.2.1 SN-Terminated MCG Bearer Transmission<br>• 4.1.2.2 Uplink Data Transmission Path Selection<br>• 4.1.2.3 NSA UE Scheduling Protection Based on MCG Cell Load<br>• 4.1.3.1 TDM Power Control<br>• 4.1.3.2 Network-Coordinated Dynamic UE Power Sharing<br>• 4.1.3.3 PC2 UE Uplink NR Power Configuration Maximization<br>• 4.2 Network Analysis<br>• 4.3 Requirements<br>• 4.4 Operation and Maintenance |
| LTE FDD | LNOFD-151333 | EN-DC Optimal Carrier Selection | 5 EN-DC Optimal Carrier Selection |
| LTE FDD | MRFD-171222 | NSA/SA Selection Based on User Experience (LTE FDD) | • 4.1.1.2 Blind PSCell Addition for Experience-based Fallback UEs<br>• 4.2 Network Analysis<br>• 4.3 Requirements<br>• 4.4 Operation and Maintenance |

| RAT | Feature ID | Feature Name | Chapter/Section |
|---|---|---|---|
| LTE TDD | MRFD-151233 | EN-DC Performance Enhancement (LTE TDD) | • **4.1.1.1 Blind PSCell Addition for EPS Fallback UEs**<br>• **4.1.2.1 SN-Terminated MCG Bearer Transmission**<br>• **4.1.2.2 Uplink Data Transmission Path Selection**<br>• **4.1.2.3 NSA UE Scheduling Protection Based on MCG Cell Load**<br>• **4.1.3.1 TDM Power Control**<br>• **4.1.3.2 Network-Coordinated Dynamic UE Power Sharing**<br>• **4.1.3.3 PC2 UE Uplink NR Power Configuration Maximization**<br>• **4.2 Network Analysis**<br>• **4.3 Requirements**<br>• **4.4 Operation and Maintenance** |
| LTE TDD | TDLNOFD-151504 | EN-DC Optimal Carrier Selection | **5 EN-DC Optimal Carrier Selection** |
| LTE TDD | MRFD-171232 | NSA/SA Selection Based on User Experience (LTE TDD) | • **4.1.1.2 Blind PSCell Addition for Experience-based Fallback UEs**<br>• **4.2 Network Analysis**<br>• **4.3 Requirements**<br>• **4.4 Operation and Maintenance** |

| RAT | Feature ID | Feature Name | Chapter/Section |
| --- | --- | --- | --- |
| NR | MRFD-151263 | EN-DC Performance Enhancement (NR) | • **4.1.1.1 Blind PSCell Addition for EPS Fallback UEs**<br>• **4.1.2.1 SN-Terminated MCG Bearer Transmission**<br>• **4.1.2.2 Uplink Data Transmission Path Selection**<br>• **4.1.2.3 NSA UE Scheduling Protection Based on MCG Cell Load**<br>• **4.1.3.1 TDM Power Control**<br>• **4.1.3.2 Network-Coordinated Dynamic UE Power Sharing**<br>• **4.1.3.3 PC2 UE Uplink NR Power Configuration Maximization**<br>• **4.2 Network Analysis**<br>• **4.3 Requirements**<br>• **4.4 Operation and Maintenance** |
| NR | MRFD-171262 | NSA/SA Selection Based on User Experience (NR) | • **4.1.1.2 Blind PSCell Addition for Experience-based Fallback UEs**<br>• **4.2 Network Analysis**<br>• **4.3 Requirements**<br>• **4.4 Operation and Maintenance** |

# 2.4 Differences

## Differences Between LTE FDD and LTE TDD

| Function Name | Difference | Chapter/Section |
|---|---|---|
| Blind PSCell addition for EPS fallback UEs | None | 4.1.1.1 Blind PSCell Addition for EPS Fallback UEs |
| Blind PSCell addition for experience-based fallback UEs | None | 4.1.1.2 Blind PSCell Addition for Experience-based Fallback UEs |
| SN-terminated MCG bearer transmission | None | 4.1.2.1 SN-Terminated MCG Bearer Transmission |
| Uplink data transmission path selection | None | 4.1.2.2 Uplink Data Transmission Path Selection |
| NSA UE scheduling protection based on MCG cell load | None | 4.1.2.3 NSA UE Scheduling Protection Based on MCG Cell Load |
| TDM power control | Supported only in LTE FDD | 4.1.3.1 TDM Power Control |
| Network-coordinated dynamic UE power sharing | None | 4.1.3.2 Network-Coordinated Dynamic UE Power Sharing |
| PC2 UE uplink NR power configuration maximization | None | 4.1.3.3 PC2 UE Uplink NR Power Configuration Maximization |
| EN-DC optimal carrier selection | None | 5 EN-DC Optimal Carrier Selection |

## Differences Between NR FDD and NR TDD

| Function Name | Difference | Chapter/Section |
| --- | --- | --- |
| Blind PSCell addition for EPS fallback UEs | None | **4.1.1.1 Blind PSCell Addition for EPS Fallback UEs** |
| Blind PSCell addition for experience-based fallback UEs | None | **4.1.1.2 Blind PSCell Addition for Experience-based Fallback UEs** |
| SN-terminated MCG bearer transmission | None | **4.1.2.1 SN-Terminated MCG Bearer Transmission** |
| Uplink data transmission path selection | None | **4.1.2.2 Uplink Data Transmission Path Selection** |
| NSA UE scheduling protection based on MCG cell load | None | **4.1.2.3 NSA UE Scheduling Protection Based on MCG Cell Load** |
| TDM power control | Supported only in NR TDD | **4.1.3.1 TDM Power Control** |
| Network-coordinated dynamic UE power sharing | None | **4.1.3.2 Network-Coordinated Dynamic UE Power Sharing** |
| PC2 UE uplink NR power configuration maximization | None | **4.1.3.3 PC2 UE Uplink NR Power Configuration Maximization** |
| EN-DC optimal carrier selection | None | **5 EN-DC Optimal Carrier Selection** |

## Differences Between NSA and SA

| Function Name | Difference | Chapter/Section |
| --- | --- | --- |
| Blind PSCell addition for EPS fallback UEs | Supported only in NSA networking | **4.1.1.1 Blind PSCell Addition for EPS Fallback UEs** |
| Blind PSCell addition for experience-based fallback UEs | Supported only in NSA networking | **4.1.1.2 Blind PSCell Addition for Experience-based Fallback UEs** |
| SN-terminated MCG bearer transmission | Supported only in NSA networking | **4.1.2.1 SN-Terminated MCG Bearer Transmission** |
| Uplink data transmission path selection | Supported only in NSA networking | **4.1.2.2 Uplink Data Transmission Path Selection** |
| NSA UE scheduling protection based on MCG cell load | Supported only in NSA networking | **4.1.2.3 NSA UE Scheduling Protection Based on MCG Cell Load** |
| TDM power control | Supported only in NSA networking | **4.1.3.1 TDM Power Control** |
| Network-coordinated dynamic UE power sharing | Supported only in NSA networking | **4.1.3.2 Network-Coordinated Dynamic UE Power Sharing** |
| PC2 UE uplink NR power configuration maximization | Supported only in NSA networking | **4.1.3.3 PC2 UE Uplink NR Power Configuration Maximization** |
| EN-DC optimal carrier selection | Supported only in NSA networking | **5 EN-DC Optimal Carrier Selection** |

## Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

| Function Name | Difference | Chapter/Section |
|---|---|---|
| Blind PSCell addition for EPS fallback UEs | None | **4.1.1.1 Blind PSCell Addition for EPS Fallback UEs** |
| Blind PSCell addition for experience-based fallback UEs | None | **4.1.1.2 Blind PSCell Addition for Experience-based Fallback UEs** |
| SN-terminated MCG bearer transmission | None | **4.1.2.1 SN-Terminated MCG Bearer Transmission** |
| Uplink data transmission path selection | Supported only in low frequency bands of NR | **4.1.2.2 Uplink Data Transmission Path Selection** |
| NSA UE scheduling protection based on MCG cell load | None | **4.1.2.3 NSA UE Scheduling Protection Based on MCG Cell Load** |
| TDM power control | Supported only in low frequency bands of NR | **4.1.3.1 TDM Power Control** |
| Network-coordinated dynamic UE power sharing | Supported only in low frequency bands of NR | **4.1.3.2 Network-Coordinated Dynamic UE Power Sharing** |
| PC2 UE uplink NR power configuration maximization | None | **4.1.3.3 PC2 UE Uplink NR Power Configuration Maximization** |
| EN-DC optimal carrier selection | None | **5 EN-DC Optimal Carrier Selection** |

# 3 Overview

This document describes carrier management enhancement in NSA DC, data split enhancement in NSA DC, uplink power control enhancement in NSA DC, and EN-DC optimal carrier selection based on *EPC-based NSA Basics*.

# 4 EN-DC Performance Enhancement

## 4.1 Principles

### 4.1.1 Carrier Management Enhancement in NSA DC

#### 4.1.1.1 Blind PSCell Addition for EPS Fallback UEs

Different blind PSCell addition procedures can be triggered for EPS fallback UEs (UEs that fall back to E-UTRAN), depending on the **EPS_FALLBACK_BLIND_SCG_ADD_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoExtSwitch*** parameter.

- If this option is selected, the eNodeB obtains NG-RAN CGI information from the *UE History Information* IE in the Handover Request message and then queries the NR NCL (which contains the information about external cells), regardless of whether the **NSA_BLIND_SCG_ADDITION_SWITCH** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter is selected. If the **NrExternalCell**.***NrNetworkingOption*** parameter of the NR cell from which the UE falls back is set to **SA_NSA** or **UNLIMITED** and the UE supports EN-DC on the corresponding frequency, the eNodeB sends an SCG addition request.
- If this option is deselected and the **NSA_BLIND_SCG_ADDITION_SWITCH** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter is selected, blind PSCell addition is performed. For details, see **5.1.3 Blind PSCell Addition**. If blind PSCell addition fails, PSCell configuration based on data volume is triggered. A measurement procedure is started and then measurement-based PSCell configuration is performed. For details, see *EPC-based NSA Basics*.

#### 4.1.1.2 Blind PSCell Addition for Experience-based Fallback UEs

Different PSCell addition procedures can be triggered for UEs that fall back from SA networking to NSA networking due to poor user experience in different situations.

Different types of cells are controlled by different parameter options. LTE FDD cells are controlled by the **LTE_FDD_NSA_SA_DL_SEL_OPT_SW** or

**LTE_FDD_NSA_SA_UL_SEL_OPT_SW** option of the **EnodebAlgoExtSwitch**.***MultiNetworkingOptionOptSw*** parameter. LTE TDD cells are controlled by the **LTE_TDD_NSA_SA_DL_SEL_OPT_SW** option of the **EnodebAlgoExtSwitch**.***MultiNetworkingOptionOptSw*** parameter. NR cells are controlled by the **NSA_SA_DL_SEL_OPT_SW** or **NSA_SA_UL_SEL_OPT_SW** option of the **gNodeBParam**.***NetworkingOptionOptSw*** parameter, and the **NSA_SA_DL_SEL_OPT_SW** or **NSA_SA_UL_SEL_OPT_SW** option of the **gNBOperator**.***OperatorInterRatPolicySw*** parameter.

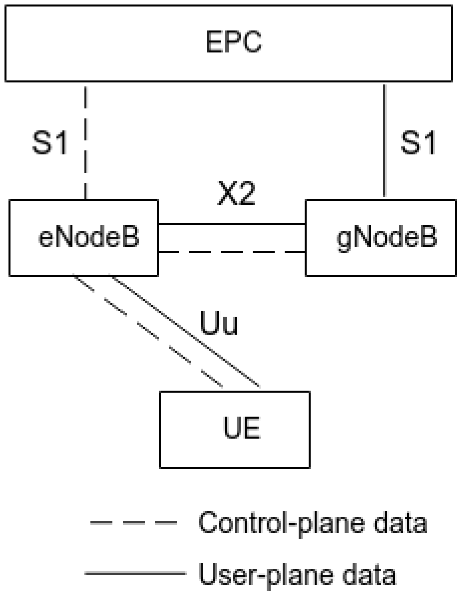
- If the related options on LTE and NR sides are selected, regardless of whether the **NSA_BLIND_SCG_ADDITION_SWITCH** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter on the LTE side is selected, the eNodeB determines EN-DC band combinations based on the HANDOVER REQUEST message, initiates an SCG addition, and sends an SCG Addition message including NR CA carrier information to the NR side. During an SA-to-NSA handover for an NSA UE, whether a PSCell can be directly added for the UE is controlled by the **DIRECT_NR_HO_TO_EN_DC_SW** option of the **EnodebAlgoExtSwitch**.***NrHandoverAlgoSwitch*** parameter.
  – If this option is selected and the UE has required capabilities, a PSCell is added during the SA-to-NSA handover to complete a direct switching from NR to EN-DC.
  – If this option is deselected, a PSCell can be added for the NSA UE only after the SA-to-NSA handover.
- If the related options on LTE and NR sides are deselected and the **NSA_BLIND_SCG_ADDITION_SWITCH** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter on the LTE side is selected, blind PSCell addition is performed. For details, see **5.1.3 Blind PSCell Addition**. If blind PSCell addition fails, PSCell configuration based on data volume is triggered. A measurement procedure is started and then measurement-based PSCell configuration is performed. For details, see *EPC-based NSA Basics*.

## 4.1.2 Data Split Enhancement in NSA DC

### 4.1.2.1 SN-Terminated MCG Bearer Transmission

**When the NR coverage is discontinuous, UEs frequently add or release SCGs at the NR coverage edge. As a result, PDCP is frequently switched between the gNodeB and the eNodeB. To address this issue, the SN-terminated MCG bearer function is introduced to Option 3x for user-plane data, as shown in Figure 4-1. This function reduces the impact of signaling on the core network.**

**Figure 4-1** SN-terminated MCG bearer transmission architecture




This function takes effect when the following options on the LTE and NR sides are both selected.

- On the LTE side: **SN_TERMN_MCG_BEARER_SWITCH** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter
- On the NR side: **SN_TERMN_MCG_BEARER_SWITCH** option of the **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** parameter

**NOTE**

- When the local switch setting, license, or cell status changes, the function changes from valid to invalid or from invalid to valid. In this case, it takes some time (less than 5 minutes) before the peer end receives the change in the local end. Before the peer end receives the change, the following symptoms occur:
  - When the local function changes from valid to invalid, the peer end still attempts to initiate an SN-terminated MCG bearer request during this period. The local end rejects the request and, as a result, the number of SgNB modification failures increases.
  - When the local function changes from invalid to valid, the peer end does not initiate an SN-terminated MCG bearer request during this period but the local end does. The peer end rejects the request and, as a result, the number of SgNB modification failures increases.
- This function requires the selection of both the **SN_TERMN_MCG_BEARER_SWITCH** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter and the **SN_TERMN_MCG_BEARER_SWITCH** option of the **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** parameter. Under these settings, this function can take effect only when neighboring NR cells are configured as external cells on the LTE side for handovers to LTE cells.

This function can be enabled in the following scenarios:

- The SCG is released when NR coverage is limited.
- The SgNB initiates an SCG release when the inactivity timer on the NR side expires.

- The SCG is released when a VoLTE service is initiated.
- The SCG is released based on the SCG failure message reported by the UE.

When the NSA UE is in the SN-terminated MCG bearer state, the PDCP layer is on the NR side. When the UE is in the LTE-only state, the PDCP layer is on the LTE side.

When the NSA UE is in the SN-terminated MCG bearer state, it does not require NSA DC uplink power control, data split, or interference avoidance.

When the NSA UE is in the SN-terminated MCG bearer state, blind SCG addition is supported during an LTE cell handover. For details, see **5.1.3 Blind PSCell Addition**. If the target MeNB and the SgNB have an LTE-NR X2 link between them and the target cell also supports SN-terminated MCG bearer transmission after an LTE cell handover, the NSA UE is still in the SN-terminated MCG bearer state; otherwise, the NSA UE transits to the LTE-only state.

When the NSA UE is in the SN-terminated MCG bearer state, the NR coverage status can be obtained through B1 measurement. For details, see *EPC-based NSA Basics*. If B1 measurement reports cannot be obtained from the NR side for two consecutive measurement periods, the NSA UE transits to the LTE-only state to avoid occupying the LTE-NR X2 bandwidth for a long time. To prevent SN-terminated MCG bearer UEs from occupying too much LTE-NR X2 bandwidth, the **gNodeBParam**.***MaxX2TransRate*** parameter is used to specify the maximum X2 bandwidth that can be occupied. If the NSA UE is performing VoLTE services, it does not transit to the LTE-only state due to the preceding reasons, preventing the PDCP bearer from being switched between the gNodeB and the eNodeB.

NOTE

- When the transmission load of the main control board on the LTE side is high, the LTE side does not send an SN-terminated MCG bearer request; in addition, if receiving such a request from the peer end, it rejects the request, which increases the number of SgNB modification failures.
- When the UMPTb is used as the main control board on the NR side, SN-terminated MCG bearer transmission is not supported.

### 4.1.2.2 Uplink Data Transmission Path Selection

## Basic Principles

**If NSA UEs do not support uplink data split in NSA DC scenarios, uplink services can be carried only on the LTE or NR side. Uplink experience will vary with UE movement, as there are differences between LTE and NR in factors such as bandwidth, subframe configuration, and load. Therefore, the optimal uplink data transmission path needs to be selected to improve uplink user experience by estimating the uplink LTE and NR data rates based on factors such as air interface status and load.**

This function requires the selection of both the **NSA_DC_UL_PATH_SELECTION_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter on the LTE side and the **NSA_DC_UL_PATH_SELECTION_SW** option of the **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** parameter on the NR side.

For NSA UEs with uplink data transmitted only on the SCG side:

1. The uplink SINR on the NR side is determined as follows:
   - Assume that the previous uplink NR SINR evaluation result is "poor". Then, the SINR is measured for five consecutive times at an interval of **NRDUCellSrsMeas**.***NsaUlPathSelSinrTimeToTrig*** (ms). If all the SINR values are greater than **NRDUCellSrsMeas**.***NsaUlPathSelSinrLowThld*** + **NRDUCellSrsMeas**.***NsaUlPathSelSinrHyst***, the next-layer evaluation is required. Otherwise, the final evaluation result of this time is "poor".

     In the next-layer evaluation, if all the SINR values are greater than **NRDUCellSrsMeas**.***NsaUlPathSelSinrHighThld*** + **NRDUCellSrsMeas**.***NsaUlPathSelSinrHyst***, the final evaluation result of this time is "good". Otherwise, the result is "average".
   - Assume that the previous uplink NR SINR evaluation result is "good". Then, the SINR is measured for five consecutive times at an interval of **NRDUCellSrsMeas**.***NsaUlPathSelSinrTimeToTrig*** (ms). If all the SINR values are less than **NRDUCellSrsMeas**.***NsaUlPathSelSinrHighThld*** – **NRDUCellSrsMeas**.***NsaUlPathSelSinrHyst***, the next-layer evaluation is required. Otherwise, the final evaluation result of this time is "good".

     In the next-layer evaluation, if all the SINR values are less than **NRDUCellSrsMeas**.***NsaUlPathSelSinrLowThld*** – **NRDUCellSrsMeas**.***NsaUlPathSelSinrHyst***, the final evaluation result of this time is "poor". Otherwise, the result is "average".
   - Assume that the initial or previous uplink NR SINR evaluation result is "average". Then, the SINR is measured for five consecutive times at an interval of **NRDUCellSrsMeas**.***NsaUlPathSelSinrTimeToTrig*** (ms). If all the SINR values are greater than **NRDUCellSrsMeas**.***NsaUlPathSelSinrHighThld*** + **NRDUCellSrsMeas**.***NsaUlPathSelSinrHyst***, the final evaluation result of this time is "good". If all the SINR values are less than **NRDUCellSrsMeas**.***NsaUlPathSelSinrLowThld*** – **NRDUCellSrsMeas**.***NsaUlPathSelSinrHyst***, the final evaluation result of this time is "poor". Otherwise, the result is "average".
2. Different policies are used for different evaluation results:
   - If the evaluation result is "poor", an uplink data split mode change is triggered. The eNodeB sends an RRC connection reconfiguration message to the UE, instructing the UE to send data on the LTE side.
   - If the evaluation result is "good", the eNodeB sends an RRC connection reconfiguration message to the UE, instructing the UE to use the original uplink data split mode.
   - If the evaluation result is "average" and the UE has large packets, the estimated rates on the two sides are compared. If the comparison result meets the conditions listed in **Table 4-1**, processing is performed accordingly. In other cases, no change is performed.

**Table 4-1** Processing after comparing the estimated data rates of LTE and NR sides

| Option | Condition | Processing |
|---|---|---|
| Option 3 | Estimated uplink LTE data rate > Estimated uplink NR data rate x **NsaDcMgmtConfig**.***NsaDcUlPathToLteRateCoeff*** | The uplink data split mode is changed.<br>The eNodeB sends an RRC connection reconfiguration message to the UE, instructing the UE to send data on the LTE side. |
| | Estimated uplink LTE data rate < Estimated uplink NR data rate x **NsaDcMgmtConfig**.***NsaDcUlPathToNrRateCoeff*** | The eNodeB sends an RRC connection reconfiguration message to the UE, instructing the UE to use the original uplink data split mode. |
| Option 3x | Estimated uplink LTE data rate > Estimated uplink NR data rate x **NRDUCellSrsMeas**.***NsaUlPathToLteRateRatio*** | The uplink data split mode is changed.<br>The eNodeB sends an RRC connection reconfiguration message to the UE, instructing the UE to send data on the LTE side. |
| | Estimated uplink LTE data rate < Estimated uplink NR data rate x **NRDUCellSrsMeas**.***NsaUlPathToNrRateRatio*** | The eNodeB sends an RRC connection reconfiguration message to the UE, instructing the UE to use the original uplink data split mode. |

For NSA UEs with uplink data transmitted in other modes:

- Assume that the measured uplink NR SINR is less than **NRDUCellSrsMeas**.***NsaSplitUlPathSelSinrThld*** – **NRDUCellSrsMeas**.***NsaUlPathSelSinrHyst*** for five consecutive times at an interval specified by **NRDUCellSrsMeas**.***NsaUlPathSelSinrTimeToTrig*** (ms). In this case, the uplink data split mode is changed. The eNodeB sends an RRC connection reconfiguration message to the UE, instructing the UE to send data on the LTE side.
- Assume that the measured uplink NR SINR is greater than **NRDUCellSrsMeas**.***NsaSplitUlPathSelSinrThld*** + **NRDUCellSrsMeas**.***NsaUlPathSelSinrHyst*** for five consecutive times at an interval specified by **NRDUCellSrsMeas**.***NsaUlPathSelSinrTimeToTrig*** (ms). In this case, the eNodeB sends an RRC connection reconfiguration message to the UE, instructing the UE to use the original uplink data split mode.

**NOTE**

Before uplink data transmission path selection is enabled, when the uplink coverage on the NR side is limited, the UE sends an SCGFailureInformationNR message to notify the LTE side of the NR SCG radio link failure. After uplink data transmission path selection is enabled, when a UE has data to transmit in the uplink and the uplink coverage on the NR side is limited, the uplink data transmission path of the UE is switched to the LTE side with better signal quality. In this case, the number of SCGFailureInformationNR messages that the UE sends on the LTE side, the number of SgNB Release Request messages that the LTE side sends to the NR side, and the values of the following counters on the NR side decrease:

- **N.NsaDc.SgNB.Rel** - **N.NsaDc.SgNB.Rel.MeNBTrigger.NormalRel**
- **N.NsaDc.SgNB.Rel.MeNBTrigger.AbnormRel.ScgFail**
- **N.NsaDc.SgNB.Rel.MeNBTrigger.AbnormRel.ScgFail.RAProblem**
- **N.NsaDc.SgNB.Rel.MeNBTrigger.AbnormRel.ScgFail.T310Expiry**
- **N.NsaDc.SgNB.Rel.MeNBTrigger.AbnormRel.ScgFail.RlcMaxNumRetx**
- **N.NsaDc.SgNB.Rel.MeNBTrigger.AbnormRel.ScgFail.RecfgFail**
- **N.NsaDc.SgNB.Rel.MeNBTrigger.AbnormRel.ScgFail.SyncRecfgFail**

After uplink data transmission path selection is enabled, when a UE has data to receive in the downlink and the uplink coverage on the NR side is limited, if the NR side does not receive RLC status reports from the UE for 32 consecutive times, the NR side sends an SgNB Release Required message with the cause value of "Radio Connection With UE Lost" to the LTE side. In this case, the values of the following counters on the NR side increase:

- **N.NsaDc.SgNB.AbnormRel**
- **N.NsaDc.SgNB.AbnormRel.Radio**
- **N.NsaDc.SgNB.AbnormRel.Radio.UeLost**

## Application Limitations

- This function is not triggered if uplink data is transmitted only on the MCG side.
- It is recommended that the **SchedulerCtrlPowerSwitch** option of the **CellAlgoSwitch**.***UlSchSwitch*** parameter be selected on the LTE side. Otherwise, there is a possibility that NSA UEs cannot be determined as large-packet UEs, affecting uplink data transmission path selection.
- It is recommended that the **EnodebAlgoExtSwitch**.***NrScgInactivityRelStrategy*** parameter be set to **REL_UPON_LTE_INACTVTY_TMR_EXPN** on the LTE side. Otherwise, the SgNB may be added again based on data volume after being released.
- In LTE FDD and NR spectrum sharing scenarios, the **LteNrSpctShrCellGrp**.***LteNrSpctShrLtePriResRatio*** parameter on the LTE side cannot be set to a value greater than 80%. Otherwise, uplink data transmission may fail to be switched back to NR after being switched to the LTE side.
- When the **SPKT_UL_PATH_BACK_TO_NR_SW** option of the **NRDUCellAlgoSwitch**.***NsaDcAlgoSwitch*** parameter is selected on the NR side and the uplink NR SINR evaluation result is "average", the uplink data transmission path of non-large-packet UEs is directly switched to the NR side for transmission, and no estimated rate comparison between the two sides is performed during the penalty time specified by the **NRDUCellSrsMeas**.***NsaSPktUlBackToNrPenaltyTime*** parameter.
- Assume that the **UL_KEEP_IN_LTE_AFTER_HO_SW** option of the **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** parameter is selected on the NR side.

In addition, assume that uplink fallback to LTE has been triggered or the LTE side has been selected by the uplink data transmission path selection function for a UE. In this case, if the UE's NR cell changes, the UE's uplink remains in the LTE state after the change.

- During the initial X2 setup phase, there is a delay (less than 5 minutes) for exchanging switch status information between the local and peer ends. During this period, uplink data transmission path selection does not take effect for newly admitted NSA UEs.
- After the switch status of the local end changes, the peer end has to wait for a period of time (less than 5 minutes) before receiving the change information. Before the peer end receives the change information, there may be the situations described in **Table 4-2**.

**Table 4-2** Possible situations before the peer end receives the switch status change information of the local end

| Option | NR: NSA_DC_UL_PATH_SELECTION_SW Option of NRCellNsaDcConfig.*NsaDcAlgoSwitch* | LTE: NSA_DC_UL_PATH_SELECTION_SW Option of NsaDcMgmtConfig.*NsaDcAlgoSwitch* | Possible Situation |
| --- | --- | --- | --- |
| Option 3x | On | On > Off | NSA UEs already admitted and NSA UEs newly admitted during this period are both affected.<br>When the uplink NR SINR is medium, the NR side still attempts to initiate uplink data transmission path selection. As the switch on the LTE side is turned off, the attempts are rejected and the selection fails. When the uplink NR SINR is high or low, uplink data transmission path selection is performed properly. |

| Option | NR: NSA_DC_UL_PATH_SELECTION_SW Option of NRCellNsaDcConfig.*NsaDcAlgoSwitch* | LTE: NSA_DC_UL_PATH_SELECTION_SW Option of NsaDcMgmtConfig.*NsaDcAlgoSwitch* | Possible Situation |
|---|---|---|---|
| | Off > On | On | NSA UEs already admitted are not affected. However, NSA UEs newly admitted during this period are affected. When the uplink NR SINR is medium, the NR side still attempts to initiate uplink data transmission path selection. As the function on the LTE side does not take effect during this period, the selection fails. When the uplink NR SINR is high or low, uplink data transmission path selection is performed properly. |
| | In other scenarios | | NSA UEs already admitted are not affected. Uplink data transmission path selection does not take effect for NSA UEs newly admitted during this period. |

| Option | NR: NSA_DC_UL_PATH_SELECTION_SW Option of NRCellNsaDcConfig.*NsaDcAlgoSwitch* | LTE: NSA_DC_UL_PATH_SELECTION_SW Option of NsaDcMgmtConfig.*NsaDcAlgoSwitch* | Possible Situation |
|---|---|---|---|
| Option 3 | On | On > Off | NSA UEs already admitted and NSA UEs newly admitted during this period are both affected. No matter whether the uplink NR SINR is high, medium, or low, the NR side still attempts to initiate uplink data transmission path selection before receiving the switch status change from the LTE side. As the switch on the LTE side has been turned off, the messages from the NR side are rejected and the selection fails. |
| | Off > On | On | NSA UEs already admitted are not affected. However, NSA UEs newly admitted during this period are affected. No matter whether the uplink NR SINR is high, medium, or low, the NR side still attempts to initiate uplink data transmission path selection. As the function on the LTE side does not take effect during this period, the selection fails. |

| Option | NR: NSA_DC_UL_PATH_SELECTION_SW Option of NRCellNsaDcConfig.*NsaDcAlgoSwitch* | LTE: NSA_DC_UL_PATH_SELECTION_SW Option of NsaDcMgmtConfig.*NsaDcAlgoSwitch* | Possible Situation |
|---|---|---|---|
| | In other scenarios | | NSA UEs already admitted are not affected. Uplink data transmission path selection does not take effect for NSA UEs newly admitted during this period. |

### 4.1.2.3 NSA UE Scheduling Protection Based on MCG Cell Load

**In NSA DC, when an MCG cell is heavily loaded and LTE UEs are not fully scheduled due to preemption of physical resource block (PRB) resources by NSA UEs, NSA UE scheduling protection based on MCG cell load can be used to reduce the impact on LTE services and increase the LTE user-perceived rate.**

For example, the total rate of the MCG cell is 100 Mbit/s, and the expected rates of both LTE-only UEs and NSA UEs are 70 Mbit/s. If the **CellDlschAlgo**.***McgHighLoadThreshold*** parameter is set to **100**, this function is disabled and the rates of LTE-only UEs and NSA UEs are both 50 Mbit/s. If the **CellDlschAlgo**.***McgHighLoadThreshold*** parameter is set to a value other than **100**, LTE-only UEs can be preferentially scheduled and their maximum rate is 70 Mbit/s, but the maximum rate of NSA UEs is only 30 Mbit/s.

- When the eNodeB determines that the PRB usage of the current MCG cell is greater than or equal to the value of **CellDlschAlgo**.***McgHighLoadThreshold***, the eNodeB adjusts the downlink scheduling priority of non-GBR services of NSA UEs in the current MCG cell to the lowest.
- When the eNodeB determines that the PRB usage of the current MCG cell is less than the value of **CellDlschAlgo**.***McgHighLoadThreshold*** minus 10%, the eNodeB restores the downlink scheduling priority of non-GBR services of NSA UEs in the current MCG cell.

**NOTE**

- Scheduling priority adjustment is performed only for non-GBR services with QCIs other than 5 or 69. The scheduling priorities of GBR services are not adjusted.
- To prevent ping-pong priority adjustment, the offset value 10% is added when the eNodeB determines whether the PRB usage of the current LTE cell is less than the specified threshold.

## 4.1.3 Uplink Power Control Enhancement in NSA DC

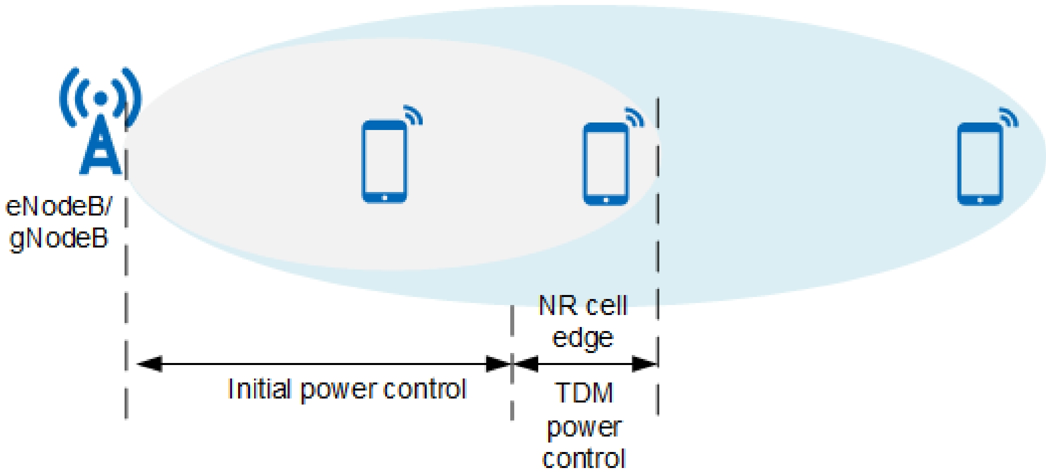
#### 4.1.3.1 TDM Power Control

**The eNodeB and the gNodeB can control the uplink transmit power of NSA UEs in the NSA DC state. Uplink power control consists of initial power control and time division multiplexing (TDM) power control.**

Initial power control is performed for NSA UEs that initially access the network, as shown in **Figure 4-2**. For details, see *EPC-based NSA Basics*.

**TDM power control is performed when trigger conditions are met in networking scenarios with both LTE FDD and NR TDD. In this case, UEs can use all power to send data to the LTE and NR sides in TDM mode, improving the uplink coverage of LTE and NR.**

**Figure 4-2** Uplink power control in NSA DC




## Basic Principles

TDM power control takes effect when all the following options are selected: (1) **NSA_DC_ENH_UL_POWER_CONTROL_SW** and **TDM_SWITCH** options of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter on the LTE side; (2) **NSA_DC_ENH_UL_POWER_CONTROL_SW** option of the **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** parameter on the NR side. After this function takes effect, the LTE and NR sides separately check whether a UE meets the conditions for triggering TDM power control, that is, whether the UE is located at the NR cell edge. When either side determines that the UE supports TDM power control and meets the trigger conditions, TDM power control is triggered. When both the LTE and NR sides determine that the UE meets the exit conditions, TDM power control is stopped. **Table 4-3** describes the trigger and exit conditions of TDM power control.

**Table 4-3** Trigger and exit conditions of TDM power control

| LTE/NR Side | Trigger Condition | Exit Condition |
|---|---|---|
| On the LTE side | Uplink SINR of the UE on the LTE side ≤ **NsaDcMgmtConfig**.***NsaTdmPcTrigSinrThld*** | Uplink SINR of the UE on the LTE side > **NsaDcMgmtConfig**.***NsaTdmPcTrigSinrThld*** + 7 |
| On the NR side | Uplink SINR of the UE on the NR side + **NRDUCellSrsMeas**.***NsaTdmPcTrigSinrHyst*** ≤ **NRDUCellSrsMeas**.***NsaTdmPcTrigSinrThld*** | Uplink SINR of the UE on the NR side – **NRDUCellSrsMeas**.***NsaTdmPcTrigSinrHyst*** > **NRDUCellSrsMeas**.***NsaTdmPcTrigSinrThld*** |

This function requires time synchronization between LTE FDD and NR TDD in NSA DC and calculates the relative frame offset between LTE and NR to determine the TDM pattern that can take effect. This function supports relative frame offsets of 0 ms and 3 ms, with the latter indicating that the frame offset on the LTE side is 3 ms earlier than that on the NR side. Calculating the relative frame offset between LTE and NR requires the selection of both the **LNR_RELATIVE_FRM_OFS_ADAPT_SW** option of the **NsaDcAlgoParam**.***NsaDcAlgoSwitch*** parameter on the LTE side and the **LNR_RELATIVE_FRM_OFS_ADAPT_SW** option of the **gNodeBParam**.***NsaDcOptSwitch*** parameter on the NR side. The LTE side sends its frame offset to the NR side, and the NR side calculates the relative frame offset.

The TDM pattern is finally determined by the NR side, regardless of whether TDM power control is triggered by the LTE or NR side. In addition, the TDM pattern can be selected only from SA0 to SA6 of LTE TDD. For details about TDM patterns, see section 6.2.2 "Message definitions" in 3GPP TS 36.331 V15.7.0. The NR side sends the TDM pattern to the LTE side through an SgNB Modification Required message. Upon reception of this message, the LTE side sends the TDM pattern to the UE. Then, TDM power control takes effect on both the UE and base station sides.

When TDM power control is enabled, it is recommended that the **TDM_CHG_WITH_INTRA_CELL_HO_SW** option of the **NsaDcAlgoParam**.***NsaDcAlgoSwitch*** parameter be selected. Under this setting, an intra-MeNB-cell handover is triggered when there is a TDM status change, ensuring that the TDM status of the MeNB is consistent with that of the NSA UE.

## TDM-Pattern Selection Policies

In networking scenarios with both LTE FDD and NR TDD, the LTE side needs to avoid conflict with NR uplink slots. In addition, the NR side needs to disable some self-contained slots to ensure the number of available LTE uplink subframes. The NR side determines the TDM pattern based on its slot configuration and the following policies.

**NOTE**

In the following figures, ⊘ indicates "Prohibited".

- As shown in **Figure 4-3** through **Figure 4-6**, when the relative frame offset between LTE FDD and NR is 0 ms, the **gNodeBParam**.***NsaDcResCoordScenario*** parameter must be set to **SYNC_NSA_DC_0_FRAME_OFS**. Under this setting, if TDM power control is triggered, the NR side configures the TDM pattern based on the relative frame offset of 0 ms.

**Figure 4-3** Example of TDM pattern when NR uses 4:1 slot configuration (relative frame offset of 0 ms)

| Harqoffset=3 | | Subframe number | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 0 | | 1 | | 2 | | 3 | | 4 | | 5 | | 6 | | 7 | | 8 | | 9 | |
| TDM-Pattern:sa1 | 1ms | U | | U | | D | | D | | S | | U | | U | | D | | D | | S | |
| LTE FDD | 1ms | U | | U | | ⊘ | | ⊘ | | ⊘ | | U | | U | | ⊘ | | ⊘ | | ⊘ | |
| NR TDD | 0.5ms | D | D | D | ⊘ (S) | U | D | D | D | S | U | D | D | D | ⊘ (S) | U | D | D | D | S | U |

**Figure 4-4** Example of TDM pattern when NR uses single-period 8:2 slot configuration (relative frame offset of 0 ms)

| Harqoffset=3 | | Subframe number | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 0 | | 1 | | 2 | | 3 | | 4 | | 5 | | 6 | | 7 | | 8 | | 9 | |
| TDM-Pattern:sa0 | 1ms | U | | U | | U | | D | | S | | U | | U | | U | | D | | S | |
| LTE FDD | 1ms | U | | U | | U | | ⊘ | | ⊘ | | U | | U | | U | | ⊘ | | ⊘ | |
| NR TDD | 0.5ms | D | D | D | D | D | D | D | S | U | U | D | D | D | D | D | D | D | S | U | U |

**Figure 4-5** Example of TDM pattern when NR uses dual-period 7:3 slot configuration (relative frame offset of 0 ms)

| Harqoffset=3 | | Subframe number | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 0 | | 1 | | 2 | | 3 | | 4 | | 5 | | 6 | | 7 | | 8 | | 9 | |
| TDM-Pattern:sa1 | 1ms | U | | U | | D | | D | | S | | U | | U | | D | | D | | S | |
| LTE FDD | 1ms | U | | U | | ⊘ | | ⊘ | | ⊘ | | U | | U | | ⊘ | | ⊘ | | ⊘ | |
| NR TDD | 0.5ms | D | D | D | ⊘ (S) | U | D | D | S | U | U | D | D | D | ⊘ (S) | U | D | D | S | U | U |

**Figure 4-6** Example of TDM pattern when NR uses dual-period 8:2 slot configuration (relative frame offset of 0 ms)

| Harqoffset=7 | | Subframe number | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 0 | | 1 | | 2 | | 3 | | 4 | | 5 | | 6 | | 7 | | 8 | | 9 | |
| TDM-Pattern:sa6 | 1ms | U | | U | | D | | S | | U | | U | | D | | D | | S | | U | |
| LTE FDD | 1ms | U | | U | | ⊘ | | ⊘ | | U | | U | | ⊘ | | ⊘ | | ⊘ | | U | |
| NR TDD | 0.5ms | D | D | D | ⊘ (S) | U | U | D | D | D | D | D | D | D | S | U | U | D | D | D | D |

- As shown in **Figure 4-7**, **Figure 4-8**, and **Figure 4-9**, when the relative frame offset between LTE FDD and NR is 3 ms, the **gNodeBParam**.***NsaDcResCoordScenario*** parameter must be set to **SYNC_NSA_DC_3Pl2Mi_FRAME_OFS**. Under this setting, if TDM power control is triggered, the NR side configures the TDM pattern based on the relative frame offset of 3 ms.

**Figure 4-7** Example of TDM pattern when NR uses 4:1 slot configuration (relative frame offset of 3 ms)

| Harqoffset=4 | | Subframe number | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| TDM-Pattern:sa2 | 1ms | S | U | D | D | D | S | U | D | D | D |
| LTE FDD | 1ms | ⊘ | U | ⊘ | ⊘ | ⊘ | ⊘ | U | ⊘ | ⊘ | ⊘ |
| NR TDD | 0.5ms | U D | D D | S U | D D | D S | U D | D D | S U | D D | D S |

**Figure 4-8** Example of TDM pattern when NR uses single-period 8:2 slot configuration (relative frame offset of 3 ms)

| Harqoffset=7 | | Subframe number | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| TDM-Pattern:sa6 | 1ms | U | U | D | S | U | U | D | D | S | U |
| LTE FDD | 1ms | U | U | ⊘ | ⊘ | U | U | ⊘ | ⊘ | ⊘ | U |
| NR TDD | 0.5ms | D D | D ⊘(S) | U U | D D | D D | D D | D S | U U | D D | D D |

**Figure 4-9** Example of TDM pattern when NR uses dual-period 7:3 slot configuration (relative frame offset of 3 ms)

| Harqoffset=4 | | Subframe number | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| TDM-Pattern:sa2 | 1ms | S | U | D | D | D | S | U | D | D | D |
| LTE FDD | 1ms | ⊘ | U | ⊘ | ⊘ | ⊘ | ⊘ | U | ⊘ | ⊘ | ⊘ |
| NR TDD | 0.5ms | U D | D ⊘(S) | U U | D D | D S | U D | D ⊘(S) | U U | D D | D S |

## Deployment Constraints

- In NSA DC, after TDM power control takes effect in an LTE FDD cell, the PUCCH in the cell always uses format 3, which increases the number of RBs occupied by the PUCCH. As a result, the number of RBs available for the PUSCH decreases, and the overall cell throughput may decrease.
- If the **Cell.*UePowerMax*** parameter value configured on the LTE side is less than the maximum transmit power supported by the UE (for example, 23 dBm), the UE's maximum transmit power on the LTE side cannot reach 23 dBm in TDM power control mode. As a result, TDM power control offers lower uplink coverage gains for LTE in NSA DC.
- TDM power control does not take effect for NSA UEs that support power class 2. For details about UE power class, see section 4.2 "UE Capability Parameters" in 3GPP TS 38.306 V15.8.0.
- If time is not synchronized between LTE FDD and NR TDD cells in NSA networking, there is a possibility that LTE and NR data is transmitted in the uplink at the same time. In this case, LTE or NR services may be affected because the UE behavior cannot be predicted.
- Assume that the power saving BWP function is enabled for UEs (by selecting the **BWP2_SWITCH** option of the **NRDUCellUePwrSaving.*BwpPwrSavingSw***

parameter). When the relative frame offset between LTE FDD and NR is 0 ms and the NR slot assignment is single-period 4:1 or dual-period 7:3, the UEs send SRSs in slots 3 and 13 on the NR side; in this case, some transmit power in the corresponding LTE uplink subframes is used. When the relative frame offset between LTE FDD and NR is 3 ms and the NR slot assignment is single-period 8:2, the UEs send SRSs in slot 17 on the NR side; in this case, some transmit power in the corresponding LTE uplink subframes is used.

- After TDM power control takes effect for a UE, the UE can send data on the LTE side only in the uplink subframes indicated in the TDM pattern. If the LTE PRACH time-domain position is not in the uplink subframes indicated in the TDM pattern, the UE cannot perform random access during a handover or resynchronization on the LTE side. If the UE is out of synchronization in TDM mode and the LTE PRACH time-domain position is not in the uplink subframes indicated in the TDM pattern, resynchronization cannot be performed. In such case, the UE re-accesses the LTE network through RRC connection reestablishment. To avoid the preceding situations, LTE PRACH configuration must be restricted when TDM power control is enabled. In TDM patterns shown in **Figure 4-3** through **Figure 4-9**, subframe 1 is configured to allow transmission. To enable LTE to implement resynchronization in TDM mode, the time-domain position of the PRACH in an LTE cell must include subframe 1. That is, the LTE cell must meet any of the following conditions:
  - The **RACHCfg**.***PrachConfigIndexCfgInd*** parameter is set to **NOT_CFG**, the **ENodeBAlgoSwitch**.***PrachTimeStagSwitch*** parameter is set to **OFF**, and the **RachAdjSwitch** option of the **CellAlgoSwitch**.***RachAlgoSwitch*** parameter is deselected.
  - The **RACHCfg**.***PrachConfigIndexCfgInd*** parameter is set to **CFG**, and the **RACHCfg**.***PrachConfigIndex*** parameter is set to **0**, **3**, **6**, **9**, **13**, **14**, **16**, **19**, **22**, **25**, **29**, **32**, **35**, **38**, **41**, **45**, **48**, **51**, **54**, or **57**.

### 4.1.3.2 Network-Coordinated Dynamic UE Power Sharing

If an NSA UE supports dynamic power sharing, the UE power must be preferentially allocated to the LTE side and then to the NR side, according to section 7.6 "Dual connectivity" in 3GPP TS 38.213 V16.1.0. If most or all of the UE power is allocated to the LTE side but only little or even no power is allocated to the NR side, uplink data cannot be sent on the NR side and the SCG will be released.

In subframes with both LTE and NR, power control and coordinated scheduling are performed to ensure that there is available uplink power on the NR side. In scenarios with both LTE and NR TDD, this function supports the following slot assignments (specified by the **NRDUCell**.***SlotAssignment*** parameter) for NR TDD cells: 4:1, single-period 8:2, and dual-period 8:2, 7:3, and 2:3.

Assume that both the **NSA_DC_COOPERATION_DPS_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter on the LTE side and the **NSA_DC_COOPERATION_DPS_SW** option of the **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** parameter on the NR side are selected. If the UE does not support dynamic power sharing, the maximum uplink transmit power assigned by the eNodeB to the LTE and NR sides of the UE is specified by **NsaDcMgmtConfig**.***NsaDcUeMcgUlMaximumPower*** and **NsaDcMgmtConfig**.***NsaDcUeScgUlMaximumPower***, respectively. If the UE supports dynamic power sharing:

- In non-time-synchronization scenarios, the maximum uplink transmit power assigned by the eNodeB to the LTE and NR sides of the UE is the value of **NsaDcMgmtConfig**.***NsaDcUeMcgUlMaximumPower*** and 23 dBm, respectively.
- In time synchronization scenarios:
  - For the combination of LTE FDD and NR TDD:
    - If the ratio of downlink slots to uplink slots on the NR side is 7:3 or 2:3, the maximum uplink transmit power assigned by the eNodeB to the LTE and NR sides of the UE is 20 dBm and 23 dBm, respectively.
    - If the ratio of downlink slots to uplink slots on the NR side is 4:1 or 8:2, the maximum uplink transmit power assigned by the eNodeB to the LTE and NR sides of the UE is 23 dBm and 23 dBm respectively when the UE is at the LTE cell edge, or 20 dBm and 23 dBm respectively when the UE is not at the edge.
  - For the combination of LTE FDD and NR FDD or the combination of LTE TDD and NR FDD:

    The maximum uplink transmit power assigned by the eNodeB to the LTE and NR sides of the UE is 20 dBm and 23 dBm, respectively.
  - For the combination of LTE TDD using SA2 and NR TDD:
    - If the relative frame offset between LTE TDD and NR TDD is 0 and the ratio of downlink slots to uplink slots on the NR side is 4:1, 7:3, dual-period 8:2, or 2:3, the maximum uplink transmit power assigned by the eNodeB to the LTE and NR sides of the UE is 20 dBm and 23 dBm, respectively.
    - If the relative frame offset between LTE TDD and NR TDD is 0 and the ratio of downlink slots to uplink slots on the NR side is single-period 8:2, the maximum uplink transmit power assigned by the eNodeB to the LTE and NR sides of the UE is 23 dBm and 23 dBm, respectively.

Assume that the **NSA_DC_COOPERATION_DPS_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter on the LTE side and the **NSA_DC_COOPERATION_DPS_SW** option of the **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** parameter on the NR side are not both selected.

- If the UE supports dynamic power sharing:
  - If the **NSA_DC_DPS_CONTROL_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter on the LTE side is selected, the maximum uplink transmit power assigned by the eNodeB to the LTE and NR sides of the UE is 23 dBm and 23 dBm, respectively. In this case, if most or all of the UE power is allocated to the LTE side but only little or even no power is allocated to the NR side, uplink data cannot be sent on the NR side and the SCG will be released.
  - If the **NSA_DC_DPS_CONTROL_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter on the LTE side is deselected, the maximum uplink transmit power assigned by the eNodeB to the LTE and NR sides of the UE is the value of **NsaDcMgmtConfig**.***NsaDcUeMcgUlMaximumPower*** and 23 dBm, respectively.

- If the UE does not support dynamic power sharing, the maximum uplink transmit power assigned by the eNodeB to the LTE and NR sides of the UE is specified by **NsaDcMgmtConfig**.***NsaDcUeMcgUlMaximumPower*** and **NsaDcMgmtConfig**.***NsaDcUeScgUlMaximumPower***, respectively.

NOTE

The **NSA_DC_DPS_CONTROL_SW** and **NSA_DC_COOPERATION_DPS_SW** options of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter on the LTE side cannot be both selected.

### 4.1.3.3 PC2 UE Uplink NR Power Configuration Maximization

According to section 6.2B "Transmitter power for DC" in 3GPP TS 38.101-3 V16.10.0, the maximum total uplink transmit power on the LTE and NR sides of an NSA UE that supports power class 2 (PC2) is 26 dBm, with the maximum uplink transmit power being 23 dBm on the LTE side and 26 dBm on the NR side. After PC2 UE uplink NR power configuration maximization takes effect, the NR side increases the configured uplink power based on the UE-reported capability to maximize the total transmit power of PC2 UEs. This function depends on the band combination capabilities reported by UEs. It is recommended that this function be enabled when the n41/n77/n78/n79/n257/n258/n260/n261 band is used in NSA networking.

This function takes effect when the **PC2_UE_NR_UL_PWR_CFG_MAX_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter on the LTE side and the **PC2_UE_NR_UL_PWR_CFG_MAX_SW** option of the **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** parameter on the NR side are both selected and an NSA UE supports PC2. After this function takes effect, the maximum uplink transmit power delivered by the eNodeB to the UE is as follows:

- If the UE does not support dynamic power sharing, the maximum uplink transmit power on both the LTE and NR sides is 23 dBm.
- If the UE supports dynamic power sharing, the maximum uplink transmit power on the LTE side is 23 dBm and that on the NR side is 26 dBm.

The configuration of the maximum total uplink transmit power of NSA UEs is controlled by the **NRDUCellSelConfig**.***MaxUeTransmitPower*** parameter on the NR side. Only when this parameter is set to a value greater than or equal to **26**, the **PC2_UE_NR_UL_PWR_CFG_MAX_SW** option of the **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** parameter can be selected.

For UE compatibility considerations, a blacklist mechanism can be applied to this function. When the **PC2_UE_NR_UL_PWR_CFG_MAX_OFF** option of the **gNBUeCompatSw**.***BlacklistCtrlSwitch*** parameter on the NR side is selected, this function does not take effect for blacklisted UEs.

**NOTE**

Due to the switch negotiation mechanism, the LTE side sends the switch change information to the NR side every 5 minutes. As such, there is a delay between the time when the switch on the LTE side is modified and the time when the function effective state changes.

When a switch is modified,

- For UEs that initially access, are handed over to, or have their RRC connections reestablished in the cell before the function takes effect, the original switch setting takes effect.
- For UEs that initially access, are handed over to, or have their RRC connections reestablished in the cell after the function takes effect, the new switch setting takes effect. Batch reconfiguration is not performed for existing UEs.

If both the PC2 UE uplink NR power configuration maximization function and the network-coordinated dynamic UE power sharing function are enabled:

- If the UE supports PC2, the PC2 UE uplink NR power configuration maximization function preferentially takes effect.
- If the UE does not support PC2, the network-coordinated dynamic UE power sharing function takes effect.

# 4.2 Network Analysis

## 4.2.1 Benefits

- SN-terminated MCG bearer transmission

  This function prevents frequent switching of the NSA UE bearer between the gNodeB and the eNodeB when NR coverage is discontinuous, reducing the impact of excessive signaling on the core network.

- Uplink data transmission path selection
  - In the typical scenario where LTE FDD 1.8 GHz and NR TDD 3.5 GHz are used on the same site, assume that LTE uplink interference and uplink load are light, that is, the LTE uplink bandwidth is greater than or equal to 10 MHz and the average uplink PRB usage is less than 20%. For UEs moving at low speeds and performing uplink large-packet services, the uplink data transmission path selection function significantly improves uplink experience. For UEs in non-cell-center areas where the NR cell RSRP is less than –100 dBm, this function increases the average uplink UE throughput by more than 20% compared with uplink data transmission only on the SCG side.
  - When the uplink throughput of LTE is equivalent to that of NR (for example, in LTE FDD + NR FDD scenarios) or when the uplink experience of LTE is worse than that of NR (for example, in LTE TDD scenarios), uplink data transmission is seldom or even never switched to LTE and the function gains are not obvious.

  The gains provided by the uplink data transmission path selection function depend on LTE and NR cell conditions such as the frequency band, bandwidth, UE quantity, uplink load, interference, and RSRP. The gains are higher if the number of NR UEs is larger but the number of LTE UEs is smaller, the NR RSRP is lower but the LTE RSRP is higher, and the LTE uplink load and interference are lighter.

- Network-coordinated dynamic UE power sharing

  This function increases the uplink transmit power of NSA UEs by 0 dB to 3 dB and accordingly increases the uplink throughput and downlink TCP throughput.

- PC2 UE uplink NR power configuration maximization

  After this function takes effect, the maximum NR uplink coverage is expanded and the maximum used-perceived rate increases for CEUs. The gains provided by this function depend on the uplink LTE transmit power of PC2 UEs. A lower uplink LTE transmit power used by PC2 UEs results in higher maximum gains. The actual gains depend on the UE implementation. This function only changes the maximum power configuration on the base station.

## 4.2.2 Impacts

The impacts between this function and other functions are described in **4.3.2.3 Function Impacts**.

- SN-terminated MCG bearer transmission has the following network impacts:
  - X2 traffic between LTE and NR increases, S1 throughput on the NR side increases, and S1 throughput on the LTE side decreases.
  - SN-terminated MCG bearer UEs face longer user-plane delays due to X2 delays.
  - The number of NSA UEs in the DC state increases, that is, the value of the **L.Traffic.User.NsaDc.PCell.Avg** counter increases. This is because SN-terminated MCG bearer UEs are also NSA UEs according to 3GPP specifications.
  - The values of counters related to SCG modification increase. Such counters include **L.NsaDc.SCG.Mod.Required.Att** and **N.NsaDc.SgNB.Mod.Required.Att**.
  - On the LTE side, the values of counters related to SgNB addition and deletion decrease. Such counters include **L.NsaDc.SgNB.Add.Att**, **L.NsaDc.SgNB.Rmv.Att**, **L.NsaDc.SgnbTrig.SgNB.NormRel**, and **L.NsaDc.MenbTrig.SgNB.NormRel**. KPIs calculated based on the preceding counters will also change.
  - On the NR side, counters related to SgNB addition and deletion are incremented in more scenarios. When an NSA UE transits from the split bearer state to the SN-terminated MCG bearer state and the SCG is deleted, an SgNB deletion counter is incremented. When an NSA UE transits from the split bearer state to the SN-terminated MCG bearer state and an SCG is added, an SgNB addition counter is incremented. Therefore, after SN-terminated MCG bearer transmission is enabled, related counters are no longer comparable between the NR and LTE sides. For example, **N.NsaDc.SgNB.Add.Att** and **L.NsaDc.SgNB.Add.Att** are not comparable, or **N.NsaDc.SgNB.Rel** and **L.NsaDc.SgNB.Rmv.Att** are not comparable.
- After TDM power control takes effect for an NSA UE, the UE can send data on the LTE side only in the uplink subframes indicated in the configured TDM pattern. This has the following impacts on the network:
  - The uplink throughput of the NSA UE may decrease.

- If the LTE PRACH time-domain position is not in the uplink subframes indicated in the TDM pattern, the NSA UE cannot perform random access on the LTE side, affecting handover or resynchronization on this side.

- The uplink data transmission path selection function has the following network impacts:
  - The number of uplink packets lost on the LTE side (**L.Traffic.UL.PktLoss.Loss**) and the number of uplink packets lost on the NR side (**N.PDCP.UL.TrfSDU.RxPacket.Loss**) may increase if the following conditions are met: (1) uplink fallback to LTE is enabled first; (2) then, uplink data transmission path selection is enabled; (3) the value of **NRDUCellSrsMeas**.***NsaUlPathSelSinrLowThld*** or **NRDUCellSrsMeas**.***NsaSplitUlPathSelSinrThld*** is decreased.
  - The proportion of downlink traffic at the cell edge may increase as the uplink data transmission performance at the cell edge improves. This may decrease the downlink data rates of NR UEs and increase the proportion of user data distributed to the LTE side.
  - The uplink data volume on the LTE side may increase as NR UEs running uplink services at the cell edge fall back to LTE. In addition, the increase in the number of LTE CEUs may decrease the LTE uplink rate and increase the packet loss rate.
  - The number of SgNB Modification Required messages sent by the NR side increases, and the probability of procedure conflicts increases. As a result, the SgNB modification success rate is likely to decrease.
  - The service drop rate on the LTE side may increase as the number of RRC connection reconfiguration messages sent on the LTE side increases.
  - The rate of abnormal SgNB releases triggered by the SgNB increases, but the total rate of abnormal SgNB releases triggered by the MeNB and SgNB decreases.

- The PC2 UE uplink NR power configuration maximization function has the following network impacts:

  The maximum available power on the NR side increases by 3 dB for PC2 UEs. Affected by the dynamic power sharing function and the maximum total uplink transmit power supported by UEs, the actual power on the NR side increases by 0 dB to 3 dB, which may increase the background noise and cause KPI changes.

# 4.3 Requirements

## 4.3.1 Licenses

On the LTE side:

| RAT | Feature ID | Feature Name | Model | Sales Unit |
| --- | --- | --- | --- | --- |
| LTE FDD | MRFD-151223 | EN-DC Performance Enhancement (LTE FDD) | LT1SENDCCE0F | Per Cell |
| LTE TDD | MRFD-151233 | EN-DC Performance Enhancement (LTE TDD) | LT1SENDCCE0T | Per Cell |

On the NR side:

| RAT | Feature ID | Feature Name | Model | Sales Unit |
| --- | --- | --- | --- | --- |
| NR | MRFD-151263 | EN-DC Performance Enhancement (NR) | NR0SENDCCE00 | Per Cell |

**NOTE**

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

## 4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

### 4.3.2.1 Prerequisite Functions

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| LTE FDD<br>LTE TDD | NSA DC capability | **NSA_DC_CAPABILITY_SWITCH** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter | *EPC-based NSA Basics* | The **NSA_DC_CAPABILITY_SWITCH** option needs to be selected if blind PSCell addition for EPS fallback UEs is required. |
| LTE FDD | NSA DC capability | **NSA_DC_CAPABILITY_SWITCH** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter | *EPC-based NSA Basics* | The **NSA_DC_CAPABILITY_SWITCH** option needs to be selected if TDM power control is required. |
| LTE FDD<br>LTE TDD | NSA DC capability | **NSA_DC_CAPABILITY_SWITCH** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter | *EPC-based NSA Basics* | The **NSA_DC_CAPABILITY_SWITCH** option needs to be selected if uplink data transmission path selection is required. |
| LTE FDD<br>LTE TDD | SRS resource management | **SRSCfg**.***SrsCfgInd*** | *Physical Channel Resource Management* in eRAN feature documentation | Uplink data transmission path selection in NSA scenarios requires the following configurations:<br>• The **SRSCfg**.***SrsCfgInd*** parameter is set to **BOOLEAN_TRUE**.<br>• The **Cell**.***HighSpeedFlag*** parameter is set to **LOW_SPEED**.<br>• The **NsaDcAlgoParam**.***HoNsaBandCombSelectPolicy*** parameter is set to **LTE_SCC_NR_SCG_FIRST**. |
| LTE FDD<br>LTE TDD | High speed mobility | **Cell**.***HighSpeedFlag*** | *High Speed Mobility* in eRAN feature documentation | |
| LTE FDD<br>LTE TDD | HO NSA Band Combination Selection Policy | **NsaDcAlgoParam**.***HoNsaBandCombSelectPolicy*** | None | |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| NR FDD<br>Low-frequency NR TDD<br>High-frequency NR TDD | UE transmit power control | **NRDUCellSelConfig**.***MaxUeTransmitPower*** | *Power Control* in 5G RAN feature documentation | The PC2 UE uplink NR power configuration maximization function requires that the **NRDUCellSelConfig**.***MaxUeTransmitPower*** parameter be set to a value greater than or equal to **26**. |

### 4.3.2.2 Mutually Exclusive Functions

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| LTE FDD<br>LTE TDD | PSCell management for VoLTE UEs | **NsaDcMgmtConfig**.***VolteUeScgMgmtStrategy*** | *EPC-based NSA Basics* | The **NsaDcMgmtConfig**.***VolteUeScgMgmtStrategy*** parameter cannot be set to **VOLTE_PREFERRED** if blind PSCell addition for EPS fallback UEs is enabled. |
| LTE FDD | Ultra-low-latency scheduling for inter-eNodeB CA based on relaxed backhaul | **INTER_ENB_ULTRA_LOW_LATENCY_SW** option of the **MultiCarrUnifiedSch**.***MultiCarrierUnifiedSchEnSw*** parameter | *Multi-carrier Unified Scheduling* in eRAN feature documentation | Ultra-low-latency scheduling for inter-eNodeB CA based on relaxed backhaul is not compatible with TDM. |
| NR TDD | SRS carrier switching | **SRS_CARRIER_SWITCHING_SW** option of the **NRDUCellCarrMgmt**.***CaEnhancedAlgoSwitch*** parameter | *CA Experience Improvement* in 5G RAN feature documentation | SRS carrier switching is not compatible with TDM power control. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| NR FDD<br>Low-frequency NR TDD | Uplink fallback to LTE | **UL_FALLBACK_TO_LTE_SWITCH** option of the **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** parameter | *EPC-based NSA Basics* | Uplink data transmission path selection is not compatible with uplink fallback to LTE. |
| LTE FDD | Absence of SRS configuration for LTE UEs | **LTE_UE_SRS_NOT_CONFIG_SW** option of the **SpectrumCloud**.***SpectrumCloudEnhSwitch*** parameter | *LTE FDD and NR Dynamic Spectrum Sharing* | Absence of SRS configuration for LTE UEs is not compatible with uplink data transmission path selection. |
| LTE FDD | LTE and NR SRS resource allocation optimization | **LTE_NR_SRS_ALLOC_OPT_SW** option of the **LteNrSpctShrCellGrp**.***LteNrSpctShrSwitch*** parameter | *LTE FDD and NR Dynamic Spectrum Sharing*<br>*LTE FDD and NR Hybrid Dynamic Spectrum Sharing* | LTE and NR SRS resource allocation optimization is not compatible with uplink data transmission path selection. |
| NR FDD | LTE FDD and NR Flash Dynamic Spectrum Sharing | **LTE_NR_FDD_SPCT_SHR_SW** option of the **NRDUCellAlgoSwitch**.***SpectrumCloudSwitch*** parameter | *LTE FDD and NR Dynamic Spectrum Sharing* | LTE FDD and NR Flash Dynamic Spectrum Sharing is not compatible with TDM power control when **NRDUCellTrp**.***TxRxMode*** is set to **32T32R**. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| NR FDD | Hybrid DSS Based on Asymmetric Bandwidth | **LTE_NR_FDD_SPCT_SHR_SW** option of the **NRDUCellAlgoSwitch**.***SpectrumCloudSwitch*** parameter and **LTE_NR_FDD_SPCT_SHR_ASYM_SW** option of the **NRDUCellAlgoSwitch**.***SpectrumCloudEnhSwitch*** parameter | *LTE FDD and NR Hybrid Dynamic Spectrum Sharing* | Hybrid DSS Based on Asymmetric Bandwidth is not compatible with TDM power control when **NRDUCellTrp**.***TxRxMode*** is set to **32T32R**. |

### 4.3.2.3 Function Impacts

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| LTE FDD<br>LTE TDD | Traffic model-based performance optimization | **DlPacketLenAwareSchSw** option of the **CellAlgoSwitch**.***DlSchSwitch*** parameter | *Downlink Scheduling* in eRAN feature documentation | The throughput of LTE UEs decreases, and the throughput of NSA UEs with SN-terminated MCG bearers increases. |
| LTE FDD<br>LTE TDD | Downlink non-GBR packet bundling | **NonGbrBundlingSwitch** option of the **CellAlgoSwitch**.***DlSchSwitch*** parameter | *Downlink Scheduling* in eRAN feature documentation | The throughput of LTE UEs increases, and the throughput of NSA UEs with SN-terminated MCG bearers decreases. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| LTE TDD | Downlink user-perceived rate optimization for heavily loaded cells | **CellDlSchExtParam**.***DlHeavyLoadOptPrbThld*** | *Downlink Scheduling* in eRAN feature documentation | NSA UE scheduling protection based on MCG cell load affects priority calculation. When downlink user-perceived rate optimization for heavily loaded cells takes effect, NSA UE scheduling protection based on MCG cell load does not take effect. |
| LTE FDD | Downlink-load-based scheduling policy adaptation | **DL_LOAD_BASED_SCH_POL_ADAPT_SW** option of the **CellDlSchExtParam**.***DlSchExtSwitch*** parameter | *Downlink Scheduling* in eRAN feature documentation | NSA UE scheduling protection based on MCG cell load affects priority calculation. When the downlink-load-based scheduling policy adaptation function is in effect, NSA UE scheduling protection based on MCG cell load does not take effect. |
| LTE FDD | Downlink-load-based scheduling priority optimization | **DL_LOAD_BASED_SCH_PRI_OPT_SW** option of the **CellDlSchExtParam**.***DlSchExtSwitch*** parameter | *Downlink Scheduling* in eRAN feature documentation | NSA UE scheduling protection based on MCG cell load affects priority calculation. When the downlink-load-based scheduling priority optimization function is in effect, NSA UE scheduling protection based on MCG cell load does not take effect. |
| LTE FDD<br>LTE TDD | Delay-based differentiated scheduling for extended QCIs | **QciPara**.***DlPdbForExtendedQci*** | *Downlink Scheduling* in eRAN feature documentation | The throughput of LTE UEs increases, and the throughput of NSA UEs with SN-terminated MCG bearers decreases. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| LTE FDD<br>LTE TDD | Breathing Pilot | **BreathingPilotSwitch** option of the **CellDlschAlgo**.***BreathingPilotAlgoSwitch*** parameter | *Breathing Pilot* in eRAN feature documentation | After Breathing Pilot is enabled, the downlink data volume of NSA UEs with SN-terminated MCG bearers may be lower than the downlink data volume threshold for large-packet services and consequently such UEs are identified as non-large-packet UEs. As a result, the gains brought by Breathing Pilot are affected. |
| LTE FDD | Downlink massive CA | **DlMassiveCaSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter | *Carrier Aggregation* in eRAN feature documentation | PUCCH format 3 is always used in an FDD cell with TDM power control having taken effect for NSA DC. However, PUCCH format 5 is used in a cell with downlink massive CA enabled. Therefore, TDM power control and downlink massive CA cannot take effect at the same time. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| LTE FDD | Downlink 5CC aggregation | **CaDl5CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter | *Carrier Aggregation* in eRAN feature documentation | In TDM mode, ACK/NACK feedback for multiple downlink subframes needs to be sent in the same uplink subframe. ACK/NACK feedback for all CCs may not be supported in downlink 5CC aggregation. Therefore, the maximum number of downlink CCs allowed in TDM mode is limited under certain uplink-downlink subframe configurations. Specifically, the maximum numbers under SA0 to SA6 are 5, 5, 4, 3, 3, 2, and 5, respectively. If TDM is required but the number of aggregated CCs exceeds the maximum number of CCs allowed in TDM mode, the excess SCCs will be deleted. SCCs can be added based on data volume but the total number of downlink CCs cannot exceed the maximum number. |

<table>
<tr><th>RAT</th><th>Function Name</th><th>Function Switch</th><th>Reference</th><th>Description</th></tr>
<tr><td>LTE FDD</td><td>Air interface latency optimization</td><td><ul><li>SRI_PERIOD_OPT_SW option of the CellQciPara.QciAlgoSwitch parameter</li><li>LowDelayServiceOptSwitch option of the CellAlgoSwitch.ServiceDiffSwitch parameter</li></ul></td><td>Air Interface Latency Optimization in eRAN feature documentation</td><td>After TDM takes effect, the SR period is fixed at 80 ms. TDM does not take effect in the following scenarios:<ul><li>The SRI_PERIOD_OPT_SW option of the CellQciPara.QciAlgoSwitch parameter is selected.</li><li>The LowDelayServiceOptSwitch option of the CellAlgoSwitch.ServiceDiffSwitch parameter is selected, and the CellQciPara.LowLatencyFlag parameter corresponding to the QCI is set to TRUE.</li></ul></td></tr>
</table>

<table>
<tr><th>RAT</th><th>Function Name</th><th>Function Switch</th><th>Reference</th><th>Description</th></tr>
<tr><td>LTE FDD</td><td>Fixed SRI Reporting Period</td><td>PUCCHCfg.FixedSriPeriod</td><td>Physical Channel Resource Management in eRAN feature documentation</td><td rowspan="2">After TDM takes effect, the SRI period is fixed at 80 ms. TDM does not take effect in the following scenarios:<ul><li>The PUCCHCfg.FixedSriPeriod parameter is configured.</li><li>The PUCCHCfg.SriPeriodAdaptive parameter is set to QCIADAPTIVE.</li><li>The Cell.DlBandWidth parameter is set to CELL_BW_N6, that is, the LTE cell bandwidth is 1.4 MHz.</li></ul></td></tr>
<tr><td>LTE FDD</td><td>SRI reporting period adaptation</td><td>PUCCHCfg.SriPeriodAdaptive</td><td>Physical Channel Resource Management in eRAN feature documentation</td></tr>
<tr><td>LTE FDD</td><td>TTI bundling</td><td>TtiBundlingSwitch option of the CellAlgoSwitch.UlSchSwitch parameter</td><td>VoLTE in eRAN feature documentation<br>Video Experience Optimization in eRAN feature documentation</td><td>TTI bundling and TDM cannot take effect simultaneously.</td></tr>
<tr><td>LTE FDD</td><td>Semi-persistent scheduling</td><td>SpsSchSwitch option of the CellAlgoSwitch.UlSchSwitch parameter<br>SpsSchSwitch option of the CellAlgoSwitch.DlSchSwitch parameter</td><td>VoLTE in eRAN feature documentation</td><td>Semi-persistent scheduling and TDM cannot take effect simultaneously.</td></tr>
</table>

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| LTE FDD | Short TTI | **SHORT_TTI_SW** option of the **CellShortTtiAlgo**.***SttiAlgoSwitch*** parameter | *Short TTI (FDD)* in eRAN feature documentation | Short TTI and TDM cannot take effect simultaneously. |
| LTE FDD | Virtual 4T4R | **Virtual4T4RSwitch** option of the **CellAlgoSwitch**.***EmimoSwitch*** parameter | *Virtual 4T4R (FDD)* in eRAN feature documentation | If virtual 4T4R has been enabled, TDM does not take effect. |
| LTE FDD | Uplink MU-MIMO | **UlVmimoSwitch** option of the **CellAlgoSwitch**.***UlSchSwitch*** parameter | *MIMO* in eRAN feature documentation | • When TDM is in progress, uplink MU-MIMO does not take effect.<br>• If TDM needs to be triggered while uplink MU-MIMO is in progress, uplink MU-MIMO does not take effect. |
| LTE FDD | Uplink SU-MIMO | **ULSUMIMO2LayersSwitch** option of the **CellAlgoSwitch**.***UlSuMimoAlgoSwitch*** parameter | *MIMO* in eRAN feature documentation | If uplink SU-MIMO has been enabled, TDM does not take effect. |
| LTE FDD | UL CoMP | **UlJointReceptionSwitch** option of the **CellAlgoSwitch**.***UplinkCompSwitch*** parameter | *UL CoMP* in eRAN feature documentation | • When TDM is in progress, UL CoMP does not take effect.<br>• If TDM needs to be triggered while UL CoMP is in progress, UL CoMP must stop first. |
| LTE FDD | eMBMS | **CellMBMSCfg**.***MBMSSwitch*** | *eMBMS* in eRAN feature documentation | If eMBMS has taken effect, TDM does not take effect. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| LTE FDD | Uplink CA in EN-DC | **NSA_LTE_UL_2CC_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoExtSwitch*** parameter | *EPC-based NSA Basics* | If both uplink CA in EN-DC and TDM power control are enabled, TDM power control preferentially takes effect. |
| LTE FDD<br>LTE TDD | RAN sharing with common carrier | **CellAlgoSwitch**.***RanShareModeSwitch*** | *RAN Sharing* in eRAN feature documentation | If dynamic sharing is configured for RAN sharing with common carrier, uplink data transmission path selection does not take effect. |
| LTE FDD<br>LTE TDD | WBB | **WBBMBB_USER_PRB_UP_LMT_SWITCH** option of the **CellAlgoSwitch**.***SpecUserAlgoSwitch*** parameter, and **CellWttxParaCfg**.***PrbUpLimitCtrlMode*** parameter | *WBB* in eRAN feature documentation | If both control over the maximum PRB usage of WBB or MBB UEs and tight control over the maximum PRB usage are enabled, uplink data transmission path selection does not take effect. |
| LTE FDD | LNR CRS muting | **LNR_CRS_MUTE_SW** option of the **CellAlgoExtSwitch**.***LnrZeroBufferzoneSw*** parameter | *LTE and NR Zero Bufferzone* in eRAN feature documentation | After LNR CRS muting is enabled, the downlink data volume of NSA UEs with SN-terminated MCG bearers may be lower than the downlink data volume threshold for large-packet services and consequently such UEs are identified as non-large-packet UEs. As a result, the gains brought by LNR CRS muting are affected. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| LTE FDD | LTE key event assurance | **LTE_BIG_EVENT_ASSURANCE_SW** option of the **LteNrSpctShrCellGrp**.***LteNrSpctShrSwitch*** parameter | *LTE FDD and NR Dynamic Spectrum Sharing* | If LTE key event assurance is enabled, uplink data transmission path selection does not take effect. |
| LTE FDD | Uplink full-antenna reception | **UL_COVERAGE_BOOST_SW** option of the **SectorSplitGroup**.***SectorSplitSwitch*** parameter | *Massive MIMO Enhancements (FDD)* in eRAN feature documentation | After uplink full-antenna reception takes effect, the uplink coverage on the LTE side improves, so the number of uplink data split mode changes may vary for NSA UEs that do not support uplink data split. This affects uplink data transmission path selection. |
| LTE FDD<br>LTE TDD<br>NR FDD<br>Low-frequency NR TDD | Initial power control in NSA DC | • **NsaDcMgmtConfig**.***NsaDcUeMcgUlMaximumPower***<br>• **NsaDcMgmtConfig**.***NsaDcUeScgUlMaximumPower*** | *EPC-based NSA Basics* | When the PC2 UE uplink NR power configuration maximization function takes effect for PC2 UEs, initial power control in NSA DC does not take effect. |
| LTE FDD<br>LTE TDD | Dynamic power sharing in NSA DC | **NSA_DC_DPS_CONTROL_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter | *EPC-based NSA Performance Enhancement* | When the PC2 UE uplink NR power configuration maximization function takes effect for PC2 UEs, dynamic power sharing in NSA DC does not take effect. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| LTE FDD<br>LTE TDD | Voice service UE power increase in NSA networking | **NSA_VOLTE_UL_POWER_BOOST_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoExtSwitch*** parameter | *VoLTE* in eRAN feature documentation | When the PC2 UE uplink NR power configuration maximization function takes effect for PC2 UEs, voice service UE power increase in NSA networking does not take effect. |
| Low-frequency NR TDD | Downlink intra-FR inter-band CA | **INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter | *Carrier Aggregation* in 5G RAN feature documentation | TDM power control allows for the LTE-NR relative frame offset of 3 ms. If the frame offset of an NR cell is changed and the downlink intra-FR inter-band CA feature is enabled for the NR cell, the frame offset of the PSCell must also be changed because this feature requires that the absolute value of the frame offset difference between NR carriers be less than or equal to 625 Ts.<br>For details about the requirements of downlink intra-FR inter-band CA on the frame offset difference between the PCell and SCells, see "Downlink Intra-FR Inter-Band CA" in *Carrier Aggregation* in 5G RAN feature documentation. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| NR FDD<br>Low-frequency NR TDD | DRX | **BASIC_DRX_SW** option of the **NRDUCellUePwrSaving.*NrDuCellDrxAlgoSwitch*** parameter | *DRX* in 5G RAN feature documentation | If DRX is enabled and the **FULL_BUFFER_UE_EXIT_DRX_SW** option of the **NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch*** parameter is selected, DRX cannot take effect for a UE when uplink data transmission path selection takes effect for this UE (as a UE selected by this path selection function is identified as a full-buffer UE). |
| Low-frequency NR TDD | UL and DL Decoupling | **NRDUCellAlgoSwitch**.***UlDlDecouplingSwitch*** | *UL and DL Decoupling* in 5G RAN feature documentation | If UL and DL Decoupling in SUL independent deployment mode has taken effect, TDM power control does not take effect. The TDM pattern corresponding to UL and DL Decoupling in SUL independent deployment mode is used. |
| Low-frequency NR TDD | UL and DL Decoupling | **NRDUCellAlgoSwitch**.***UlDlDecouplingSwitch*** | *UL and DL Decoupling* in 5G RAN feature documentation | If UL and DL Decoupling is enabled in an NR cell, uplink data transmission path selection does not take effect. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| Low-frequency NR TDD | UL and DL Decoupling | ● LTE: **TDM_SWITCH** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter<br>● NR: **NRDUCellAlgoSwitch**.***UlDlDecouplingSwitch*** | *UL and DL Decoupling* in 5G RAN feature documentation | When UL and DL Decoupling is in effect and the base station performs uplink scheduling in TDM mode, network-coordinated dynamic UE power sharing does not take effect. |
| Low-frequency NR TDD | Uplink fallback to LTE | **UL_FALLBACK_TO_LTE_SWITCH** option of the **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** parameter | *EPC-based NSA Basics* | If TDM is enabled, uplink fallback to LTE does not take effect. |
| Low-frequency NR TDD | Uplink data transmission path selection | ● LTE: **NSA_DC_UL_PATH_SELECTION_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter<br>● NR: **NSA_DC_UL_PATH_SELECTION_SW** option of the **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** parameter | *EPC-based NSA Performance Enhancement* | If TDM is enabled, uplink data transmission path selection does not take effect. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| LTE FDD<br>Low-frequency NR TDD | Secondary intermodulation interference avoidance | • LTE: **INTERFERENCE_AVOID_SW** and **TDM_SWITCH** options of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter<br>• NR: **CROSS_MDLT_INTRF_AVOID_SW** option of the **NRDUCellAlgoSwitch**.***NsaDcAlgoSwitch*** parameter | *EPC-based NSA Basics* | If secondary intermodulation interference avoidance has taken effect, network-coordinated dynamic UE power sharing does not take effect. |
| LTE FDD<br>Low-frequency NR TDD | Secondary intermodulation interference avoidance | • LTE: **INTERFERENCE_AVOID_SW** and **TDM_SWITCH** options of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter<br>• NR: **CROSS_MDLT_INTRF_AVOID_SW** option of the **NRDUCellAlgoSwitch**.***NsaDcAlgoSwitch*** parameter | *EPC-based NSA Basics* | If secondary intermodulation interference avoidance has taken effect, uplink data transmission path selection does not take effect. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| LTE FDD<br>Low-frequency NR TDD | TDM power control | • LTE: **NSA_DC_ENH_UL_POWER_CONTROL_SW** and **TDM_SWITCH** options of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter<br>• NR: **NSA_DC_ENH_UL_POWER_CONTROL_SW** option of the **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** parameter | *EPC-based NSA Performance Enhancement* | If network-coordinated dynamic UE power sharing has taken effect, TDM power control does not take effect. |
| LTE FDD<br>NR FDD | Uplink single-side transmission | • LTE: **TDM_SWITCH** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter<br>• NR: **SINGLE_UL_NSA_SW** option of the **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** parameter | *EPC-based NSA Basics* | If uplink single-side transmission has taken effect, network-coordinated dynamic UE power sharing does not take effect. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| LTE FDD<br>NR FDD | Uplink single-side transmission | • LTE: **TDM_SWITCH** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter<br>• NR: **SINGLE_UL_NSA_SW** option of the **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** parameter | *EPC-based NSA Basics* | If uplink single-side transmission has taken effect, uplink data transmission path selection does not take effect. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| LTE FDD<br>LTE TDD<br>NR FDD<br>Low-frequency NR TDD | Secondary harmonic interference avoidance | • LTE: **INTERFERENCE_AVOID_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter, and **NsaDcMgmtConfig**.***HarmonicIntrfAvoidRange*** parameter<br>• NR: **HARMONIC_INTRF_AVOID_SW** option of the **NRDUCellAlgoSwitch**.***NsaDcAlgoSwitch*** parameter, and **NRDUCellNsaDcConfig**.***HarmonicIntrfAvoidRange*** parameter | *EPC-based NSA Basics* | If secondary harmonic interference avoidance has taken effect, uplink data transmission path selection does not take effect. |
| NR FDD<br>Low-frequency NR TDD | Cloud X services | **CLOUD_VR_SERVICE_SWITCH** option of the **NRCellAlgoSwitch**.***CloudVrSwitch*** parameter | *Ultimate Cloud X Experience* in 5G RAN feature documentation | When uplink data transmission path selection takes effect on the uplink path of the Cloud X bearer, uplink data transmission is redirected to the MCG. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| LTE FDD<br>LTE TDD<br>NR FDD<br>Low-frequency NR TDD | Network-coordinated dynamic UE power sharing | • LTE: **NSA_DC_COOPERATION_DPS_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter<br>• NR: **NSA_DC_COOPERATION_DPS_SW** option of the **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** parameter | *EPC-based NSA Performance Enhancement* | When the PC2 UE uplink NR power configuration maximization function takes effect for PC2 UEs, network-coordinated dynamic UE power sharing does not take effect. |

## 4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

### Base Station Models

- PC2 UE uplink NR power configuration maximization

  On the LTE side, the compatible base stations are as follows:

  3900 and 5900 series base stations (macro base stations) excluding the BTS5929R. 5900 series base stations must be configured with a BBU model other than the BBU5900C.

  On the NR side, the compatible base stations are as follows:

  3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910, and 5900 series base stations must be configured with a BBU model other than the BBU5900C.

- Other functions

  On the LTE side, the compatible base stations are as follows:

  – 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R. 5900 series base stations must be configured with a BBU model other than the BBU5900C.

– DBS3900 LampSite and DBS5900 LampSite

On the NR side, the compatible base stations are as follows:

– 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910, and 5900 series base stations must be configured with a BBU model other than the BBU5900C.
– DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

## Boards

The compatible boards are listed below.

| RAT | Board Type | Board Name | Option 3 | Option 3x |
|---|---|---|---|---|
| LTE | Main control board | UMPTb | Supported | Supported |
| LTE | Main control board | UMPTe | Supported | Supported |
| LTE | Main control board | UMPTg/UMPTga | Supported | Supported |
| LTE | Main control board | UMPTha | Supported | Supported |
| LTE | Baseband processing unit | All UBBP boards | Supported | Supported |
| NR | Main control board | UMPTe | Supported | Supported |
| NR | Main control board | UMPTg/UMPTga | Supported | Supported |
| NR | Main control board | UMPTha | Supported | Supported |
| NR | Baseband processing unit | UBBPg | Supported | Supported |

| RAT | Board Type | Board Name | Option 3 | Option 3x |
|---|---|---|---|---|
| NR | Baseband processing unit | UBBPfw1 | Supported | Supported |
| NR | Baseband processing unit | UBBPh | Supported | Supported |

### RF Modules

This function does not depend on RF modules.

## 4.3.4 Networking

For details, see the networking requirements in *EPC-based NSA Basics*.

## 4.3.5 Cells

On the LTE side:

None

On the NR side:

TDM power control:

- The **NRDUCellPrach**.*PrachConfigurationIndex* parameter must be set to **0**, **4**, **8**, **12**, **16**, **19**, or **65535** when TDM power control is enabled and all of the following conditions are met:
  - The **NRDUCell**.***DuplexMode*** parameter is set to **CELL_FDD**.
  - The **LNR_RELATIVE_FRM_OFS_ADAPT_SW** option of the **gNodeBParam**.***NsaDcOptSwitch*** parameter is selected.
  - The **NRDUCellAlgoSwitch**.***NsaTdmPatternTypeStrategy*** parameter is set to **LTE_UL_PREFERRED**.
- The **NRDUCellPrach**.*PrachConfigurationIndex* parameter must be set to **1**, **2**, **3**, **5**, **6**, **7**, **9**, **10**, **11**, **13**, **14**, **15**, **17**, **18**, or **20** when TDM power control is enabled and all of the following conditions are met:
  - The **NRDUCell**.***DuplexMode*** parameter is set to **CELL_FDD**.
  - The **LNR_RELATIVE_FRM_OFS_ADAPT_SW** option of the **gNodeBParam**.***NsaDcOptSwitch*** parameter is selected.
  - The **NRDUCellAlgoSwitch**.***NsaTdmPatternTypeStrategy*** parameter is set to **NR_UL_PREFERRED**.

### 4.3.6 Others

- UE
  - UEs must support NSA DC specified in 3GPP Release 15.
  - UEs must have subscribed to LTE and NR services.
  - UEs must match gNodeB and eNodeB versions.
  - TDM-based intermodulation interference avoidance, uplink power control, and uplink single-side transmission require UEs to support TDM.
- EPC
  - The EPC must be CloudEPC to support Option 3 and Option 3x.
  - The EPC must support NSA DC. If the core network is provided by Huawei, see *WSFD-021101 5G NSA (Opt.3) Dual Connectivity Management* for details.
  - If NSA DC is enabled on an eNodeB, the connected MMEs need to support NSA DC. If a connected MME does not support NSA DC, the **MmeCapInfo**.***MmeNsaDcCapability*** parameter for this MME must be set to **NOT_SUPPORT**.
- Clock

  TDM requires time synchronization between LTE FDD and NR.

## 4.4 Operation and Maintenance

### 4.4.1 Data Configuration

#### 4.4.1.1 Data Preparation

The following table describes the parameters that must be set in the **NsaDcMgmtConfig** MO on the LTE side to enable blind SCG addition for EPS fallback UEs.

| Parameter Name | Parameter ID | Setting Notes |
| --- | --- | --- |
| NSA DC Algorithm Extension Switch | **NsaDcMgmtConfig**.***NsaDcAlgoExtSwitch*** | Select the **EPS_FALLBACK_BLIND_SCG_ADD_SW** option. |

The following table describes the parameters that must be set in the **EnodebAlgoExtSwitch** MO on the LTE side and the **gNodeBParam** and **gNBOperator** MOs on the NR side to configure uplink and downlink blind PSCell addition for experience-based fallback UEs.

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| Multi Networking Option Opt Switch | LTE: **EnodebAlgoExtSwitch**.***MultiNetworkingOptionOptSw*** | • For the uplink, select the **LTE_FDD_NSA_SA_UL_SEL_OPT_SW** option.<br>• For the downlink, select the **LTE_FDD_NSA_SA_DL_SEL_OPT_SW** or **LTE_TDD_NSA_SA_DL_SEL_OPT_SW** option. |
| Networking Option Opt Sw | NR: **gNodeBParam**.***NetworkingOptionOptSw*** | • For the uplink, select the **NSA_SA_UL_SEL_OPT_SW** option.<br>• For the downlink, select the **NSA_SA_DL_SEL_OPT_SW** option. |
| Operator Inter-RAT Policy Switch | NR: **gNBOperator**.***OperatorInterRatPolicySw*** | • For the uplink, select the **NSA_SA_UL_SEL_OPT_SW** option.<br>• For the downlink, select the **NSA_SA_DL_SEL_OPT_SW** option. |
| Handover to NR Algo Switch | LTE: **EnodebAlgoExtSwitch**.***NrHandoverAlgoSwitch*** | Select the **DIRECT_NR_HO_TO_EN_DC_SW** option. |

The following table describes the parameters that must be set in the **CellDlSchAlgo** MO on the LTE side to configure NSA UE scheduling protection based on MCG cell load.

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| MCG High Load Threshold | **CellDlschAlgo**.***McgHighLoadThreshold*** | The value **60** is recommended. |

The following table describes the parameters that must be set in the **NsaDcMgmtConfig** MO on the LTE side and the **NRCellNsaDcConfig** MO on the NR side to configure SN-terminated MCG bearer transmission.

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| NSA DC Algorithm Switch | LTE: **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** | Select the **SN_TERMN_MCG_BEARER_SWITCH** option. |
| NSA DC Algorithm Switch | NR: **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** | Select the **SN_TERMN_MCG_BEARER_SWITCH** option. |

The following table describes the parameters that must be set in the **NsaDcMgmtConfig** MO on the LTE side and the **NRCellNsaDcConfig** and **NRDUCellSrsMeas** MOs on the NR side to configure uplink data transmission path selection.

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| NSA DC Algorithm Switch | LTE: **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** | Select the **NSA_DC_UL_PATH_SELECTION_SW** option. |
| NSA DC Algorithm Switch | NR: **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** | Select the **NSA_DC_UL_PATH_SELECTION_SW** option. |
| NSA DC UL Path Change to LTE Rate Coeff | LTE: **NsaDcMgmtConfig**.***NsaDcUlPathToLteRateCoeff*** | The value **400** is recommended. |
| NSA DC UL Path Change to NR Rate Coeff | LTE: **NsaDcMgmtConfig**.***NsaDcUlPathToNrRateCoeff*** | The value **200** is recommended. |
| NSA Split UL Path Select SINR Threshold | NR: **NRDUCellSrsMeas**.***NsaSplitUlPathSelSinrThld*** | The value **-30** is recommended. |
| NSA UL Path Select SINR High Threshold | NR: **NRDUCellSrsMeas**.***NsaUlPathSelSinrHighThld*** | The value **150** is recommended. |
| NSA UL Path Select SINR Hyst | NR: **NRDUCellSrsMeas**.***NsaUlPathSelSinrHyst*** | The value **30** is recommended. |

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| NSA UL Path Select SINR Low Threshold | NR: **NRDUCellSrsMeas**.***NsaUlPathSelSinrLowThld*** | The value **-30** is recommended. |
| NSA UL Path Select SINR Time-to-Trigger | NR: **NRDUCellSrsMeas**.***NsaUlPathSelSinrTimeToTrig*** | The value **10** is recommended. |
| NSA UL Path Change to LTE Rate Ratio | NR: **NRDUCellSrsMeas**.***NsaUlPathToLteRateRatio*** | The value **400** is recommended. |
| NSA UL Path Change to NR Rate Ratio | NR: **NRDUCellSrsMeas**.***NsaUlPathToNrRateRatio*** | The value **200** is recommended. |

The following table describes the parameters that must be set in the **NsaDcMgmtConfig** MO on the LTE side and the **NRCellNsaDcConfig** and **NRDUCellSrsMeas** MOs on the NR side to configure uplink power control.

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| NSA DC Algorithm Switch | LTE: **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** | Select both the **NSA_DC_ENH_UL_POWER_CONTROL_SW** and **TDM_SWITCH** options. |
| NSA DC Algorithm Switch | NR: **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** | Select the **NSA_DC_ENH_UL_POWER_CONTROL_SW** option. |

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| NSA TDM Power Control Trigger SINR Thld | LTE: **NsaDcMgmtConfig**.***NsaTdmPcTrigSinrThld***<br>NR: **NRDUCellSrsMeas**.***NsaTdmPcTrigSinrThld*** | • On the LTE side, the value **-62** is recommended.<br>• On the NR side, the value **-60** is recommended.<br>It is not recommended that the LTE and NR parameters be set to a large value (for example, **100**). If a large value is used, TDM power control will be triggered in the cell center where the power is not limited. A TDM status change triggers an intra-cell handover. After the UE accesses the NR cell again, the SRS power increases, leading to ping-pong switching of the TDM status. |
| NSA TDM Power Control Trigger SINR Hyst | NR: **NRDUCellSrsMeas**.***NsaTdmPcTrigSinrHyst*** | The value **30** is recommended. |
| NSA DC Algorithm Switch | LTE: **NsaDcAlgoParam**.***NsaDcAlgoSwitch*** | Select the **LNR_RELATIVE_FRM_OFS_ADAPT_SW** option. |
| NSA DC Optimization Switch | NR: **gNodeBParam**.***NsaDcOptSwitch*** | Select the **LNR_RELATIVE_FRM_OFS_ADAPT_SW** option. |

The following table describes the parameters that must be set in the **NsaDcMgmtConfig** MO on the LTE side and the **NRCellNsaDcConfig** MO on the NR side to configure network-coordinated dynamic UE power sharing.

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| NSA DC Algorithm Switch | LTE: **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** | Select the **NSA_DC_COOPERATION_DPS_SW** option. Set this parameter based on the operator's network plan. |

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| NSA DC Algorithm Switch | NR: **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** | Select the **NSA_DC_COOPERATION_DPS_SW** option. Set this parameter based on the operator's network plan. |

The following table describes the parameters that must be set in the **NsaDcMgmtConfig** MO on the LTE side and the **NRCellNsaDcConfig** and **gNBUeCompat** MOs on the NR side to configure the PC2 UE uplink NR power configuration maximization function.

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| NSA DC Algorithm Switch | LTE: **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** | Select the **PC2_UE_NR_UL_PWR_CFG_MAX_SW** option if the PC2 UE uplink NR power configuration maximization function is required. |
| NSA DC Algorithm Switch | NR: **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** | Select the **PC2_UE_NR_UL_PWR_CFG_MAX_SW** option if the PC2 UE uplink NR power configuration maximization function is required. |
| Blacklist Control Switch | NR: **gNBUeCompatSw**.***BlacklistCtrlSwitch*** | Select the **PC2_UE_NR_UL_PWR_CFG_MAX_OFF** option if the PC2 UE uplink NR power configuration maximization function needs to be prohibited for blacklisted UEs. |

### 4.4.1.2 Using MML Commands

Before using MML commands, refer to **4.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

## Activation Command Examples

On the eNodeB side

```
//(Optional) Enabling blind PSCell addition for EPS fallback UEs (if required)
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoExtSwitch=EPS_FALLBACK_BLIND_SCG_ADD_SW-1;
//(Optional) Enabling uplink and/or downlink blind PSCell addition for experience-based fallback UEs (if
required)
MOD ENODEBALGOEXTSWITCH: MultiNetworkingOptionOptSw=LTE_FDD_NSA_SA_DL_SEL_OPT_SW-1;
MOD ENODEBALGOEXTSWITCH: MultiNetworkingOptionOptSw=LTE_TDD_NSA_SA_DL_SEL_OPT_SW-1;
MOD ENODEBALGOEXTSWITCH: MultiNetworkingOptionOptSw=LTE_FDD_NSA_SA_UL_SEL_OPT_SW-1;
MOD ENODEBALGOEXTSWITCH: NrHandoverAlgoSwitch=DIRECT_NR_HO_TO_EN_DC_SW-1;
//(Optional) Enabling NSA UE scheduling protection based on MCG cell load (if required)
MOD CELLDLSCHALGO: LocalCellId=21, McgHighLoadThreshold=60;
//(Optional) Enabling SN-terminated MCG bearer transmission (if required)
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoSwitch=SN_TERMN_MCG_BEARER_SWITCH-1;
//(Optional) Enabling uplink data transmission path selection (if required)
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoSwitch=NSA_DC_UL_PATH_SELECTION_SW-1;
//(Optional) Enabling power control enhancement in NSA networking (if required)
MOD NSADCMGMTCONFIG: LocalCellId=21,
NsaDcAlgoSwitch=NSA_DC_ENH_UL_POWER_CONTROL_SW-1&TDM_SWITCH-1, NsaTdmPcTrigSinrThld=-62;
MOD NSADCALGOPARAM: NsaDcAlgoSwitch=LNR_RELATIVE_FRM_OFS_ADAPT_SW-1;
//(Optional) Enabling network-coordinated dynamic UE power sharing (if required)
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoSwitch=NSA_DC_COOPERATION_DPS_SW-1;
//(Optional) Enabling the PC2 UE uplink NR power configuration maximization function (if required)
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoSwitch=PC2_UE_NR_UL_PWR_CFG_MAX_SW-1;
```

On the gNodeB side

```
//(Optional) Enabling uplink and/or downlink blind PSCell addition for experience-based fallback UEs
served by a specific operator (if required)
MOD GNODEBPARAM: NetworkingOptionOptSw=NSA_SA_DL_SEL_OPT_SW-1;
MOD GNODEBPARAM: NetworkingOptionOptSw=NSA_SA_UL_SEL_OPT_SW-1;
MOD GNBOPERATOR: OperatorId=0, OperatorName="5G", Mcc="302", Mnc="220",
OperatorType=PRIMARY_OPERATOR, NrNetworkingOption=SA_NSA,
OperatorInterRatPolicySw=NSA_SA_DL_SEL_OPT_SW-1;
MOD GNBOPERATOR: OperatorId=0, OperatorName="5G", Mcc="302", Mnc="220",
OperatorType=PRIMARY_OPERATOR, NrNetworkingOption=SA_NSA,
OperatorInterRatPolicySw=NSA_SA_UL_SEL_OPT_SW-1;
//(Optional) Enabling SN-terminated MCG bearer transmission (if required)
MOD NRCELLNSADCCONFIG: NrCellId=7, NsaDcAlgoSwitch=SN_TERMN_MCG_BEARER_SWITCH-1;
//(Optional) Enabling uplink data transmission path selection (if required)
MOD NRCELLNSADCCONFIG: NrCellId=7, NsaDcAlgoSwitch=NSA_DC_UL_PATH_SELECTION_SW-1;
MOD NRDUCELLSRSMEAS: NrDuCellId=120, NsaUlPathSelSinrHighThld=150,
NsaUlPathSelSinrLowThld=-30, NsaSplitUlPathSelSinrThld=-30, NsaUlPathSelSinrHyst=30,
NsaUlPathSelSinrTimeToTrig=10, NsaUlPathToLteRateRatio=400, NsaUlPathToNrRateRatio=200;
//(Optional) Enabling power control enhancement in NSA networking (if required)
MOD NRCELLNSADCCONFIG: NrCellId=7, NsaDcAlgoSwitch=NSA_DC_ENH_UL_POWER_CONTROL_SW-1;
MOD NRDUCELLSRSMEAS: NrDuCellId=120, NsaTdmPcTrigSinrThld=-60, NsaTdmPcTrigSinrHyst=30;
MOD GNODEBPARAM: NsaDcOptSwitch=LNR_RELATIVE_FRM_OFS_ADAPT_SW-1;
//(Optional) Enabling network-coordinated dynamic UE power sharing (if required)
MOD NRCELLNSADCCONFIG: NrCellId=7, NsaDcAlgoSwitch=NSA_DC_COOPERATION_DPS_SW-1;
//(Optional) Enabling the PC2 UE uplink NR power configuration maximization function (if required)
MOD NRCELLNSADCCONFIG: NrCellId=7, NsaDcAlgoSwitch=PC2_UE_NR_UL_PWR_CFG_MAX_SW-1;
//(Optional) Disabling the PC2 UE uplink NR power configuration maximization function for blacklisted UEs
(if required)
MOD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL,
BlacklistCtrlSwitch=PC2_UE_NR_UL_PWR_CFG_MAX_OFF-1;
```

## Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

On the eNodeB side

```
//(Optional) Disabling blind PSCell addition for EPS fallback UEs
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoExtSwitch=EPS_FALLBACK_BLIND_SCG_ADD_SW-0;
//(Optional) Disabling uplink or downlink blind PSCell addition for experience-based fallback UEs
MOD ENODEBALGOEXTSWITCH: MultiNetworkingOptionOptSw=LTE_FDD_NSA_SA_DL_SEL_OPT_SW-0;
MOD ENODEBALGOEXTSWITCH: MultiNetworkingOptionOptSw=LTE_TDD_NSA_SA_DL_SEL_OPT_SW-0;
MOD ENODEBALGOEXTSWITCH: MultiNetworkingOptionOptSw=LTE_FDD_NSA_SA_UL_SEL_OPT_SW-0;
MOD ENODEBALGOEXTSWITCH: NrHandoverAlgoSwitch=DIRECT_NR_HO_TO_EN_DC_SW-0;
//(Optional) Disabling NSA UE scheduling protection based on MCG cell load
MOD CELLDLSCHALGO: LocalCellId=21, McgHighLoadThreshold=100;
//(Optional) Disabling SN-terminated MCG bearer transmission
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoSwitch=SN_TERMN_MCG_BEARER_SWITCH-0;
//(Optional) Disabling uplink data transmission path selection
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoSwitch=NSA_DC_UL_PATH_SELECTION_SW-0;
//(Optional) Disabling power control enhancement in NSA networking
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoSwitch=NSA_DC_ENH_UL_POWER_CONTROL_SW-0;
//(Optional) Disabling network-coordinated dynamic UE power sharing
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoSwitch=NSA_DC_COOPERATION_DPS_SW-0;
//(Optional) Disabling the PC2 UE uplink NR power configuration maximization function
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoSwitch=PC2_UE_NR_UL_PWR_CFG_MAX_SW-0;
```

On the gNodeB side

```
//(Optional) Disabling uplink or downlink blind PSCell addition for experience-based fallback UEs served by
a specific operator
MOD GNODEBPARAM: NetworkingOptionOptSw=NSA_SA_DL_SEL_OPT_SW-0;
MOD GNODEBPARAM: NetworkingOptionOptSw=NSA_SA_UL_SEL_OPT_SW-0;
MOD GNBOPERATOR: OperatorId=0, OperatorName="5G", Mcc="302", Mnc="220",
OperatorType=PRIMARY_OPERATOR, NrNetworkingOption=SA_NSA,
OperatorInterRatPolicySw=NSA_SA_DL_SEL_OPT_SW-0;
MOD GNBOPERATOR: OperatorId=0, OperatorName="5G", Mcc="302", Mnc="220",
OperatorType=PRIMARY_OPERATOR, NrNetworkingOption=SA_NSA,
OperatorInterRatPolicySw=NSA_SA_UL_SEL_OPT_SW-0;
//(Optional) Disabling SN-terminated MCG bearer transmission
MOD NRCELLNSADCCONFIG: NrCellId=7, NsaDcAlgoSwitch=SN_TERMN_MCG_BEARER_SWITCH-0;
//(Optional) Disabling uplink data transmission path selection
MOD NRCELLNSADCCONFIG: NrCellId=7, NsaDcAlgoSwitch=NSA_DC_UL_PATH_SELECTION_SW-0;
//(Optional) Disabling power control enhancement in NSA networking
MOD NRCELLNSADCCONFIG: NrCellId=7, NsaDcAlgoSwitch=NSA_DC_ENH_UL_POWER_CONTROL_SW-0;
//(Optional) Disabling network-coordinated dynamic UE power sharing
MOD NRCELLNSADCCONFIG: NrCellId=7, NsaDcAlgoSwitch=NSA_DC_COOPERATION_DPS_SW-0;
//(Optional) Disabling the PC2 UE uplink NR power configuration maximization function
MOD NRCELLNSADCCONFIG: NrCellId=7, NsaDcAlgoSwitch=PC2_UE_NR_UL_PWR_CFG_MAX_SW-0;
//(Optional) Turning off the blacklist control switch for the PC2 UE uplink NR power configuration
maximization function
MOD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL,
BlacklistCtrlSwitch=PC2_UE_NR_UL_PWR_CFG_MAX_OFF-0;
```

#### 4.4.1.3 Using the MAE-Deployment

- Fast batch activation

  This function can be batch activated using the Feature Operation and Maintenance function of the MAE-Deployment. For detailed operations, see the following section in the MAE-Deployment product documentation or online help: **MAE-Deployment Operation and Maintenance** > **MAE-Deployment Guidelines** > **Enhanced Feature Management** > **Feature Operation and Maintenance**.

- Single/Batch configuration

  This function can be activated for a single base station or a batch of base stations on the MAE-Deployment. For detailed operations, see Feature Configuration Using the MAE-Deployment.

  **NOTE**

  Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

### 4.4.2 Activation Verification

#### Monitoring Counters

Counters related to NSA DC can be subscribed on the MAE-Access in one-click mode.

- Check whether NSA UE scheduling protection based on MCG cell load has taken effect.

  If the data volume distributed to the LTE side (**N.PDCP.Vol.DL.X2U.TrfPDU.Tx**) decreases when the downlink PRB usage of the LTE cell (**L.ChMeas.PRB.DL.Used.Avg** / **L.ChMeas.PRB.DL.Avail** x 100%) exceeds a preset threshold, this function has taken effect.
- Check whether SN-terminated MCG bearer transmission has taken effect.

  If the values of the following counters are not 0, this function has taken effect. If the value of any counter is always 0, this function has not taken effect.

| Counter ID | Counter Name | NE |
|---|---|---|
| 1526758902 | **L.NsaDc.E-RAB.SplitToMcg** | eNodeB |
| 1911822835 | **N.User.gNodeB.SNTrmt.MCGBearer.Avg** | gNodeB |
| 1911822836 | **N.User.gNodeB.SNTrmt.MCGBearer.Max** | gNodeB |

- Check whether uplink data transmission path selection has taken effect.

  If the values of the following counters are not 0, this function has taken effect. If the value of any counter is always 0, this function has not taken effect.

| Counter ID | Counter Name | NE |
|---|---|---|
| 1911827153 | **N.NsaDc.ULChangetoMeNB** | gNodeB |
| 1911827152 | **N.NsaDc.UL.PDCP.Rx.X2Split.PDU.Volume.ULChangetoMeNB** | gNodeB |

#### Message Tracing

1. Log in to the MAE-Access. Choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**. The **Signaling Trace Management** window is displayed.
2. Trace UE random access.

   Choose **Trace Type** > **LTE** > **Application Layer** > **Uu Interface Trace**. You can observe that the UE sends an RRC_CONN_SETUP_CMP message to the eNodeB to initiate an LTE access procedure.
3. (Optional) Check whether TDM power control has taken effect.

   – Choose **Trace Type** > **LTE** > **Application Layer** > **Uu Interface Trace**. Check whether the eNodeB sends a UE an RRC_CONN_RECFG message with a *tdm-PatternConfig-r15* IE. If so, this function has taken effect.
   – Choose **Trace Type** > **LTE** > **Application Layer** > **X2 Interface Trace**. Check that the gNodeB sends the eNodeB an SgNB Modification Required message with IEs *SgNB Resource Coordination Information* > *UL Coordination Information* that includes a bitmap indicating a TDM pattern.
4. (Optional) Check whether network-coordinated dynamic UE power sharing has taken effect.

   Choose **Trace Type** > **LTE** > **Application Layer** > **X2 Interface Trace**. Check whether the gNodeB sends the eNodeB an EN-DC Private Vendor Configuration Transfer message with a *gNB Cell Configuration Indication 1* IE whose bit 2 is set to 1. If so, the function has taken effect.
5. (Optional) Check the effective status of the PC2 UE uplink NR power configuration maximization function.
   – Choose **Trace Type** > **LTE** > **Application Layer** > **Uu Interface Trace**. Check the *p-MaxEUTRA-r15* IE contained in the RRC_CONN_RECFG message sent from the eNodeB to the UE. The IE value indicates the maximum uplink LTE transmit power configuration allowed by the eNodeB.
   – Choose **Trace Type** > **LTE** > **Application Layer** > **Uu Interface Trace**. Check the *p-NR-FR1* IE contained in the RRC_CONN_RECFG message sent from the eNodeB to the UE. The IE value indicates the maximum uplink NR transmit power configuration allowed by the eNodeB.
6. (Optional) Check whether uplink data transmission path selection has taken effect.
   – Choose **Trace Type** > **LTE** > **Application Layer** > **X2 Interface Trace**. Check whether the value of the *pdcp-Config* > *PrimaryPath* > *cellGroup* IE in the SgNB Modification Required message sent from the gNodeB to the eNodeB is **0**. If so, the function has taken effect.
   – Choose **Trace Type** > **LTE** > **Application Layer** > **Uu Interface Trace**. Check whether the eNodeB sends a UE an RRC_CONN_RECFG message with IEs *pdcp-Config* > *PrimaryPath* > *cellGroup* that is set to 0. If so, the function has taken effect.

## 4.4.3 Network Monitoring

None

# 5 EN-DC Optimal Carrier Selection

## 5.1 Principles

### 5.1.1 NSA PCC Anchoring

#### PCC Anchoring Procedure for NSA UEs in Connected Mode

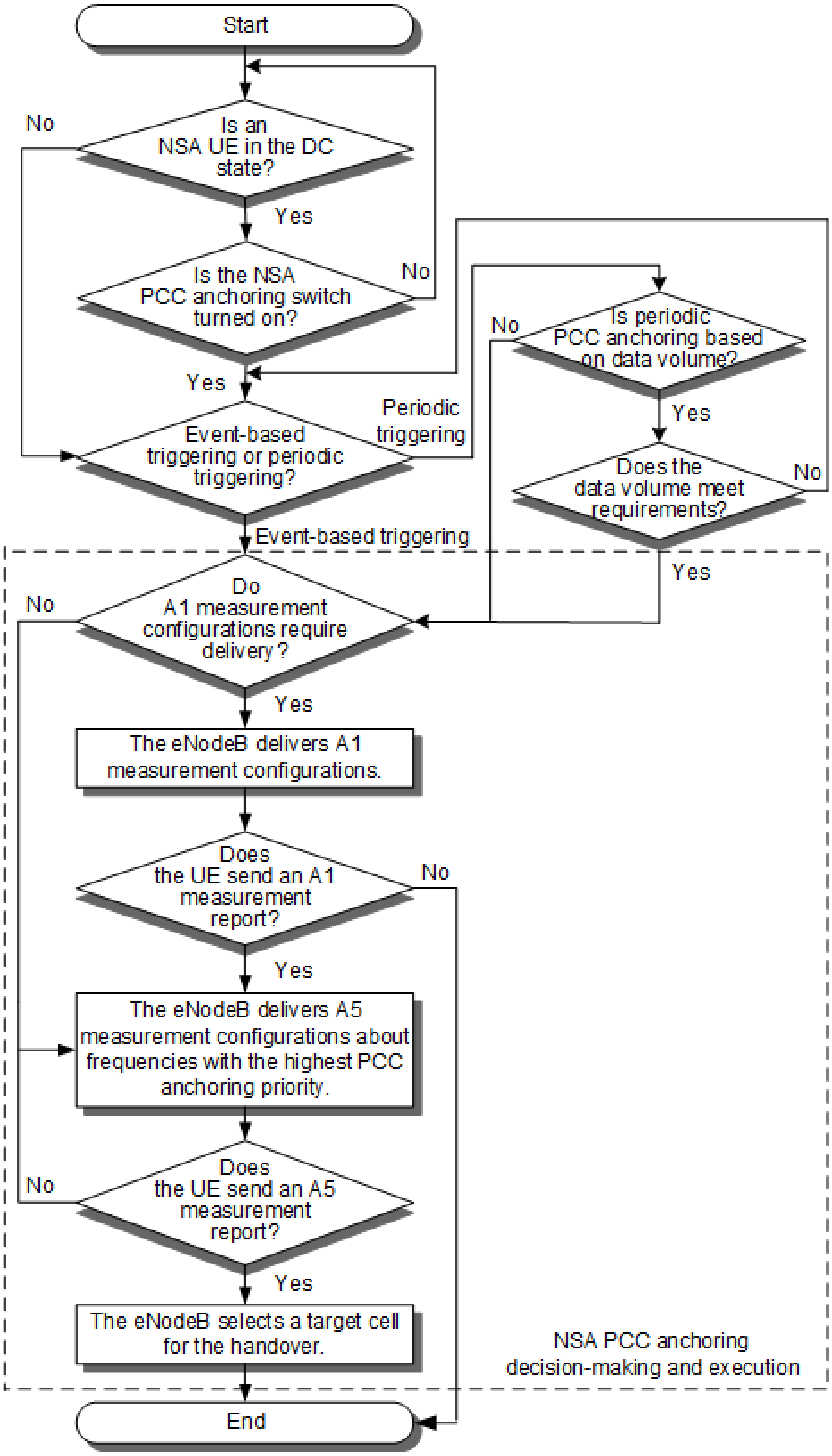
For NSA UEs in connected mode, the NSA PCC anchoring policy is determined by the **NSA_PCC_ANCHORING_SWITCH** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter as well as the **NsaDcMgmtConfig**.***NsaDcPccAnchoringPolicy*** parameter.

- If the **NSA_PCC_ANCHORING_SWITCH** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter is deselected, the PCC anchoring policy is the same as that for LTE carrier management in connected mode. For details, see *Carrier Aggregation* in eRAN feature documentation.
- If the **NSA_PCC_ANCHORING_SWITCH** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter is selected, the **NsaDcMgmtConfig**.***NsaDcPccAnchoringPolicy*** parameter is set to **DEFAULT**, and an NSA UE is camping on a lower-priority PCC or an NSA-DC-incapable PCC, the UE will be handed over to a higher-priority or an NSA-DC-capable PCC through NSA PCC anchoring. The PCC anchoring priority of a frequency is specified by the **PccFreqCfg**.***NsaPccAnchoringPriority*** parameter.

If there are UEs incompatible with NSA PCC anchoring on the network, NSA PCC anchoring can be forbidden for blacklisted UEs on the eNodeB side. Specifically, when the **NSA_PCC_ANCHORING_FORBID_SW** option of the **UeCompat**.***BlacklistControlExtSwitch2*** parameter is selected, NSA PCC anchoring for RRC_CONNECTED UEs is forbidden for blacklisted UEs. If this option is deselected, no differentiated measures are taken for UEs.

**Figure 5-1** shows the PCC anchoring procedure.

**Figure 5-1** PCC anchoring procedure for NSA UEs in connected mode




The procedure is described as follows:

1. The eNodeB checks whether the NSA UE is in the DC state.
   - If so, the procedure continues at **2**.
   - If not, the procedure continues at **3**.
2. The eNodeB checks whether the PCC anchoring switch for NSA UEs in the DC state is turned on.

   The **NSA_DC_STATE_PCC_ANCHORING_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter specifies whether PCC anchoring can be triggered for NSA UEs in the DC state.
   - If this option is selected, PCC anchoring can be triggered for NSA UEs in the DC state.
   - If this option is deselected, PCC anchoring cannot be triggered for NSA UEs in the DC state. The procedure starts again at **1**.
3. The eNodeB determines whether the triggering of PCC anchoring is based on events or at intervals.
   - Event-based triggering: PCC anchoring is triggered when the NSA UE transits from idle mode to connected mode, or performs a necessary incoming handover (for example, coverage-based inter-frequency handover) or an incoming RRC connection reestablishment.
   - Periodic triggering: PCC anchoring is triggered at intervals specified by the **NsaDcMgmtConfig**.***ScgAdditionInterval*** parameter when the **PERIODIC_PCC_ANCHORING_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter is selected.

     Whether periodic triggering of PCC anchoring is based on data volume is controlled by the **VOLUME_BASED_PERIODIC_TRIG_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter and the **VOLUME_BASED_PCC_ANCHORING_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoExtSwitch*** parameter.
     - When the NSA UE is not in the DC state:

       If the **VOLUME_BASED_PERIODIC_TRIG_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter is selected, periodic triggering of PCC anchoring is based on data volume. Otherwise, periodic triggering of PCC anchoring is not based on data volume.
     - When the NSA UE is in the DC state:
       - If the **VOLUME_BASED_PCC_ANCHORING_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoExtSwitch*** parameter is deselected, or if this option is selected but the **VOLUME_BASED_PERIODIC_TRIG_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter is deselected, then periodic triggering of PCC anchoring is not based on data volume.
       - If both the **VOLUME_BASED_PCC_ANCHORING_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoExtSwitch*** parameter and the **VOLUME_BASED_PERIODIC_TRIG_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter are selected, then periodic triggering of PCC anchoring is based on data volume.

The **NsaDcMgmtConfig**.***ScgAdditionBufferLenThld*** and **NsaDcMgmtConfig**.***ScgAdditionBufferDelayThld*** parameters determine whether the data volume meets requirements:

- When the **NsaDcMgmtConfig**.***ScgAdditionBufferLenThld*** parameter is set to **0** and the UE is involved in data transmission, PCC anchoring is triggered.
- When the **NsaDcMgmtConfig**.***ScgAdditionBufferLenThld*** parameter is not set to **0**, PCC anchoring is triggered if both of the following conditions are met:
  - Data volume buffered at the RLC layer > max (Uu data rate at the RLC layer x **NsaDcMgmtConfig**.***ScgAdditionBufferDelayThld***, **NsaDcMgmtConfig**.***ScgAdditionBufferLenThld***)
  - First packet delay at the RLC layer > **NsaDcMgmtConfig**.***ScgAdditionBufferDelayThld***

The **SCG_ADD_PCC_ANCHOR_VOL_OPT_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoExtSwitch*** parameter can be used to specify whether the eNodeB deducts the data volume of PDCP status report packets when making a PCC anchoring decision based on data volume.

The **INSTANT_JUDGEMENT_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoExtSwitch*** parameter can be used to specify whether the eNodeB uses instantaneous millisecond-level values or filtered second-level values for estimating the data volume buffered at the RLC layer when making a PCC anchoring decision based on data volume.

NOTE

Assume that periodic triggering of PCC anchoring and LTE mobility load balancing (MLB) are both enabled. If the base station hands over the NSA UE to a cell with a lower PCC anchoring priority (a non-zero priority) by means of MLB and then periodically triggers PCC anchoring, ping-pong handovers will occur. To prevent NSA UEs from being transferred due to LTE MLB, it is recommended that the **CellMlbUeSel**.***NsaDcUeSelectionStrategy*** parameter be set to **LTE_UE_PREFERRED** or a high handover threshold for load balancing be set for NSA UEs in certain handover policy groups when periodic triggering of PCC anchoring is enabled.

If frequency-priority-based inter-frequency handover takes effect, the NSA UE may be handed over from an NSA PCell to a cell operating on a non-NSA-PCC frequency, causing the NSA UE to fail to perform NSA DC services. To prevent this issue, it is recommended that the **FREQ_PRI_SEL_NON_PCC_FILTER_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoEnhSwitch*** parameter be selected to enable the "non-NSA-PCC frequency filtering during frequency-priority-based frequency selection" function. With this function, the eNodeB filters out non-NSA-PCC frequencies during frequency-priority-based frequency selection for NSA UEs. Non-NSA-PCC frequency is a frequency for which the **PccFreqCfg**.***NsaPccAnchoringPriority*** parameter is set to **0** or for which the **PccFreqCfg**.***PccDlEarfcn*** parameter is not specified. For details about frequency-priority-based inter-frequency handover, see *Mobility Management in Connected Mode* in eRAN feature documentation.

If the **HO_TRG_FREQ_FORBID_MEAS_FLAG** option of the **EutranInterNFreq**.***AggregationAttribute*** parameter is selected for a neighboring frequency, this neighboring frequency will not be selected as the target frequency during NSA PCC anchoring for NSA UEs in connected mode. For details, see *Mobility Management in Connected Mode* in eRAN feature documentation.

4. The NSA PCC anchoring decision-making and execution procedure is performed.
   a. The eNodeB delivers A1 measurement configurations for anchoring.

      The A1 threshold is specified by the **NsaDcMgmtConfig**.***NsaDcPccAnchorA1RsrpThld*** parameter. If the parameter is set to **255**, the eNodeB directly delivers A5 measurement configurations without sending A1 measurement configurations, and the procedure continues at **4.b**. If the parameter is set to a value other than **255**, the eNodeB delivers A1 measurement configurations.

      - If the UE sends an A1 measurement report, the base station directly delivers A5 measurement configurations and the procedure continues at **4.b**.
      - If the UE does not send an A1 measurement report, PCC anchoring is not triggered.
   b. The eNodeB selects target frequencies for delivery of A5 measurement configurations.

      The base station obtains the intersection of the Multi-Radio Dual Connectivity (MR-DC) capability information reported by the UE and the PCC and SCG frequency association information defined in the **NrScgFreqConfig** MO. If the intersection contains frequencies that support NSA DC, the eNodeB delivers A5 measurement configurations for the anchor frequencies with the highest priority. Otherwise, NSA PCC anchoring is not performed. For event A5, threshold 1 is –43 dBm and threshold 2 is specified by the **PccFreqCfg**.***NsaDcPccA4RsrpThld*** or

**PccFreqCfg**.***NsaDcPccA4RsrqThld*** parameter. The A5 reporting quantity can be RSRP or RSRQ, depending on the value of the **IntraRatHoComm**.***InterFreqHoA4TrigQuan*** parameter. The time to trigger for event A5 is specified by the **InterFreqHoGroup**.***InterFreqHoA4TimeToTrig*** parameter.

- If the UE sends an A5 measurement report, the eNodeB triggers NSA PCC anchoring, transferring the UE to a neighboring cell with a higher NSA PCC anchoring priority.
- If the UE does not send an A5 measurement report, the eNodeB selects the anchor frequencies with the second highest priority and delivers A5 measurement configurations. This process is repeated until the UE sends an A5 measurement report. Then the eNodeB triggers NSA PCC anchoring, transferring the UE to a neighboring cell with a higher NSA PCC anchoring priority.

NOTE

- During an intra-base-station handover from a non-anchor cell to an anchor cell, if the eNodeB has queried the NSA UE's MR-DC capability in the non-anchor cell, the eNodeB delivers B1 measurement configurations immediately after the UE is handed over to the anchor cell. If the eNodeB has not queried the NSA UE's MR-DC capability in the non-anchor cell, it queries the MR-DC capability after an incoming handover to the anchor cell. The eNodeB delivers B1 measurement configurations only after the timer specified by **NsaDcMgmtConfig**.***ScgAdditionInterval*** expires.
- If the **NSA_DC_CAPABILITY_SWITCH** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter on the LTE side is deselected, the base station still queries the NSA UE's MR-DC capability.
- The **A2_HO_SCG_ADD_PCC_ANCHOR_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoExtSwitch*** parameter on the LTE side can be used to specify whether to allow the triggering of NSA PCC anchoring after receiving A2 measurement reports related to coverage-based handovers from NSA UEs. If this option is selected, the triggering of NSA PCC anchoring is allowed. If this option is deselected, the triggering of NSA PCC anchoring is not allowed.

If the target cell for NSA PCC anchoring does not meet the conditions for NSA DC, the UE cannot perform NSA DC services after being handed over to the target cell; in addition, unnecessary handover delay will be introduced.

An eNodeB notifies neighboring eNodeBs that NSA DC is not supported by sending a message over X2 interfaces when it meets any of the following conditions:

- The UMPTa or a board of an earlier version is used as the main control board of the eNodeB.
- The **NSA_DC_CAPABILITY_SWITCH** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter is deselected for all cells of the eNodeB.

During an inter-base-station NSA PCC anchoring procedure, the eNodeB that receives such an X2 message can determine whether to initiate NSA PCC anchoring to the eNodeB that sends this message, depending on the **NSA_CAPB_BASED_PCC_ANCHOR_SW** option of the **EnodebAlgoExtSwitch**.***NsaDcAlgoSwitch*** parameter. If this option is selected, the former eNodeB filters out the latter eNodeB's candidate cells for the anchoring after receiving measurement reports from an NSA UE, and does not initiate the anchoring to the latter eNodeB. If this option is deselected, the eNodeB directly performs NSA PCC anchoring, without making the preceding decision.

The UE, if moving at a high speed, is prone to handover failures due to fast signal attenuation. This affects user experience. If this is the case, it is recommended that the **NSA_ANCHOR_HO_FORBID_SW** option of the **CellOpHoCfg**.***HighMobiUeHoForbidSw*** parameter be selected to enable the "prohibition of NSA PCC anchoring for high-mobility UEs" function. After this function takes effect, the eNodeB considers a UE moving at a speed higher than 30 km/h as a high-mobility UE and does not perform NSA PCC anchoring for the UE. Under near-NLOS propagation conditions, this function may take effect for UEs that do not move at a high speed because they are identified as high-mobility UEs. As a result, UEs served by frequencies with low PCC anchoring priorities cannot be transferred to a frequency with a high PCC anchoring priority. (NLOS is short for non-line of sight.)

## PCC Anchoring Procedure for NSA UEs in Idle Mode

For NSA UEs transiting from connected mode to idle mode, NSA PCC anchoring is controlled by the **IDLE_MODE_NSA_PCC_ANCHORING_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter.

- If this option is deselected, the eNodeB sends the priorities determined by carrier management for UEs in idle mode through an *IMMCI* IE in an RRC Connection Release message to the UE. The PCC anchoring policy is the same as that for LTE carrier management in idle mode. For details, see *Carrier Aggregation* in eRAN feature documentation.
- If this option is selected, the eNodeB obtains the intersection of the EN-DC capability information reported by the UE, and the PCC and SCG frequency association information defined in the **NrScgFreqConfig** MO, treats the NSA PCC anchoring priorities as dedicated priorities, sorts the frequencies in descending order of priority, and notifies the UE through the *IMMCI* IE in the RRC Connection Release message. The **IDLE_NSA_PCC_ANCHORING_OPT_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoExtSwitch*** parameter controls the filling of PCC frequencies in the *IMMCI* IE in the RRC Connection Release message. The PCC anchoring priority of a frequency is specified by the **PccFreqCfg**.***NsaPccAnchoringPriority*** parameter. For details about the *IMMCI* IE, see section 6.3.4 "CellReselectionPriority" in 3GPP TS 36.331 V16.0.0.
  - If this option is selected and there are LTE frequencies assigned common cell-reselection priorities but not NSA PCC anchoring priorities or LTE frequencies assigned both common cell-reselection priorities and NSA PCC anchoring priorities but not meeting NSA PCC anchoring conditions, then these frequencies are appended to those with NSA PCC anchoring priorities and arranged based on their common cell-reselection priorities.
  - If this option is deselected, only the frequencies in the intersection of the EN-DC capability information reported by the UE and the PCC and SCG frequency association information defined in the **NrScgFreqConfig** MO are filled in the *IMMCI* IE of the RRC Connection Release message.

  If dedicated cell reselection priorities based on the serving cell's PLMN IDs are configured on the eNodeB, the NSA PCC anchoring policy for UEs in idle mode is specified by the **NsaDcMgmtConfig**.***IdleModeNsaPccAnchorPolicy*** parameter.

– If this parameter is set to **DEFAULT**, the PLMN-based priorities are used because these priorities take precedence over the NSA PCC anchoring priorities. For details about cell reselection using PLMN-based priorities, see *RAN Sharing* in eRAN feature documentation.
– If this parameter is set to **PRIOR_TO_PLMN**, the NSA PCC anchoring priorities are preferentially used because these priorities take precedence over the PLMN-based priorities.

Upon reception of the dedicated priority information, the UE preferentially reselects to a frequency with the highest priority. For details about cell reselection based on dedicated priorities, see *Idle Mode Management* in eRAN feature documentation.

NOTE

- Assume that the **IDLE_MODE_NSA_PCC_ANCHORING_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter is selected. When the RRC Connection Release message sent by the MeNB to a UE needs to carry NSA PCC anchoring priorities, the MeNB performs sorting in the order of SPID-specific priorities, NSA PCC anchoring priorities, and MLB priorities if all these priorities are configured on the MeNB side.
- If the eNodeB has not obtained the NSA DC capability information of the UE when sending the RRC Connection Release message, the RRC Connection Release message does not carry the NSA PCC anchoring priorities. An example scenario is that the eNodeB releases the RRC connection after a UE in idle mode initiates a tracking area update (TAU).
- To enable some UEs in idle mode to display the NR indication, the *upperLayerIndication* IE of the serving cell's PLMN needs to be broadcast in SIB2 on the LTE side. This function is controlled by the **NsaDcMgmtConfig**.***UpperLayerIndicationSwitch*** parameter and the **UPPER_LYR_IND_NO_BROADCAST_SW** option of the **CnOperator**.***OperatorFunSwitch*** parameter.
- If the **HO_TRG_FREQ_FORBID_MEAS_FLAG** option of the **EutranInterNFreq**.***AggregationAttribute*** parameter is selected for a neighboring frequency, this neighboring frequency will not be selected as the target frequency during NSA PCC anchoring for NSA UEs in idle mode. For details, see *Mobility Management in Connected Mode* in eRAN feature documentation.

## 5.1.2 NSA PCC Anchoring Enhancement

NSA PCC anchoring enhancement includes the following functions: NSA PCC anchoring based on LTE load and uplink coverage, enhanced NSA PCC anchoring based on LTE load, NSA PCC anchoring based on NR coverage, and NSA PCC anchoring based on virtual grids. **Table 5-1** describes the processing when two of the functions are both enabled.

**Table 5-1** Processing when two of the functions are both enabled

| Function A | Function B | Processing |
|---|---|---|
| NSA PCC anchoring based on LTE load and uplink coverage | NSA PCC anchoring based on NR coverage | NSA PCC anchoring based on NR coverage takes effect. However, the high-load cells defined in function A are filtered out from the received measurement report. |

| Function A | Function B | Processing |
| --- | --- | --- |
| Enhanced NSA PCC anchoring based on LTE load | NSA PCC anchoring based on NR coverage | NSA PCC anchoring based on NR coverage takes effect. However, filtering is performed for the received measurement report:<br>● If the NSA UE camps on an ultra-high-load cell, only the ultra-high-load cells are filtered out from the measurement report (the high-load cells defined in function A are not filtered out).<br>● If the NSA UE camps on a non-ultra-high-load cell, both the high-load and ultra-high-load cells defined in function A are filtered out from the measurement report. |
| NSA PCC anchoring based on LTE load and uplink coverage | Enhanced NSA PCC anchoring based on LTE load | Enhanced NSA PCC anchoring based on LTE load takes effect. |

## NSA PCC Anchoring Based on LTE Load and Uplink Coverage

When the **NSA_PCC_ANCHORING_SWITCH** and **NSA_DC_FLEXIBLE_PCC_ANCHOR_SW** options of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter are selected, NSA PCC anchoring can be based on LTE load and uplink coverage.

- Based on LTE load: During anchor selection for an NSA UE, high-load cells in the measurement report are filtered out. The high-load threshold is specified by the **NsaDcMgmtConfig**.***UserNumberHighLoadThld*** parameter. If the number of RRC_CONNECTED UEs in a cell is greater than this threshold, the cell is considered to be a high-load cell. If there are non-high-load cells in the measurement report, anchoring is performed. If all cells in the measurement report are high-load cells, the frequency with the second highest PCC anchoring priority is selected for measurement. This operation is repeated until a qualified anchor is found or the frequency to be measured has the same PCC anchoring priority as the frequency that the UE camps on.

  **NOTE**

  To enable load-based NSA PCC anchoring, the **NSA_PCC_ANCHORING_SWITCH** and **NSA_DC_FLEXIBLE_PCC_ANCHOR_SW** options of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter must also be selected in the target cell.

- Based on uplink coverage: When the uplink coverage of the NSA PCC deteriorates, LTE spectrum coordination is triggered to switch the NSA PCC to an SCC with good uplink coverage. LTE spectrum coordination in NSA DC is not controlled by the LTE spectrum coordination feature switch. For details about LTE spectrum coordination, see in eRAN feature documentation.

NOTE

To enable LTE spectrum coordination for NSA UEs, it is recommended that the **HoAdmitSwitch** option of the **CellMlbHo**.***MlbMatchOtherFeatureMode*** parameter be deselected, the **HoWithSccCfgSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter be selected, and then the **NSA_DC_FLEXIBLE_PCC_ANCHOR_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter be selected.

## Enhanced NSA PCC Anchoring Based on LTE Load

When there are ultra-high-load or high-load cells with high NSA PCC anchoring priorities and light-load cells with low NSA PCC anchoring priorities, it is recommended that the "enhanced NSA PCC anchoring based on LTE load" function be enabled to reduce the load of cells with high NSA PCC anchoring priorities. This function is enabled by selecting the **LOAD_BASED_PCC_ANCHOR_ENH_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoEnhSwitch*** parameter.

Cell load is used to determine whether the PCell and candidate cells are light-, high-, or ultra-high-load cells during PCC anchoring of an NSA UE. The high-load threshold is specified by the **NsaDcMgmtConfig**.***UserNumberHighLoadThld*** parameter, and the ultra-high-load threshold is specified by the **NsaDcMgmtConfig**.***UserNumUltraHighLoadThld*** parameter. The cell status is determined as follows:

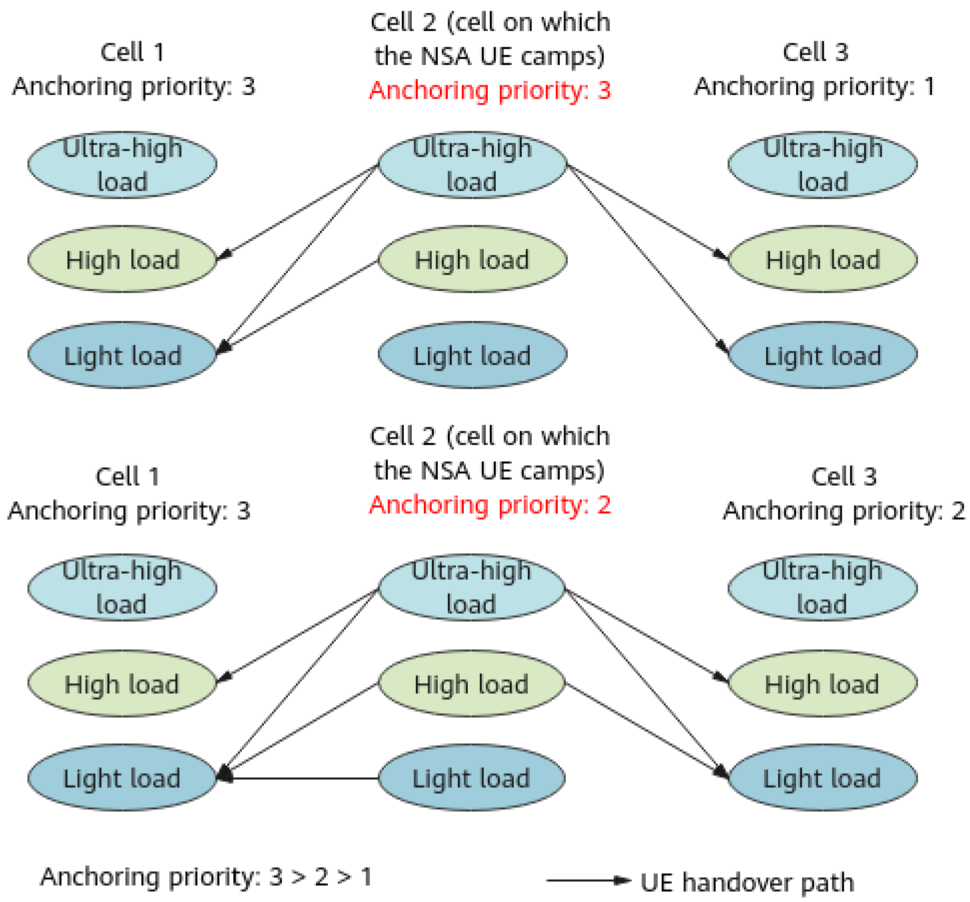
- If the number of UEs in a cell is less than or equal to the high-load threshold multiplied by 0.9, the cell is a light-load cell.
- If the number of UEs in a cell is greater than the high-load threshold and less than or equal to the ultra-high-load threshold multiplied by 0.9, the cell is a high-load cell.
- If the number of UEs in a cell is greater than the ultra-high-load threshold, the cell is an ultra-high-load cell.
- If the number of UEs in a cell is greater than the high-load threshold multiplied by 0.9 and less than or equal to the high-load threshold or the number is greater than the ultra-high-load threshold multiplied by 0.9 and less than or equal to the ultra-high-load threshold, the load status is not changed.

After this function takes effect, the NSA PCC anchoring procedure varies with the status of the PCell on which an NSA UE camps:

- If it is an ultra-high-load cell, all candidate NSA anchors are measured in descending order of priority. If a high-load or light-load cell is reported, PCC anchoring is initiated.
- If it is a high-load cell, all candidate NSA anchors that have the same or a higher priority than the PCell are measured in descending order of priority. If a light-load cell is reported, PCC anchoring is initiated.
- If it is a light-load cell, all candidate NSA anchors that have a higher priority than the PCell are measured in descending order of priority. If a light-load cell is reported, PCC anchoring is initiated.

**Figure 5-2** shows the PCC anchoring procedures of NSA UEs in cells with different load after this function is enabled.

**Figure 5-2** Enhanced NSA PCC anchoring based on LTE load




## NSA PCC Anchoring Based on NR Coverage

NSA PCC anchoring based on NR coverage takes effect when the **NSA_PCC_ANCHORING_SWITCH** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter is selected and the **NsaDcMgmtConfig**.***NsaDcPccAnchoringPolicy*** parameter is set to **BASED_ON_NR_COVERAGE**.

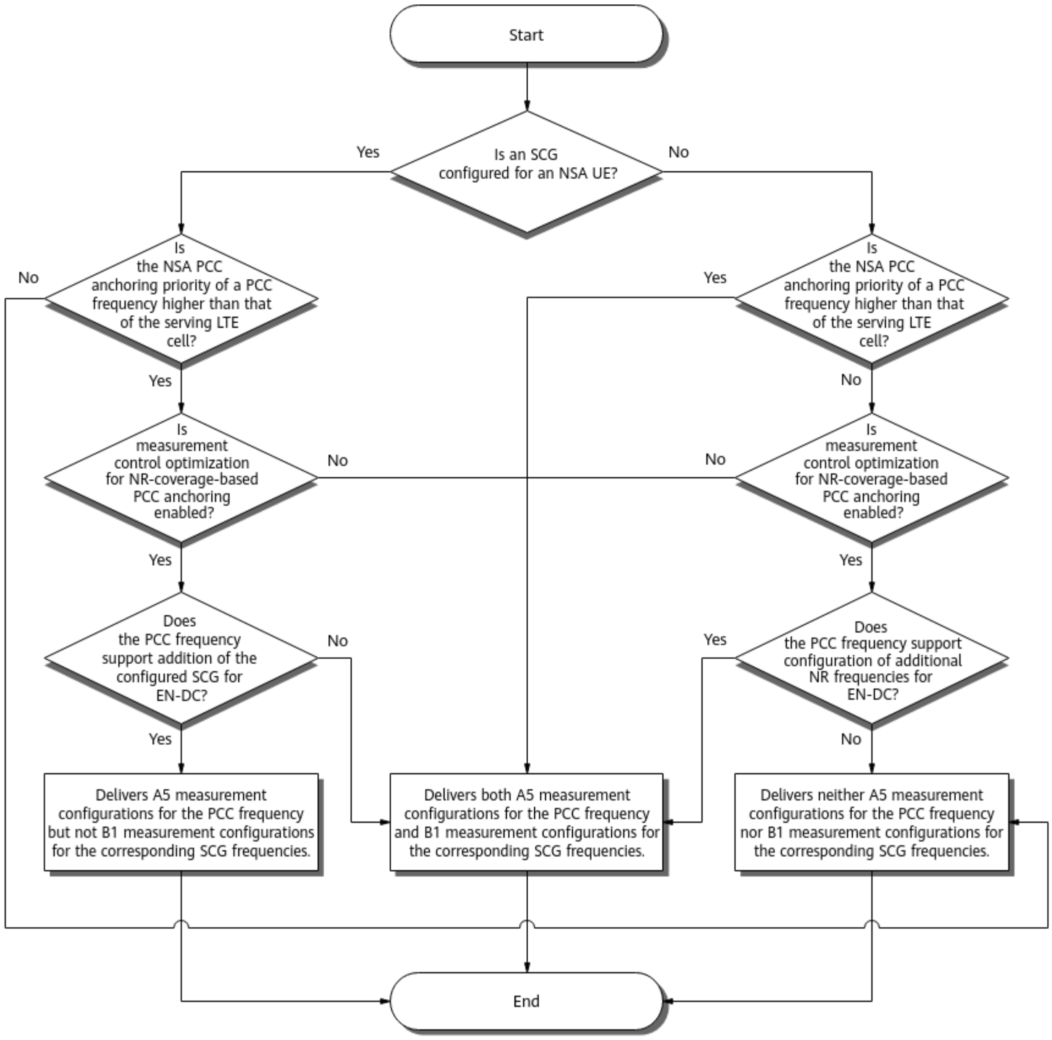
eNodeBs can perform differentiated handling on specific UEs using the blacklist mechanism. If the **NR_COV_ANCHORING_FORBID_SW** option of the **UeCompat**.***BlacklistControlExtSwitch2*** parameter is selected, NSA PCC anchoring based on NR coverage will be prohibited from taking effect for blacklisted UEs. If this option is deselected, no differentiated handling is performed on UEs.

When an NSA UE camps on an anchor cell and there is no NR coverage, the UE can be handed over to a non-anchor cell based on the non-anchor frequency priority. When an NSA UE camps on a non-anchor cell and there are anchor cells and NR coverage, the UE can be handed over to an anchor cell based on the NSA anchor frequency priority and an SCG can be added. The major operations are as follows:

1. Selecting target frequencies for measurement and delivering measurement configurations, as shown in **Figure 5-3**. Measurement control optimization for NR-coverage-based PCC anchoring is controlled by the

**NR_COV_PCC_ANCHOR_MC_OPT_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoEnhSwitch*** parameter.

**Figure 5-3** Selecting target frequencies for measurement and delivering measurement configurations




– If an SCG is configured for an NSA UE:
  - For all PCC frequencies whose NSA PCC anchoring priorities are higher than that of the serving LTE cell:
    - If measurement control optimization for NR-coverage-based PCC anchoring is disabled, the eNodeB delivers both A5 measurement configurations of all the preceding PCC frequencies and B1 measurement configurations of the corresponding SCG frequencies.
    - If measurement control optimization for NR-coverage-based PCC anchoring is enabled, the eNodeB checks each of the preceding PCC frequencies to see whether it supports addition of the configured SCG for EN-DC. If so, the eNodeB delivers only A5 measurement configurations of the PCC frequency but not B1 measurement configurations of the corresponding SCG

frequencies. If not, the eNodeB delivers both A5 measurement configurations of the PCC frequency and B1 measurement configurations of the corresponding SCG frequencies.

- For other PCC frequencies, the eNodeB delivers neither A5 measurement configurations, nor B1 measurement configurations of the corresponding SCG frequencies.

– If no SCG is configured for an NSA UE:

  - For all PCC frequencies whose NSA PCC anchoring priorities are higher than that of the serving LTE cell, the eNodeB delivers both A5 measurement configurations of these PCC frequencies and B1 measurement configurations of the corresponding SCG frequencies.
  - For other PCC frequencies:
    - If measurement control optimization for NR-coverage-based PCC anchoring is disabled, the eNodeB delivers both A5 measurement configurations of the preceding PCC frequencies and B1 measurement configurations of the corresponding SCG frequencies.
    - If measurement control optimization for NR-coverage-based PCC anchoring is enabled, the eNodeB checks each of the preceding PCC frequencies to see whether it supports configuration of additional NR frequencies (corresponding to the serving LTE frequency) for EN-DC based on UE capabilities and eNodeB configurations. If so, the eNodeB delivers both A5 measurement configurations of the PCC frequency and B1 measurement configurations of the corresponding SCG frequencies. If not, the eNodeB delivers neither of them.

To ensure that the NSA UE can quickly send A5 and B1 measurement reports for coverage evaluation, the base station delivers the minimum values of the event A5 threshold 2 and event B1 threshold, that is, –140 dBm and –156 dBm, respectively.

**NOTE**

When delivering measurement configurations of the SCG frequencies corresponding to PCC frequencies, the eNodeB excludes NR frequencies for which the **NR_DL_SCC_FREQ_ONLY_IND** option of the **NrNFreq**.***AggregationAttribute*** parameter is selected.

If measurement control optimization for NR-coverage-based PCC anchoring is enabled and the **SCG_ADD_B1_MEAS_SEQ_OPT_SW** option of the **UeCompat**.***WhitelistCtrlExtSwitch1*** parameter is selected, NR and LTE frequencies can be measured in batches for whitelisted UEs during NSA PCC anchoring based on NR coverage.

– NR frequencies that are not in the non-gap-assisted B1 measurement blacklist are classified as the first batch of NR frequencies, and others the second batch of NR frequencies.

– LTE PCC frequencies that support addition of the first batch of NR frequencies as SCG frequencies are classified as the first batch of LTE frequencies, and others the second batch of LTE frequencies.

– The first batch of LTE frequencies and the first batch of NR frequencies are combined into a candidate frequency set and then step 2 is performed. If no qualified EN-DC combinations are found after the first batch of frequencies is measured, the second batch of frequencies is measured and then step 2 is performed.

NOTE

Filtering NR frequencies based on the non-gap-assisted B1 measurement blacklist requires enabling the non-gap-assisted B1 measurement function (controlled by the **NO_GAP_B1_MEAS_SW_ON** option of the **UeCompat**.***WhiteLstCtrlSwitch*** parameter).

If the B1_NO_GAP_FREQ_COMB_ID_*X* option of the **UeCompat**.***LnrNoGapFreqCombBlacklist*** parameter is selected, the NR frequency (indicated by the **LnrFreqComb**.***DlNrArfcn*** parameter) in frequency combination *X* (indicated by the **LnrFreqComb**.***FreqCombId*** parameter) is in the non-gap-assisted B1 measurement blacklist.

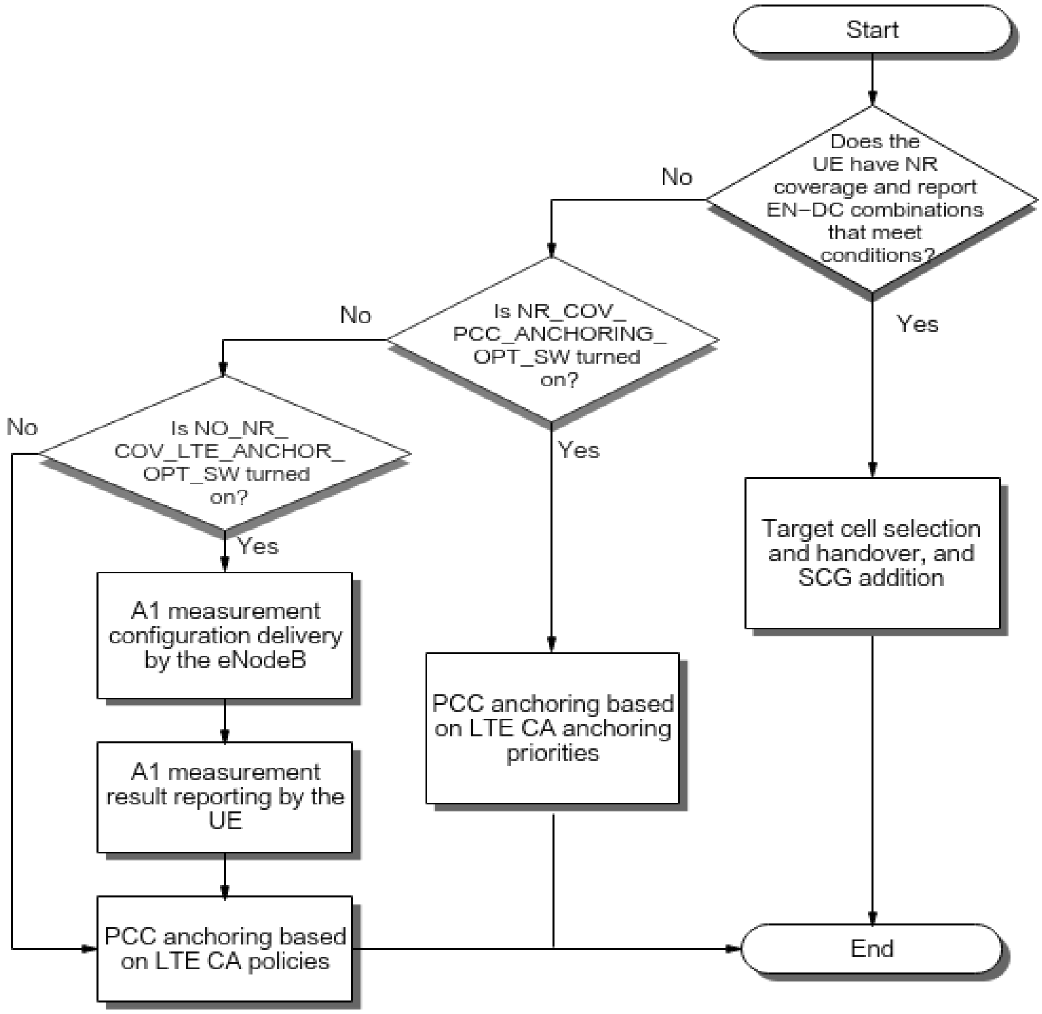
For UEs with an SCG added, if the frequencies in the B1 measurement configurations contain only NR PSCell or SCell frequencies, non-gap-assisted B1 measurement of NR SCG frequencies for NSA DC UEs can be enabled to improve UE throughputs. This function is controlled by the **NSA_DC_NR_SCG_B1_NO_GAP_SW** option of the **EnodebAlgoExtSwitch**.***NsaDcAlgoExtSwitch*** parameter.

When the **NSA_DC_NR_SCG_FBD_B1_NO_GAP_SW** option of the **UeCompat**.***BlacklistControlExtSwitch2*** parameter is selected, "non-gap-assisted B1 measurement of NR SCG frequencies for NSA DC UEs" is forbidden for blacklisted UEs.

2. A decision is made based on measurement reports, as shown in **Figure 5-4**.

**Figure 5-4** Procedure for NSA PCC anchoring based on measurement reports




The NSA UE can be handed over to a target NR cell if the UE is in the DC state before a handover is triggered on the LTE side and the target NR cell is an intra-base-station intra-frequency cell of the source NR cell. This requires the selection of the **MENB_TRIG_INTRA_SGNB_CHANGE_SW** option of the **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** parameter on the NR side.

The NSA UE can be handed over to a target NR cell if the UE is in the DC state before a handover is triggered on the LTE side and the target NR cell is an intra-base-station inter-frequency cell of the source NR cell. This requires the selection of both the **MENB_TRIG_INTRA_SGNB_CHANGE_SW** option of the **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** parameter and the **SIMU_INTRASGNB_INTERFREQ_HO_SW** option of the **gNodeBParam**.***NsaDcOptSwitch*** parameter on the NR side.

NOTE

Assume that the target NR cell is an intra-base-station inter-frequency cell of the source NR cell, the **SIMU_INTRASGNB_INTERFREQ_HO_SW** option of the **gNodeBParam**.***NsaDcOptSwitch*** parameter is deselected on the NR side, and the target LTE cell does not support EN-DC with the source NR cell. Then, SCG release will be triggered during the handover on the LTE side. To prevent this from happening, it is recommended the **INTRA_SGNB_IF_MEAS_FILTER_SW** option of the **EnodebAlgoExtSwitch**.***NsaDcAlgoSwitch*** parameter be selected on the LTE side.

After receiving a measurement report from the NSA UE, the source cell generates a candidate LTE cell list based on the reported A5 measurement report, RSRP or RSRQ threshold for A5, triggering quantity (RSRP), and reporting quantity. The RSRP or RSRQ threshold for A5 is equal to the value of the **PccFreqCfg**.***NsaDcPccA4RsrpThld*** or **PccFreqCfg**.***NsaDcPccA4RsrqThld*** parameter in the **PccFreqCfg** MO with **PccFreqCfg**.***PccDlEarfcn*** set to the EARFCN of the frequency in the event A5 measurement report. The reporting quantity is specified by the **IntraRatHoComm**.***InterFreqHoA4RprtQuan*** parameter. For details about how to generate a candidate LTE cell list, see section "Processing Measurement Reports" in *Mobility Management in Connected Mode* in eRAN feature documentation. The source cell then selects the LTE cells whose coverage areas also have NR coverage from the candidate LTE cell list based on certain factors, such as whether B1 measurement results are reported, whether the B1 measurement report meets the B1 threshold, and whether the NSA UE supports the EN-DC combination of a candidate LTE cell and the corresponding NR cell. The selected LTE cells are candidate target LTE cells for PCC anchoring, and the corresponding NR cells are candidate target NR cells of the preceding LTE cells. The B1 threshold varies depending on whether an LTE cell in the A5 measurement report is served by the same base station as the PCell.

– If the LTE cell in the A5 measurement report is not served by the same base station as the PCell, the B1 threshold equals the value of the **NrScgFreqConfig**.***NsaDcB1ThldRsrp*** parameter.
– If the LTE cell in the A5 measurement report is served by the same base station as the PCell, the B1 threshold equals the sum of the **NrScgFreqConfig**.***NsaDcB1ThldRsrp***, **NsaDcQciParamGroup**.***NsaDcB1RsrpThldOffset***, and **CellOp**.***NsaDcB1RsrpThldOffset*** parameter values.

If there are candidate target LTE cells, the source cell initiates PCC anchoring preferentially to the candidate target LTE cell with the highest NSA PCC anchoring priority (specified by the **PccFreqCfg**.***NsaPccAnchoringPriority*** parameter).

– If the highest-priority candidate target LTE cell is not served by the same base station as the PCell, the source eNodeB enters the measurement result of the highest-priority candidate NR cell corresponding to this LTE cell in the *NR Neighbor Information* IE of the Handover Request message sent during the PCC anchoring. This enables the target anchor cell to initiate an SCG addition request to this NR cell. (The SCG priority for selecting the highest-priority candidate NR cell is specified by the **NrScgFreqConfig**.***ScgDlArfcnPriority*** parameter.)

  The target anchor cell further determines whether the measurement result of the NR cell meets the B1 threshold for the target anchor cell (which equals the sum of the **NrScgFreqConfig**.***NsaDcB1ThldRsrp***, **NsaDcQciParamGroup**.***NsaDcB1RsrpThldOffset***, and **CellOp**.***NsaDcB1RsrpThldOffset*** parameter values). If the NR cell meets the threshold, it is added. If the NR cell does not meet the threshold, it is not added. In the latter case, the source base station successfully triggers NSA PCC anchoring based on NR coverage, but an SCG cannot be added after the handover to the inter-base-station LTE anchor cell.
– If the highest-priority candidate target LTE cell is served by the same base station as the PCell, the eNodeB sends an SCG addition request

preferentially to the highest-priority candidate NR cell during PCC anchoring.

If there are no candidate target LTE cells, the NSA PCC anchoring policy depends on the **NR_COV_PCC_ANCHORING_OPT_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter.

- If this option is selected, LTE CA PCC anchoring is triggered based on the **PccFreqCfg**.***PreferredPccPriority*** parameter setting. If no CA policy is configured on the LTE side, the procedure ends.
- If this option is deselected, the base station performs different procedures depending on the setting of the **NO_NR_COV_LTE_ANCHOR_OPT_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoExtSwitch*** parameter.
  - If the **NO_NR_COV_LTE_ANCHOR_OPT_SW** option is selected, the base station checks whether the RSRP of the serving cell meets the A1 threshold. If so and the procedure of NSA PCC anchoring based on NR coverage is not in progress, LTE CA PCC anchoring is triggered immediately. If no CA policy is configured on the LTE side, the procedure ends.
  - If the **NO_NR_COV_LTE_ANCHOR_OPT_SW** option is deselected, PCC anchoring is triggered based on the existing CA policies on the LTE side. If no CA policy is configured on the LTE side, the procedure ends. For details about CA policy configuration on the LTE side, see *Carrier Aggregation* in eRAN feature documentation.

**NOTE**

It is recommended that the **NR_COV_PCC_ANCHORING_OPT_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter be selected only when LTE CA PCC anchoring enhancement is enabled. For details about LTE CA PCC anchoring enhancement, see *Carrier Aggregation* in eRAN feature documentation.

If the **NR_COV_SCC_BLIND_CFG_OPT_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoEnhSwitch*** parameter is selected, LTE CA SCC selection triggered upon measurement completion during NSA PCC anchoring based on NR coverage does not add SCCs through blind configuration.

If measurement control optimization for NR-coverage-based PCC anchoring is enabled and no SCG is configured for a UE, SCG addition is triggered immediately after the base station receives a measurement report on a highest-priority SCG frequency corresponding to the current LTE PCC.

If measurement control optimization for NR-coverage-based PCC anchoring is enabled, LTE SCell configuration can also be triggered during NSA PCC anchoring based on NR coverage if the configuration conditions are met. For details about the LTE SCell configuration procedure, see *Carrier Aggregation* in eRAN feature documentation.

The priority of an NSA anchor frequency is specified by the **PccFreqCfg**.***NsaPccAnchoringPriority*** parameter. The priority of a non-anchor frequency is specified by the **PccFreqCfg**.***PreferredPccPriority*** parameter. Neighboring NR frequencies are specified by the **NrNFreq** MO.

NSA PCC anchoring based on NR coverage is triggered in the following scenarios:

- The NSA UE is not in the DC state and is in an initial access, necessary incoming handover (such as coverage-based inter-frequency handover), or RRC connection reestablishment procedure (only when the **REEST_NR_COV_ANCHOR_FORBID_SW** option of the **NsaDcAlgoParam**.***NsaDcAlgoExtSwitch*** parameter is deselected).

If both the **NSA_DC_VOLUME_BASED_SCG_ADD_SW** and **VOLUME_BASED_PERIODIC_TRIG_SW** options of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter are selected, NSA PCC anchoring based on NR coverage is triggered when the data volume reaches a threshold. The threshold setting is the same as that for periodic SCG addition based on data volume.

- The NSA UE is in the DC state and in a necessary incoming handover (such as coverage-based inter-frequency handover) procedure.
  - If all the **NSA_DC_VOLUME_BASED_SCG_ADD_SW** and **VOLUME_BASED_PERIODIC_TRIG_SW** options of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter and the **VOLUME_BASED_PCC_ANCHORING_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoExtSwitch*** parameter are selected, NSA PCC anchoring based on NR coverage is triggered only when the data volume reaches a threshold. The threshold setting is the same as that for periodic SCG addition based on data volume.
  - If the **VOLUME_BASED_PCC_ANCHORING_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoExtSwitch*** parameter is deselected, or if this option is selected but the **NSA_DC_VOLUME_BASED_SCG_ADD_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter is deselected, then the data volume is not considered for triggering of NSA PCC anchoring based on NR coverage.
- The **PERIODIC_PCC_ANCHORING_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter is selected. Under this setting, NSA PCC anchoring based on NR coverage is triggered periodically.

  Whether periodic triggering of PCC anchoring is based on data volume is controlled by the **VOLUME_BASED_PERIODIC_TRIG_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter and the **VOLUME_BASED_PCC_ANCHORING_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoExtSwitch*** parameter. The period and data volume threshold settings are the same as those for periodic SCG addition based on data volume.
  - When the NSA UE is in connected mode and no SCG is added:

    If the **VOLUME_BASED_PERIODIC_TRIG_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter is selected, the data volume is considered for periodic triggering of NSA PCC anchoring based on NR coverage. If this option is deselected, the data volume is not considered for periodic triggering of NSA PCC anchoring based on NR coverage.
  - When the NSA UE is in connected mode and an SCG is added:
    - The data volume is not considered for periodic triggering of NSA PCC anchoring based on NR coverage in the following scenarios:
      - The **VOLUME_BASED_PCC_ANCHORING_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoExtSwitch*** parameter is deselected.
      - The **VOLUME_BASED_PCC_ANCHORING_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoExtSwitch*** parameter is selected, but the **VOLUME_BASED_PERIODIC_TRIG_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter is deselected.

- The data volume is considered for periodic triggering of NSA PCC anchoring based on NR coverage if the **VOLUME_BASED_PCC_ANCHORING_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoExtSwitch*** parameter is selected and the **VOLUME_BASED_PERIODIC_TRIG_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter is selected.

- The **NR_COV_PCC_ANCHORING_DELAY_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoExtSwitch*** parameter is deselected. Under this setting, NSA PCC anchoring based on NR coverage is triggered only when the NSA UE receives an SgNB Release Required or SgNB Modification Required message with the cause value "SCG Mobility".

**NOTE**

- The SCG frequencies of the target PCC must be configured as the neighboring NR frequencies of the source PCC, but the SCG cells of the target PCC do not need to be configured as the neighboring NR cells of the source PCC.
- If the **NsaDcMgmtConfig**.***ScgAddPenaltyPeriodCount*** parameter is set to a non-zero value and an incoming RRC connection reestablishment is triggered for the NSA UE during NSA PCC anchoring based on NR coverage, the UE can be barred from entering the anchoring procedure within a period. This period equals **NsaDcMgmtConfig**.***ScgAddPenaltyPeriodCount*** x **NsaDcMgmtConfig**.***ScgAdditionInterval***.
- During PCC anchoring, if a cell with the strongest RSRP on the highest-priority frequency does not meet the conditions (for example, the cell is not configured as a neighboring cell or the cell is heavily loaded), the base station selects another qualified cell on this frequency or the second-highest-priority frequency to trigger PCC anchoring.
- When the **NR_COV_SCG_ADD_PUNISH_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoExtSwitch*** parameter is selected and the **NR_COV_PCC_ANCHORING_DELAY_SW** option is deselected, the newly added SCG cannot contain the intra-gNodeB intra-frequency cells of the previous SCG that is released due to coverage reasons.

UEs can be handed over from a Huawei eNodeB to a non-Huawei eNodeB supporting NSA. This function is enabled when both the **NSA_PCC_ANCHORING_SWITCH** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter and the **PCC_ANCHORING_SMART_DEPLOY_SW** option of the **NsaDcAlgoParam**.***NsaDcAlgoSwitch*** parameter are selected. The Huawei eNodeB determines the NSA capability of the peer non-Huawei eNodeB based on whether an *NR Neighbor Information* IE is contained in X2 messages from the peer end. The Huawei eNodeB does not trigger NSA PCC anchoring if the peer non-Huawei eNodeB does not support NSA.

**NOTE**

- Before enabling this function, check whether the Huawei eNodeB has set up an X2 interface with the peer non-Huawei eNodeB and whether it can determine the NSA capability of the peer non-Huawei eNodeB based on whether an *NR Neighbor Information* IE is contained in X2 messages from the peer end.
- When the **PCC_ANCHORING_SMART_DEPLOY_SW** option of the **NsaDcAlgoParam**.***NsaDcAlgoSwitch*** parameter is selected, the local eNodeB periodically updates the peer eNodeB status information. If the peer eNodeB status changes, the local eNodeB waits for a maximum of 30 minutes to update the status information, and then determines whether to perform NSA PCC anchoring.

The **NsaDcAlgoParam**.***LnrCoMeasLteMeasDuration*** parameter, which specifies the LTE frequency measurement duration of UEs, can be configured to effectively reduce the gap-assisted measurement time of UEs when both LTE and NR frequencies are measured.

The **NR_COV_ANCH_PINGPONG_AVOID_SW** option of the **NsaDcAlgoParam**.***NsaDcAlgoExtSwitch*** parameter can be selected to avoid ping-pong switching of NSA PCC anchors during NSA PCC anchoring based on NR coverage. If the eNodeB determines that ping-pong switching is likely to occur on an NSA UE that has been handed over to a local cell of the eNodeB based on coverage, the eNodeB will not trigger NR-coverage-based PCC anchoring for the UE.

### NSA PCC Anchoring Based on Virtual Grids

Assume that the eNodeB needs to transfer an NSA UE to an NSA PCC with a higher priority, and add a PSCell or change the PSCell. Then, the eNodeB delivers A3 measurement configurations for the camping frequency of the UE. After receiving A3 measurement results, the eNodeB queries the coverage information of candidate NSA anchor frequencies based on LTE virtual grid models and queries the coverage information of candidate SCG frequencies based on LNR virtual grid models. For details about LTE virtual grid models, see *Multi-carrier Unified Scheduling* in eRAN feature documentation. The time to trigger for event A3 is specified by the **LnrCarrierSelection**.***IntraFreqMeasA3TimeToTrig*** parameter. If both the entering condition of event A4 or A5 for NSA PCC anchoring and the entering condition of event B1 are met, the eNodeB transfers the NSA UE to the target NSA PCC, and adds a PSCell or changes the PSCell.

## 5.1.3 Blind PSCell Addition

Blind PSCell addition (also known as blind PSCell configuration) is controlled by the **NSA_BLIND_SCG_ADDITION_SWITCH** option of the **NsaDCMgmtConfig**.***NsaDcAlgoSwitch*** parameter.

The blind PSCell addition procedure is as follows:

1. The eNodeB checks whether there are blind-configurable neighboring NR cells on a high-priority neighboring NR frequency configured on the eNodeB:
   - If there is only one such NR cell, the eNodeB selects the gNodeB serving this NR cell to initiate an SgNB addition request. If there are multiple such NR cells, the eNodeB sorts the cells and selects the gNodeB serving the cell ranking at the top to initiate an SgNB addition request.
   - If there are no blind-configurable neighboring NR cells on a high-priority neighboring NR frequency but there are such cells on a low-priority neighboring NR frequency configured on the eNodeB, the eNodeB selects the high-priority neighboring NR frequency and delivers measurement configurations to the UE. If the UE does not send a measurement report on the high-priority neighboring NR frequency before the timer expires, the eNodeB selects a blind-configurable neighboring NR cell on the low-priority neighboring NR frequency to initiate blind PSCell addition.
   - If there are no blind-configurable neighboring NR cells on any neighboring NR frequency configured on the eNodeB, a measurement procedure is started and then measurement-based PSCell configuration is performed. For details, see *EPC-based NSA Basics*.

2. If blind PSCell addition fails, the following operations are performed:
   - If the preparation for blind PSCell addition fails (that is, the MeNB receives an SgNB Addition Reject message from the SgNB), the base station selects the frequencies of all neighboring NR cells that meet the conditions and initiates a measurement-based PSCell configuration procedure. For details, see *EPC-based NSA Basics*.
   - If blind PSCell addition fails, the base station directly initiates a measurement-based PSCell configuration procedure next time PSCell configuration is triggered based on data volume, without checking whether there are blind-configurable neighboring NR cells. For details, see *EPC-based NSA Basics*.

NOTE

- In Option 3x, data forwarding is not supported in the following scenarios:
  - After an NSA UE performing EN-DC services is handed over to the target MeNB through the S1 interface, blind PSCell addition is performed and then an SgNB change occurs.
  - After an NSA UE not performing EN-DC services is handed over to the target MeNB through the S1 interface, blind PSCell addition is performed.
- If the **NRDUCell**.***SupplementaryCellInd*** parameter is set to **ULDL_SUPPLEMENTARY_CELL** for a high-frequency neighboring NR cell, this cell can act as an SCell of CA. It is not recommended that this cell be configured as a blind-configurable neighboring NR cell.
- If a neighboring NR cell has an S1-U fault, the MeNB does not initiate a procedure for configuring this cell as a PSCell in a blind manner.
- If the **OVER_DISTANCE_FLAG** option of the **NrNRelationship**.***AggregationAttribute*** parameter is selected for a neighboring NR cell, the MeNB does not initiate a procedure for configuring this cell as a PSCell in a blind manner.
- During blind PSCell addition, the eNodeB filters out the NR frequencies for which the **NR_DL_SCC_FREQ_ONLY_IND** option of the **NrNFreq**.***AggregationAttribute*** parameter is selected.
- It filters out NR cells that do not support uplink transmission when the **NR_SCG_UL_NO_CHECK_SW** option of the **NsaDcAlgoParam**.***NsaDcAlgoExtSwitch*** parameter is deselected.

On the LTE side, the **NrScgFreqConfig**.***ForbiddenSpidGrpId*** parameter can be used to specify an SPID group for which a specified SCG frequency cannot be added as the frequency of an NR SCC. As the same NR frequency can be configured in the **NrScgFreqConfig** MOs of different PCCs, the same **NrScgFreqConfig**.***ForbiddenSpidGrpId*** parameter setting is recommended for this NR frequency.

During an LTE handover of an NSA UE, the target eNodeB sends PSCell addition requests to neighboring NR cells on each neighboring NR frequency in descending order of NR frequency priorities after receiving a handover request. For each neighboring NR frequency, the following situations exist:

- If blind-configurable neighboring NR cells on that NR frequency are available, the eNodeB attempts to add one of them in a blind manner.
- If measured neighboring NR cells on that NR frequency are available, the eNodeB attempts to add one of them.
- If both blind-configurable and measured neighboring NR cells on that NR frequency are available, the eNodeB proceeds according to the setting of the

**HO_WITH_SCG_BLIND_PRI_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoExtSwitch*** parameter:

- If this option is selected, the eNodeB sends a request to add a blind-configurable neighboring NR cell as a PSCell.
- If this option is deselected, the eNodeB sends a request to add a measured neighboring NR cell as a PSCell. If none of the measured neighboring NR cells can be added, for example, if neighbor relationships are missing for all measured neighboring NR cells, the eNodeB no longer attempts to add another neighboring NR cell on that NR frequency.

- If neither blind-configurable nor measured neighboring NR cells on that NR frequency are available, the eNodeB skips the frequency.

If the target eNodeB fails to send PSCell addition requests for a neighboring NR frequency, it proceeds to other neighboring NR frequencies with lower priorities. If no PSCell addition request is successfully initiated after all neighboring NR frequencies are traversed, the PSCell is not added during the handover.

# 5.2 Network Analysis

## 5.2.1 Benefits

Measurement control optimization for NR-coverage-based PCC anchoring: After this function is enabled, the number of gap-assisted measurements decreases, the SCC and SCG addition accelerates, and the user-perceived rate increases.

## 5.2.2 Impacts

The impacts between this function and other functions are described in **5.3.2.3 Function Impacts**.

Enabling the "prohibition of NSA PCC anchoring for high-mobility UEs" function may have the following impacts on the network:

- The E-RAB service drop rate decreases. E-RAB service drop rate = (**L.E-RAB.AbnormRel.eNBTot** + **L.E-RAB.AbnormRel.HOOut**) / **L.E-RAB.SuccEst** x 100
- The value of **L.RRC.ReEst.ReconfFail.Att** decreases.
- The values of the following counters decrease: **L.NsaDC.PCCAnchor.HHO.PrepAttOut**, **L.NsaDC.PCCAnchor.HHO.ExecAttOut**, and **L.NsaDC.PCCAnchor.HHO.ExecSuccOut**.

Enabling enhanced NSA PCC anchoring based on LTE load has the following impacts on the network:

- The load of ultra-high-load cells decreases, and the values of the following counters increase: **L.NsaDC.PCCAnchor.HHO.PrepAttOut**, **L.NsaDC.PCCAnchor.HHO.ExecAttOut**, and **L.NsaDC.PCCAnchor.HHO.ExecSuccOut**.
- The load of light-load cells increases.

Enabling NSA PCC anchoring based on NR coverage has the following impacts on the network:

- There is a possibility that the values of **N.NsaDc.DRB.Add.Att** and **N.NsaDc.DRB.Add.Succ** increase in scenarios where NR coverage is continuous and all LTE base stations can be anchor base stations. In addition, the value of **N.NsaDc.DRB.Rel** may increase, causing a decrease in the proportion of abnormal DRB releases for NSA UEs on the SgNB (calculated by **N.NsaDc.DRB.AbnormRel**/**N.NsaDc.DRB.Rel**).
- NSA PCC anchoring will not be performed for UEs when there is no NR coverage. Therefore, the number of NSA PCC anchoring times may decrease. The number of NSA PCC anchoring times can be observed through **L.NsaDC.PCCAnchor.HHO.ExecSuccOut**.

Enabling NR-coverage-based blind SCC configuration optimization has the following impacts on the network:

- The number of blind SCell addition attempts for CA UEs (**L.CA.DLSCell.Add.Blind.Att**) decreases.
- NSA PCC anchoring based on NR coverage does not add LTE CA SCCs in a blind manner after the measurement procedure is complete. As a result, SCCs are not added in a timely manner for UEs, resulting in slightly deteriorated user experience.

After the "non-NSA-PCC frequency filtering during frequency-priority-based frequency selection" function takes effect, the values of the following counters decrease: **L.HHO.InterFreq.FreqPri.PrepAttOut**, **L.HHO.InterFreq.FreqPri.ExecAttOut**, and **L.HHO.InterFreq.FreqPri.ExecSuccOut**.

After non-gap-assisted B1 measurement of NR SCG frequencies for NSA DC UEs is enabled, these UEs perform B1 measurement on NR SCG frequencies without relying on measurement gaps, therefore experiencing higher throughputs.

# 5.3 Requirements

## 5.3.1 Licenses

On the LTE side:

| RAT | Feature ID | Feature Name | Model | Sales Unit |
|---|---|---|---|---|
| LTE FDD | LNOFD-151333 | EN-DC Optimal Carrier Selection | LT1S0EDOCS00 | Per Cell |
| LTE TDD | TDLNOFD-151504 | EN-DC Optimal Carrier Selection | LT4SENDCSTDD | Per Cell |

On the NR side:

None

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation.

## 5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

### 5.3.2.1 Prerequisite Functions

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| LTE FDD<br>LTE TDD | NSA PCC anchoring | **NSA_PCC_ANCHORING_SWITCH** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter | *EPC-based NSA Performance Enhancement* | NSA PCC anchoring based on LTE load and uplink coverage requires that NSA PCC anchoring be enabled. |
| LTE FDD<br>LTE TDD | NSA PCC anchoring based on LTE load and uplink coverage | **NSA_DC_FLEXIBLE_PCC_ANCHOR_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter | *EPC-based NSA Performance Enhancement* | Enhanced NSA PCC anchoring based on LTE load requires that NSA PCC anchoring based on LTE load and uplink coverage be enabled. |
| LTE FDD<br>LTE TDD | NSA PCC anchoring | **NSA_PCC_ANCHORING_SWITCH** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter | *EPC-based NSA Performance Enhancement* | NSA PCC anchoring based on virtual grids requires that NSA PCC anchoring be enabled. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| LTE FDD<br>LTE TDD | Frequency-priority-based inter-frequency handover | **FREQ_PRI_HO_SW** option of the **NsaDcMgmtConfig**.***NsaDcUeLteFunActivationSw*** parameter | *NSA Mobility Management* | "Non-NSA-PCC frequency filtering during frequency-priority-based frequency selection" requires that frequency-priority-based inter-frequency handover be enabled. |

### 5.3.2.2 Mutually Exclusive Functions

None

### 5.3.2.3 Function Impacts

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| LTE FDD<br>LTE TDD | Intelligent selection of serving cell combinations | **CaSmartSelectionSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter | *Carrier Aggregation* in eRAN feature documentation | After NSA PCC anchoring is enabled, PCC anchoring for "intelligent selection of serving cell combinations" does not take effect. After NSA PCC anchors are selected, SCells are then selected based on "intelligent selection of serving cell combinations". |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| LTE FDD<br>LTE TDD | Frequency-priority-based inter-frequency handover | **FreqPriorIFHOSwitch** option of the **CellAlgoSwitch**.***FreqPriorityHoSwitch*** parameter | *Mobility Management in Connected Mode* in eRAN feature documentation | If the frequency priorities configured for frequency-priority-based inter-frequency handover are different from the NSA PCC anchoring priorities, an NSA UE that has been handed over to a high-priority NSA PCC may be handed over to a frequency with a low NSA PCC anchoring priority due to a frequency-priority-based inter-frequency handover.<br>To prevent such ping-pong handovers, you can deselect the **FREQ_PRI_HO_SW** option of the **NsaDcMgmtConfig**.***NsaDcUeLteFunActivationSw*** parameter so that frequency-priority-based inter-frequency handover does not take effect for NSA UEs. For details, see *NSA Mobility Management*. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| LTE FDD<br>LTE TDD | Service-based inter-frequency handover | **ServiceBasedInterFreq-HoSwitch** option of the **ENodeBAlgoSwitch**.***HoAlgoSwitch*** parameter and **SrvBasedInterFreq-HoSw** option of the **CellAlgoSwitch**.***HoAllowedSwitch*** parameter selected | *Mobility Management in Connected Mode* in eRAN feature documentation | Service-based inter-frequency handover may cause an NSA UE to be handed over from an NSA PCC anchor to a non-anchor carrier.<br>To prevent such handovers, you can deselect the **SERV_BASED_INTER_FREQ_HO_SW** option of the **CellQciPara**.***NsaDcOptSwitch*** parameter for a specific QCI so that service-based inter-frequency handover cannot be triggered for this QCI. For details, see *NSA Mobility Management*. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| LTE FDD<br>LTE TDD | Multi-band optimal carrier selection | **MbfcsSwitch** option of the **CellAlgoSwitch.MlbAlgoSwitch** parameter | *Multi-band Optimal Carrier Selection* in eRAN feature documentation | Multi-band optimal carrier selection may cause an NSA UE to be handed over from an NSA PCC anchor to a non-anchor carrier.<br>To prevent such handovers, you can deselect the **MBOCS_SW** option of the **NsaDcMgmtConfig**.***NsaDcUeLteFunActivationSw*** parameter so that multi-band optimal carrier selection does not take effect for NSA UEs. For details, see *NSA Mobility Management*. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| LTE FDD<br>LTE TDD | LTE spectrum coordination | **HoAdmitSwitch** option of the **CellMlbHo**.***MlbMatchOtherFeatureMode*** parameter deselected, **HoWithSccCfgSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter selected, and **SpectrumCoordinationSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter selected | *LTE Spectrum Coordination* in eRAN feature documentation | LTE spectrum coordination may cause an NSA UE to be handed over from an NSA PCC anchor to a non-anchor carrier.<br>To prevent such handovers, you can deselect the **SPCT_COORD_INTER_FREQ_HO_SW** option of the **NsaDcMgmtConfig**.***NsaDcUeLteFunActivationSw*** parameter so that LTE spectrum coordination does not take effect for NSA UEs. For details, see *NSA Mobility Management*. |
| LTE FDD<br>LTE TDD | LTE spectrum coordination enhancement | **WbbCaMultiCarrierCoordSw** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter | *LTE Spectrum Coordination* in eRAN feature documentation | NSA PCC anchoring takes precedence over LTE spectrum coordination enhancement. When both functions are enabled, only NSA PCC anchoring takes effect. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| LTE FDD<br>LTE TDD | System information broadcast | **Sib24Switch** option of the **CellSiMap**.***SiSwitch*** parameter | *Idle Mode Management* in eRAN feature documentation | If system information broadcast is enabled together with NSA PCC anchoring, then the **CellSiMap**.***SibxPeriod*** parameter (where, x = 2, 3, ...) must be greater than or equal to 16 radio frames, the **CellSiMap**.***SibTransCtrlSwitch*** parameter must be set to **OFF**, and the **CellSiMap**.***SiSchResRatio*** parameter must be set to **0**. Otherwise, NSA PCC anchoring may fail. |
| LTE FDD<br>LTE TDD | Management of neighboring NG-RAN frequency measurement flags | **NR_NFREQ_MEAS_MGMT_SW** option of the **CellAlgoExtSwitch**.***AnrOptSwitch*** parameter | *ANR Management* in eRAN feature documentation | If measurement of a neighboring frequency is disabled by setting the corresponding frequency measurement flag, the UE does not select this frequency as a target frequency for NSA PCC anchoring. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| LTE FDD | Uplink full-antenna reception | **UL_COVERAGE_BOOST_SW** option of the **SectorSplitGroup**.***SectorSplitSwitch*** parameter | *Massive MIMO Enhancements (FDD)* in eRAN feature documentation | After uplink full-antenna reception takes effect, the number of times NSA PCC anchoring is triggered by LTE uplink coverage may change because uplink coverage on the LTE side improves. This affects NSA PCC anchoring based on LTE load and uplink coverage. |
| LTE FDD<br>LTE TDD | Non-gap-assisted B1 measurement | **NO_GAP_B1_MEAS_SW_ON** option of the **UeCompat**.***WhiteLstCtrlSwitch*** parameter | *NSA Mobility Management* | After measurement control optimization for NR-coverage-based PCC anchoring is enabled, the eNodeB allows whitelisted UEs to measure NR and LTE frequencies in batches only when non-gap-assisted B1 measurement is also enabled. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| LTE FDD<br>LTE TDD | Smart carrier selection based on virtual grids | **SMART_CARRIER_SELECTION_SW** option of the **MultiCarrUnifiedSch**.***MultiCarrierUnifiedSchSw*** parameter | *Multi-carrier Unified Scheduling* in eRAN feature documentation | • On an NSA network, it is recommended that NSA PCC anchoring be enabled before smart carrier selection based on virtual grids is enabled. In this case, smart carrier selection based on virtual grids performs only SCC selection for NSA UEs but does not perform PCC anchoring for NSA UEs.<br>• If smart carrier selection based on virtual grids is enabled while NSA PCC anchoring is disabled, it performs both PCC anchoring and SCC selection for NSA UEs. In this case, the PCC is selected based on the CA capability on the LTE side.<br>– If no SCG is added for an NSA UE, it may be handed over to an LTE frequency that does not support NSA, |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| | | | | preventing SCG addition.<br>– If an SCG has been added for an NSA UE, the SCG may be changed or removed. |
| LTE FDD<br>LTE TDD | Optimization-based carrier selection | **OPT_MODEL_BASED_CARR_SEL_SW** option of the **MultiCarrUnifiedSch**.***MultiCarrierUnified-SchEnSw*** parameter | *Multi-carrier Unified Scheduling* in eRAN feature documentation | When the NSA PCC anchoring function is enabled and the NSA carrier combination selection based on user experience function is disabled in NSA networking, optimization-based carrier selection manages only SCC selection, rather than PCC anchoring, for NSA UEs. As such, the gains of optimization-based carrier selection may decrease in this scenario. |
| LTE FDD<br>LTE TDD | NSA carrier combination selection based on user experience | **NSA_LTE_FDD_SEL_OPT_SW** or **NSA_LTE_TDD_SEL_OPT_SW** option of the **EnodebAlgoExtSwitch**.***MultiNetworkingOp-tionOptSw*** parameter | *EPC-based NSA Experience Improvement* | |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| LTE FDD<br>LTE TDD | Differentiated SCG addition threshold configuration for NR CA-capable UEs | **SCG_ADD_B1_THLD_DIFF_SW** option of the **NsaDcAlgoParam**.***NsaDcAlgoExtSwitch*** parameter | *EPC-based NSA Basics* | Differentiated SCG addition threshold configuration for NR CA-capable UEs has the following impacts on NSA PCC anchoring based on NR coverage for NSA UEs that support NR FDD +TDD CA:<br>The sum of the original effective RSRP threshold and the offset specified by the **NrScgFreqConfig**.***CaUeScgAddRsrpThldOffset*** parameter is used as the actual RSRP threshold for event B1 to determine whether an SCG can be added with the candidate target LTE cell during PCC anchoring. |

| RAT | Function Name | Function Switch | Reference | Description |
|---|---|---|---|---|
| LTE FDD<br>LTE TDD | Fast SCG addition for NSA DC UEs in LTE and NR co-site co-coverage scenarios | **NsaDcMgmtConfig**.***NsaDcFastScgAddA1RsrpThld*** | *NSA Mobility Management* | During NSA PCC anchoring based on NR coverage, if the eNodeB receives an event A1 measurement report in response to the measurement configurations delivered for the fast SCG addition for NSA DC UEs in LTE and NR co-site co-coverage scenarios function, the eNodeB deletes the event A1 measurement configurations. In this case, the SCG addition procedure does not take effect. |
| LTE FDD<br>LTE TDD | FDD SCG addition for NR CA-capable UEs based on the RSRP of candidate NR TDD SCCs | **NsaDcAlgoParam**.***NrTddSccRsrpThld*** set to a value other than **255** | *EPC-based NSA Basics* | The "FDD SCG addition for NR CA-capable UEs based on the RSRP of candidate NR TDD SCCs" function does not take effect when NSA PCC anchoring based on NR coverage takes effect. |

## 5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

### Base Station Models

On the LTE side, the compatible base stations are as follows:

- 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R. 5900 series base stations must be configured with a BBU model other than the BBU5900C.
- DBS3900 LampSite and DBS5900 LampSite

On the NR side, the compatible base stations are as follows:

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910, and 5900 series base stations must be configured with a BBU model other than the BBU5900C.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

### Boards

The board requirements are as follows:

- NSA PCC anchoring

  LTE:

  – Main control boards: UMPTb and later
  – BBPs: All BBPs except the LBBPd and LBBPc allow cells on these boards to serve as NSA PCC anchors.

  **NOTE**

  Though the LBBPd and LBBPc do not allow cells on these boards to serve as NSA PCC anchors, the boards support handover from non-anchor cells to NSA PCC anchor cells on other boards. In addition, the LBBPc does not support data-volume-based handover.

  NR: no requirements

- NR-coverage-based blind SCC configuration optimization

  LTE:

  – Main control boards: UMPTb and later
  – Baseband processing units: LBBPd and later

- Blind PSCell addition

  LTE:

  – Main control boards: UMPTb and later
  – Baseband processing units: LBBPd and later

  NR: no requirements

### RF Modules

This function does not depend on RF modules.

## 5.3.4 Networking

For details, see *EPC-based NSA Basics*.

### 5.3.5 Others

- UE
  - UEs must support NSA DC specified in 3GPP Release 15.
  - UEs must have subscribed to LTE and NR services.
  - UEs must match gNodeB and eNodeB versions.
- EPC
  - The EPC must be CloudEPC to support Option 3 and Option 3x.
  - The EPC must support NSA DC. If the core network is provided by Huawei, see *WSFD-021101 5G NSA (Opt.3) Dual Connectivity Management* for details.
  - If NSA DC is enabled on an eNodeB, the connected MMEs need to support NSA DC. If a connected MME does not support NSA DC, the **MmeCapInfo**.***MmeNsaDcCapability*** parameter for this MME must be set to **NOT_SUPPORT**.

## 5.4 Operation and Maintenance

### 5.4.1 Data Configuration

#### 5.4.1.1 Data Preparation

The following table describes the parameters that must be set in the **PccFreqCfg** MO on the LTE side to configure PCC frequencies (or anchor frequencies).

**NOTE**

This MO and PCC anchoring priority do not need to be configured for non-anchor frequencies.

| Parameter Name | Parameter ID | Setting Notes |
| --- | --- | --- |
| PCC Downlink EARFCN | **PccFreqCfg**.***PccDlEarfcn*** | Set this parameter based on the operator's network plan. |
| NSA PCC Anchoring Priority | **PccFreqCfg**.***NsaPccAnchoringPriority*** | Set this parameter based on the network plan if an independent camping policy needs to be configured for NSA UEs. |

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| NSA DC PCC A4 RSRP Threshold | **PccFreqCfg**.***NsaDcPccA4RsrpThld*** | It is recommended that this parameter be set to a value greater than or equal to the coverage-based inter-frequency handover threshold for event A2 in the target cell. This prevents the triggering of a coverage-based inter-frequency handover after a handover to a target cell and the triggering of PCC anchoring after an outgoing RRC connection reestablishment, thereby preventing ping-pong handovers. |

The following table describes the parameters that must be set in the **NsaDcMgmtConfig**, **NsaDcAlgoParam**, **EnodebAlgoExtSwitch**, and **CellOpHoCfg** MOs to configure NSA PCC anchoring on the LTE side so that separate camping priorities can be configured for NSA UEs.

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| NSA DC Algorithm Switch | **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** | Select the **NSA_PCC_ANCHORING_SWITCH** option. |
| NSA DC PCC Anchoring Policy | **NsaDcMgmtConfig**.***NsaDcPccAnchoringPolicy*** | Set this parameter to **DEFAULT**. |
| NSA DC Algorithm Switch | **NsaDcAlgoParam**.***NsaDcAlgoSwitch*** | Select the **PCC_ANCHORING_SMART_DEPLOY_SW** option if both Huawei eNodeBs and non-Huawei eNodeBs are deployed. |
| NSA Protocol Compatibility Switch | **EnodebAlgoExtSwitch**.***NsaProtocolCompatSw*** | Select the **NR_NB_INFO_BASED_NSA_CAPB_SW** option if both Huawei eNodeBs and non-Huawei eNodeBs are deployed. |

| Parameter Name | Parameter ID | Setting Notes |
| --- | --- | --- |
| High Mobility UE Handover Forbidden Sw | **CellOpHoCfg**.*HighMobiUeHoForbidSw* | Select the **NSA_ANCHOR_HO_FORBID_SW** option if the "prohibition of NSA PCC anchoring for high-mobility UEs" function is required. |
| Blacklist Control Extension Switch 2 | **UeCompat**.*BlacklistControlExtSwitch2* | If there are UEs incompatible with NSA PCC anchoring on the network, select the **NSA_PCC_ANCHORING_FORBID_SW** option so that NSA PCC anchoring for RRC_CONNECTED UEs is forbidden for blacklisted UEs. |

The following table describes the parameters that must be set in the **NsaDcMgmtConfig** MO to enable periodic triggering of NSA PCC anchoring on the LTE side.

| Parameter Name | Parameter ID | Setting Notes |
| --- | --- | --- |
| NSA DC Algorithm Switch | **NsaDcMgmtConfig**.*NsaDcAlgoSwitch* | Select the **PERIODIC_PCC_ANCHORING_SW** option. |

The following table describes the parameters that must be set in the **NsaDcMgmtConfig** MO to enable data-volume-based NSA PCC anchoring for UEs in the EN-DC state by turning on the periodic data-volume-based triggering switch on the LTE side.

| Parameter Name | Parameter ID | Setting Notes |
| --- | --- | --- |
| NSA DC Algorithm Switch | **NsaDcMgmtConfig**.*NsaDcAlgoSwitch* | Select the **VOLUME_BASED_PERIODIC_TRIG_SW** option. |

The following table describes the parameters that must be set in the **NsaDcMgmtConfig** MO to enable data-volume-based NSA PCC anchoring for UEs not in the EN-DC state by turning on both the periodic data-volume-based triggering switch and the data-volume-based NSA PCC anchoring switch on the LTE side.

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| NSA DC Algorithm Switch | **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** | Select the **VOLUME_BASED_PERIODIC_TRIG_SW** option. |
| NSA DC Algorithm Extension Switch | **NsaDcMgmtConfig**.***NsaDcAlgoExtSwitch*** | Select the **VOLUME_BASED_PCC_ANCHORING_SW** option. |

The following table describes the parameters that must be set in the **NsaDcMgmtConfig** MO on the LTE side to configure NSA DC PCC anchoring for RRC_IDLE UEs.

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| NSA DC Algorithm Switch | **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** | Select the **IDLE_MODE_NSA_PCC_ANCHORING_SW** option if NSA DC PCC anchoring for RRC_IDLE UEs is required. |
| NSA DC Algorithm Extension Switch | **NsaDcMgmtConfig**.***NsaDcAlgoExtSwitch*** | Select the **IDLE_NSA_PCC_ANCHORING_OPT_SW** option if NSA PCC anchoring optimization for RRC_IDLE UEs is required. |

The following table describes the parameters that must be set in the **NsaDcMgmtConfig** MO on the LTE side to specify whether to broadcast the *upperLayerIndication* IE in SIB2.

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| Upper Layer Indication Switch | **NsaDcMgmtConfig**.***UpperLayerIndicationSwitch*** | The value **NR_NCELL_BASED_BROADCAST** is recommended. If there are UEs not supporting the R15 *upperLayerIndication* IE in the network, the value **OFF** is recommended. |

The following table describes the parameters that must be set in the **CnOperator** MO on the LTE side to specify whether to broadcast the serving PLMN *upperLayerIndication* IE in SIB2.

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| Operator Function Switch | **CnOperator**.***OperatorFunSwitch*** | Set this parameter based on the operator's network plan. If the serving PLMN *upperLayerIndication* IE needs to be broadcast in SIB2, deselect the **UPPER_LYR_IND_NO_BROADCAST_SW** option. |

The following table describes the parameters that must be set in the **NsaDcMgmtConfig**, **NsaDcAlgoParam**, and **UeCompat** MOs to configure NSA PCC anchoring enhancement on the LTE side.

<table>
<tr><th>Parameter Name</th><th>Parameter ID</th><th>Setting Notes</th></tr>
<tr><td>NSA DC Algorithm Switch</td><td>NsaDcMgmtConfig.NsaDcAlgoSwitch</td><td>Select the NSA_PCC_ANCHORING_SWITCH and NSA_DC_FLEXIBLE_PCC_ANCHOR_SW options to enable NSA PCC anchoring based on LTE load and uplink coverage.</td></tr>
<tr><td>NSA DC Algorithm Enhancement Switch</td><td>NsaDcMgmtConfig.NsaDcAlgoEnhSwitch</td><td>Select the LOAD_BASED_PCC_ANCHOR_ENH_SW option to enable enhanced NSA PCC anchoring based on LTE load.</td></tr>
<tr><td>NSA DC Algorithm Switch</td><td>NsaDcMgmtConfig.NsaDcAlgoSwitch</td><td rowspan="2">Select the NSA_PCC_ANCHORING_SWITCH option of the NsaDcMgmtConfig.NsaDcAlgoSwitch parameter and set the NsaDcMgmtConfig.NsaDcPccAnchoringPolicy parameter to BASED_ON_NR_COVERAGE to enable NSA PCC anchoring based on NR coverage.</td></tr>
<tr><td>NSA DC PCC Anchoring Policy</td><td>NsaDcMgmtConfig.NsaDcPccAnchoringPolicy</td></tr>
</table>

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| NSA DC Algorithm Extension Switch | **NsaDcAlgoParam**.***NsaDcAlgoExtSwitch*** | Select the **REEST_NR_COV_ANCHOR_FORBID_SW** option of this parameter. |
| Blacklist Control Extension Switch 2 | **UeCompat**.***BlacklistControlExtSwitch2*** | Select the **NR_COV_ANCHORING_FORBID_SW** option of this parameter if there are UEs with compatibility issues on the network. |
| User Number High Load Threshold | **NsaDcMgmtConfig**.***UserNumberHighLoadThld*** | Set this parameter based on the network plan. |
| User Number Ultra-High Load Threshold | **NsaDcMgmtConfig**.***UserNumUltraHighLoadThld*** | The value of this parameter multiplied by 0.9 must be greater than the value of the **User Number High Load Threshold** parameter. |

The following table describes the parameters that must be set in the **NsaDcMgmtConfig** and **NrNRelationship** MOs on the LTE side to enable blind SCG addition.

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| NSA DC Algorithm Switch | **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** | Select the **NSA_BLIND_SCG_ADDITION_SWITCH** option, and set **NrNRelationship**.***BlindConfigIndicator*** to **TRUE**. |
| Blind Configuration Indicator | **NrNRelationship**.***BlindConfigIndicator*** | |

The following table describes the parameters that must be set in the **NsaDcMgmtConfig** MO to enable NR-coverage-based blind SCC configuration optimization on the LTE side. This function is used to prevent blind SCC configuration during LTE CA SCC selection triggered upon measurement completion in NSA PCC anchoring based on NR coverage.

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| NSA DC Algorithm Enhancement Switch | **NsaDcMgmtConfig**.***NsaDcAlgoEnhSwitch*** | Select the **NR_COV_SCC_BLIND_CFG_OPT_SW** option. |

The following table describes the parameter that must be set in the **NsaDcMgmtConfig** MO on the LTE side to prevent NSA UEs from being handed over from NSA PCells to cells operating on non-NSA-PCC frequencies during frequency-priority-based inter-frequency handovers.

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| NSA DC Algorithm Enhancement Switch | **NsaDcMgmtConfig**.***NsaDcAlgoEnhSwitch*** | Select the **FREQ_PRI_SEL_NON_PCC_FILTER_SW** option. |

The following table describes the parameters that must be set in the **NsaDcMgmtConfig**, **UeCompat**, **EnodebAlgoExtSwitch**, and **NsaDcAlgoParam** MOs to configure optimizations for NR-coverage-based PCC anchoring on the LTE side.

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| NSA DC Algorithm Enhancement Switch | **NsaDcMgmtConfig**.***NsaDcAlgoEnhSwitch*** | Select the **NR_COV_PCC_ANCHOR_MC_OPT_SW** option. |
| White List Control Switch | **UeCompat**.***WhiteLstCtrlSwitch*** | Select the **NO_GAP_B1_MEAS_SW_ON** option when whitelisted UEs are required to measure NR and LTE frequencies in batches. |

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| NSA DC Algorithm Switch | **EnodebAlgoExtSwitch**.***NsaDcAlgoSwitch*** | Assume that the target NR cell is an intra-gNodeB inter-frequency cell of the source NR cell, the **SIMU_INTRASGNB_INTERFREQ_HO_SW** option of the **gNodeBParam**.***NsaDcOptSwitch*** parameter is deselected on the NR side, and the target LTE cell does not support EN-DC with the source NR cell. Then, SCG release will be triggered during the handover on the LTE side. To prevent this from happening, it is recommended the **INTRA_SGNB_IF_MEAS_FILTER_SW** option be selected. |
| NSA DC Algorithm Switch | **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** | To trigger LTE CA PCC anchoring based on the **PccFreqCfg**.***PreferredPccPriority*** parameter for NSA UEs outside NR coverage, select the **NR_COV_PCC_ANCHORING_OPT_SW** option. It is recommended that this option be selected only when LTE CA PCC anchoring enhancement is enabled. |
| NSA DC Algorithm Extension Switch | **NsaDcMgmtConfig**.***NsaDcAlgoExtSwitch*** | To allow LTE PCC anchoring for an NSA UE outside NR coverage only if the RSRP of the serving cell meets the event A1 threshold, select the **NO_NR_COV_LTE_ANCHOR_OPT_SW** option. |
| NSA DC Algorithm Switch | **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** | To enable the traffic-volume-based SCG addition for NSA DC UEs during initial access, incoming handovers, or incoming RRC connection reestablishment, select the **NSA_DC_VOLUME_BASED_SCG_ADD_SW** option. |

| Parameter Name | Parameter ID | Setting Notes |
| --- | --- | --- |
| NSA DC Algorithm Extension Switch | **NsaDcMgmtConfig**.***NsaDcAlgoExtSwitch*** | To exclude all intra-gNodeB intra-frequency cells of the previous SCG that is released due to coverage reasons from the newly added SCG, select the **NR_COV_SCG_ADD_PUNISH_SW** option and deselect the **NR_COV_PCC_ANCHORING_DELAY_SW** option. |
| LNR Co-Meas LTE Measure Duration | **NsaDcAlgoParam**.***LnrCoMeasLteMeasDuration*** | Set this parameter based on the network plan. |
| NSA DC Algorithm Extension Switch | **NsaDcAlgoParam**.***NsaDcAlgoExtSwitch*** | To prevent ping-pong handovers of NSA UEs for NSA PCC anchoring based on NR coverage, select the **NR_COV_ANCH_PINGPONG_AVOID_SW** option. |
| Whitelist Control Extension Switch 1 | **UeCompat**.***WhitelistCtrlExtSwitch1*** | To allow NR and LTE frequencies to be measured in batches for whitelisted UEs during NSA PCC anchoring based on NR coverage when measurement control optimization for NR-coverage-based PCC anchoring is enabled, select the **SCG_ADD_B1_MEAS_SEQ_OPT_SW** option. |
| NSA DC Algorithm Extension Switch | **EnodebAlgoExtSwitch**.***NsaDcAlgoExtSwitch*** | To enable non-gap-assisted B1 measurement of NR SCG frequencies for NSA DC UEs when the B1 measurement configuration contains only NR PSCell or SCell frequencies and is not expected to include gap configurations, select the **NSA_DC_NR_SCG_B1_NO_GAP_SW** option. |
| Blacklist Control Extension Switch 2 | **UeCompat**.***BlacklistControlExtSwitch2*** | To disable "non-gap-assisted B1 measurement of NR SCG frequencies for NSA DC UEs" for blacklisted UEs, select the **NSA_DC_NR_SCG_FBD_B1_NO_GAP_SW** option. |

The following table describes the parameters that must be set in the **NRCellNsaDcConfig** and **gNodeBParam** MOs on the NR side to configure optimizations for NSA PCC anchoring based on NR coverage.

| Parameter Name | Parameter ID | Setting Notes |
|---|---|---|
| NSA DC Algorithm Switch | **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** | • To support the intra-gNodeB cell change procedure triggered by LTE handovers when an NSA UE is in the DC state prior to an LTE handover and the target NR cell is an intra-gNodeB intra-frequency cell of the source NR cell, select the **MENB_TRIG_INTRA_SGNB_CHANGE_SW** option of the **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** parameter.<br>• To support the intra-gNodeB cell change procedure triggered by LTE handovers when an NSA UE is in the DC state prior to an LTE handover and the target NR cell is an intra-gNodeB inter-frequency cell of the source NR cell, select the **MENB_TRIG_INTRA_SGNB_CHANGE_SW** option of the **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** parameter and the **SIMU_INTRASGNB_INTERFREQ_HO_SW** option of the **gNodeBParam**.***NsaDcOptSwitch*** parameter. |
| NSA DC Optimization Switch | **gNodeBParam**.***NsaDcOptSwitch*** | |

### 5.4.1.2 Using MML Commands

Before using MML commands, refer to **5.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

## Activation Command Examples

On the eNodeB side

```
//(Optional) Enabling blind SCG addition
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoSwitch=NSA_BLIND_SCG_ADDITION_SWITCH-1;
MOD NRNRELATIONSHIP: LocalCellId=21, Mcc="262", Mnc="01", GnodebId=1, CellId=7,
BlindConfigIndicator=TRUE;
//(Optional) Configuring NSA PCC anchoring if an independent camping policy needs to be configured for
NSA UEs
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoSwitch=NSA_PCC_ANCHORING_SWITCH-1,
NsaDcPccAnchoringPolicy=DEFAULT;
MOD PCCFREQCFG: PccDlEarfcn=2950, NsaPccAnchoringPriority=2, NsaDcPccA4RsrpThld=-105;
//(Optional) Configuring blacklisted UEs if NSA PCC anchoring needs to be forbidden for blacklisted UEs in
connected mode
ADD UECOMPAT: Index=0, UeInfoType=UE_CAPABILITY, UeCapIndex=0,
BlacklistControlExtSwitch2=NSA_PCC_ANCHORING_FORBID_SW-1;
//(Optional) Enabling periodic NSA PCC anchoring (if required)
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoSwitch=PERIODIC_PCC_ANCHORING_SW-1;
//(Optional) Enabling data-volume-based NSA PCC anchoring for UEs in the EN-DC state
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoSwitch=VOLUME_BASED_PERIODIC_TRIG_SW-1;
//(Optional) Enabling data-volume-based NSA PCC anchoring for UEs not in the EN-DC state
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoSwitch=VOLUME_BASED_PERIODIC_TRIG_SW-1;
MOD NSADCMGMTCONFIG: LocalCellId=21,
NsaDcAlgoExtSwitch=VOLUME_BASED_PCC_ANCHORING_SW-1;
//(Optional) Enabling the "prohibition of NSA PCC anchoring for high-mobility UEs" function if NSA PCC
anchoring needs to be prohibited for high-mobility UEs (If a CELLOPHOCFG MO with CnOperatorId set to 0
has been configured for the local cell, the ADD CELLOPHOCFG command does not need to be executed.)
ADD CELLOPHOCFG: LocalCellId=21, CnOperatorId=0;
MOD CELLOPHOCFG: LocalCellId=21, CnOperatorId=0,
HighMobiUeHoForbidSw=NSA_ANCHOR_HO_FORBID_SW-1;
//(Optional) Enabling NSA DC PCC anchoring for RRC_IDLE UEs (if required)
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoSwitch=IDLE_MODE_NSA_PCC_ANCHORING_SW-1;
//(Optional) Enabling NSA PCC anchoring optimization for UEs in idle mode (if required)
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoExtSwitch=IDLE_NSA_PCC_ANCHORING_OPT_SW-1;
//(Optional) Enabling the upperLayerIndication IE to be broadcast in SIB2 (if required)
MOD NSADCMGMTCONFIG: LocalCellId=21, UpperLayerIndicationSwitch=NR_NCELL_BASED_BROADCAST;
//(Optional) Enabling the serving PLMN's upperLayerIndication IE to be broadcast in SIB2 (if required)
MOD CNOPERATOR: CnOperatorId=0, OperatorFunSwitch=UPPER_LYR_IND_NO_BROADCAST_SW-0;
//(Optional) Enabling NSA PCC anchoring based on LTE load and uplink coverage (if required)
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoSwitch=NSA_DC_FLEXIBLE_PCC_ANCHOR_SW-1,
UserNumberHighLoadThld=50;
//(Optional) Enabling enhanced NSA PCC anchoring based on LTE load when there are both ultra-high-load
or high-load cells with high NSA PCC anchoring priorities and light-load cells with low NSA PCC anchoring
priorities (with UserNumUltraHighLoadThld setting to 80 as an example)
MOD NSADCMGMTCONFIG: LocalCellId=21,
NsaDcAlgoEnhSwitch=LOAD_BASED_PCC_ANCHOR_ENH_SW-1, UserNumUltraHighLoadThld=80;
//(Optional) Enabling NSA PCC anchoring based on NR coverage (if required)
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoSwitch=NSA_PCC_ANCHORING_SWITCH-1,
NsaDcPccAnchoringPolicy=BASED_ON_NR_COVERAGE;
//(Optional) Prohibiting NSA PCC anchoring based on NR coverage for blacklisted UEs
ADD UECOMPAT: Index=0, UeInfoType=UE_CAPABILITY, UeCapIndex=0,
BlacklistControlExtSwitch2=NR_COV_ANCHORING_FORBID_SW-1;
//(Optional) Enabling NR-coverage-based blind SCC configuration optimization (if required)
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoEnhSwitch=NR_COV_SCC_BLIND_CFG_OPT_SW-1;
//(Optional) Forbidding NSA PCC anchoring based on NR coverage for UEs that experience incoming RRC
connection reestablishments (if required)
MOD NSADCALGOPARAM: NsaDcAlgoExtSwitch=REEST_NR_COV_ANCHOR_FORBID_SW-1;
//(Optional) Enabling measurement control optimization for NR-coverage-based PCC anchoring (if required)
```

```
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoEnhSwitch=NR_COV_PCC_ANCHOR_MC_OPT_SW-1;
//(Optional) Enabling non-gap-assisted B1 measurement if whitelisted UEs are required to measure NR and LTE frequencies in batches
MOD UECOMPAT: Index=0, UeInfoType=UE_CAPABILITY, UeCapIndex=0, WhiteLstCtrlSwitch=NO_GAP_B1_MEAS_SW_ON-1;
//(Optional) Enabling the "non-NSA-PCC frequency filtering during frequency-priority-based frequency selection" function to prevent NSA UEs from being handed over from NSA PCC anchor frequencies to non-anchor frequencies during frequency-priority-based inter-frequency handovers
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoEnhSwitch=FREQ_PRI_SEL_NON_PCC_FILTER_SW-1;
//(Optional) Configuring optimizations for NR-coverage-based PCC anchoring
//(Optional, recommended) Selecting the INTRA_SGNB_IF_MEAS_FILTER_SW option to prevent SCG release triggered during LTE handovers when the following conditions are met: (1) The target NR cell is an intra-gNodeB inter-frequency cell of the source NR cell. (2) The SIMU_INTRASGNB_INTERFREQ_HO_SW option on the NR side is deselected. (3) The target LTE cell does not support EN-DC with the source NR cell.
MOD ENODEBALGOEXTSWITCH: NsaDcAlgoSwitch=INTRA_SGNB_IF_MEAS_FILTER_SW-1;
//(Optional) Selecting the NR_COV_PCC_ANCHORING_OPT_SW option to trigger LTE CA PCC anchoring based on the PccFreqCfg.PreferredPccPriority parameter for NSA UEs outside NR coverage
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoSwitch=NR_COV_PCC_ANCHORING_OPT_SW-1;
//(Optional) Select the NO_NR_COV_LTE_ANCHOR_OPT_SW option to allow LTE PCC anchoring for an NSA UE outside NR coverage only if the RSRP of the serving cell meets the event A1 threshold
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoExtSwitch=NO_NR_COV_LTE_ANCHOR_OPT_SW-1;
//(Optional) Selecting the NSA_DC_VOLUME_BASED_SCG_ADD_SW option to enable the traffic-volume-based SCG addition for NSA DC UEs during initial access, incoming handovers, or incoming RRC connection reestablishment
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoSwitch=NSA_DC_VOLUME_BASED_SCG_ADD_SW-1;
//(Optional) Selecting the NR_COV_SCG_ADD_PUNISH_SW option and deselect the NR_COV_PCC_ANCHORING_DELAY_SW option to exclude all intra-gNodeB intra-frequency cells of the previous SCG that is released due to coverage reasons from the newly added SCG
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoExtSwitch=NR_COV_PCC_ANCHORING_DELAY_SW-0&NR_COV_SCG_ADD_PUNISH_SW-1;
//(Optional) Selecting the PCC_ANCHORING_SMART_DEPLOY_SW option if both Huawei eNodeBs and non-Huawei eNodeBs are deployed
MOD NSADCALGOPARAM: NsaDcAlgoSwitch=PCC_ANCHORING_SMART_DEPLOY_SW-1;
//(Optional) Setting the LnrCoMeasLteMeasDuration parameter based on the network plan
MOD NSADCALGOPARAM: LnrCoMeasLteMeasDuration=0;
//(Optional) Selecting the NR_COV_ANCH_PINGPONG_AVOID_SW option to prevent ping-pong handovers of NSA UEs for NSA PCC anchoring based on NR coverage
MOD NSADCALGOPARAM: NsaDcAlgoExtSwitch=NR_COV_ANCH_PINGPONG_AVOID_SW-1;
//(Optional) Selecting the SCG_ADD_B1_MEAS_SEQ_OPT_SW option to allow NR and LTE frequencies to be measured in batches for whitelisted UEs during NSA PCC anchoring based on NR coverage when measurement control optimization for NR-coverage-based PCC anchoring is enabled
MOD UECOMPAT: Index=0, UeInfoType=UE_CAPABILITY, UeCapIndex=0, WhitelistCtrlExtSwitch1=SCG_ADD_B1_MEAS_SEQ_OPT_SW-1;
//(Optional) Selecting the NSA_DC_NR_SCG_B1_NO_GAP_SW option to enable non-gap-assisted B1 measurement of NR SCG frequencies for NSA DC UEs, when the B1 measurement configuration contains only NR PSCell or SCell frequencies and is not expected to include gap configurations
MOD ENODEBALGOEXTSWITCH: NsaDcAlgoExtSwitch=NSA_DC_NR_SCG_B1_NO_GAP_SW-1;
//(Optional) Disabling "non-gap-assisted B1 measurement of NR SCG frequencies for NSA DC UEs" for blacklisted UEs
//Running the following commands if no UeCompat MO has been configured
ADD UEINFO: UeCapIndex=0, AccessStratumRelease=1, UeCategory=40, Fgi=1247894;
ADD UECOMPAT: Index=0, UeInfoType=UE_CAPABILITY, UeCapIndex=0, BlacklistControlExtSwitch2=NSA_DC_NR_SCG_FBD_B1_NO_GAP_SW-1;
//Running the following command if a UeCompat MO has been configured
MOD UECOMPAT: Index=0, UeInfoType=UE_CAPABILITY, UeCapIndex=0, BlacklistControlExtSwitch2=NSA_DC_NR_SCG_FBD_B1_NO_GAP_SW-1;
```

### On the gNodeB side

```
//(Optional) Configuring optimizations for NR-coverage-based PCC anchoring
//(Optional) Selecting the MENB_TRIG_INTRA_SGNB_CHANGE_SW option to support the intra-gNodeB cell change procedure triggered by LTE handovers when an NSA UE is in the DC state prior to an LTE handover and the target NR cell is an intra-gNodeB intra-frequency cell of the source NR cell
MOD NRCELLNSADCCONFIG: NrCellId=7, NsaDcAlgoSwitch=MENB_TRIG_INTRA_SGNB_CHANGE_SW-1;
//(Optional) Selecting the SIMU_INTRASGNB_INTERFREQ_HO_SW option to support the intra-gNodeB cell change procedure triggered by LTE handovers when an NSA UE is in the DC state prior to an LTE handover and the target NR cell is an intra-gNodeB intra-frequency cell of the source NR cell (This setting takes effect only after the MENB_TRIG_INTRA_SGNB_CHANGE_SW option is selected.)
MOD GNODEBPARAM: NsaDcOptSwitch=SIMU_INTRASGNB_INTERFREQ_HO_SW-1;
```

## Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

On the eNodeB side

```
//Disabling blind SCG addition
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoSwitch=NSA_BLIND_SCG_ADDITION_SWITCH-0;
//Disabling the "prohibition of NSA PCC anchoring for high-mobility UEs" function
MOD CELLOPHOCFG: LocalCellId=21, CnOperatorId=0,
HighMobiUeHoForbidSw=NSA_ANCHOR_HO_FORBID_SW-0;
//Disabling the configuration of an independent camping policy for NSA UEs
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoSwitch=NSA_PCC_ANCHORING_SWITCH-0;
//Disabling periodic NSA PCC anchoring
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoSwitch=PERIODIC_PCC_ANCHORING_SW-0;
//Disabling data-volume-based NSA PCC anchoring for UEs in the EN-DC state
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoSwitch=VOLUME_BASED_PERIODIC_TRIG_SW-0;
//Disabling data-volume-based NSA PCC anchoring for UEs not in the EN-DC state
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoSwitch=VOLUME_BASED_PERIODIC_TRIG_SW-0;
MOD NSADCMGMTCONFIG: LocalCellId=21,
NsaDcAlgoExtSwitch=VOLUME_BASED_PCC_ANCHORING_SW-0;
//Disabling upperLayerIndication broadcast in SIB2
MOD NSADCMGMTCONFIG: LocalCellId=21, UpperLayerIndicationSwitch=OFF;
//Disabling NSA PCC anchoring optimization for UEs in idle mode
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoExtSwitch=IDLE_NSA_PCC_ANCHORING_OPT_SW-0;
//Disabling NSA DC PCC anchoring for RRC_IDLE UEs
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoSwitch=IDLE_MODE_NSA_PCC_ANCHORING_SW-0;
//Disabling NSA PCC anchoring based on LTE load and uplink coverage
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoSwitch=NSA_DC_FLEXIBLE_PCC_ANCHOR_SW-0;
//Disabling enhanced NSA PCC anchoring based on LTE load
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoEnhSwitch=LOAD_BASED_PCC_ANCHOR_ENH_SW-0;
//Disabling NR-coverage-based blind SCC configuration optimization
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoEnhSwitch=NR_COV_SCC_BLIND_CFG_OPT_SW-0;
//Disabling measurement control optimization for NR-coverage-based PCC anchoring
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoEnhSwitch=NR_COV_PCC_ANCHOR_MC_OPT_SW-0;
//Disabling non-gap-assisted B1 measurement
MOD UECOMPAT: Index=0, UeInfoType=UE_CAPABILITY, UeCapIndex=0,
WhiteLstCtrlSwitch=NO_GAP_B1_MEAS_SW_ON-0;
//Disabling the "non-NSA-PCC frequency filtering during frequency-priority-based frequency selection"
function
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoEnhSwitch=FREQ_PRI_SEL_NON_PCC_FILTER_SW-0;
//Deselecting the INTRA_SGNB_IF_MEAS_FILTER_SW option
MOD ENODEBALGOEXTSWITCH: NsaDcAlgoSwitch=INTRA_SGNB_IF_MEAS_FILTER_SW-0;
//Deselecting the NR_COV_PCC_ANCHORING_OPT_SW option
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoSwitch=NR_COV_PCC_ANCHORING_OPT_SW-0;
//Deselecting the NO_NR_COV_LTE_ANCHOR_OPT_SW option
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoExtSwitch=NO_NR_COV_LTE_ANCHOR_OPT_SW-0;
//Deselecting the NSA_DC_VOLUME_BASED_SCG_ADD_SW option
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoSwitch=NSA_DC_VOLUME_BASED_SCG_ADD_SW-0;
//Deselecting the NR_COV_SCG_ADD_PUNISH_SW option
MOD NSADCMGMTCONFIG: LocalCellId=21, NsaDcAlgoExtSwitch=NR_COV_SCG_ADD_PUNISH_SW-0;
//Deselecting the NR_COV_ANCH_PINGPONG_AVOID_SW option
MOD NSADCALGOPARAM: NsaDcAlgoExtSwitch=NR_COV_ANCH_PINGPONG_AVOID_SW-0;
//Deselecting the SCG_ADD_B1_MEAS_SEQ_OPT_SW option
MOD UECOMPAT: Index=0, UeInfoType=UE_CAPABILITY, UeCapIndex=0,
WhitelistCtrlExtSwitch1=SCG_ADD_B1_MEAS_SEQ_OPT_SW-0;
//Deselecting the NSA_DC_NR_SCG_B1_NO_GAP_SW option
MOD ENODEBALGOEXTSWITCH: NsaDcAlgoExtSwitch=NSA_DC_NR_SCG_B1_NO_GAP_SW-0;
//Deselecting the NSA_DC_NR_SCG_FBD_B1_NO_GAP_SW option for blacklisted UEs
MOD UECOMPAT: Index=0, UeInfoType=UE_CAPABILITY, UeCapIndex=0,
BlacklistControlExtSwitch2=NSA_DC_NR_SCG_FBD_B1_NO_GAP_SW-0;
```

On the gNodeB side

```
//(Optional) Deselecting the MENB_TRIG_INTRA_SGNB_CHANGE_SW option
MOD NRCELLNSADCCONFIG: NrCellId=7, NsaDcAlgoSwitch=MENB_TRIG_INTRA_SGNB_CHANGE_SW-0;
```

```
//(Optional) Deselecting the SIMU_INTRASGNB_INTERFREQ_HO_SW option
MOD GNODEBPARAM: NsaDcOptSwitch=SIMU_INTRASGNB_INTERFREQ_HO_SW-0;
```

### 5.4.1.3 Using the MAE-Deployment

- Fast batch activation

  This function can be batch activated using the Feature Operation and Maintenance function of the MAE-Deployment. For detailed operations, see the following section in the MAE-Deployment product documentation or online help: **MAE-Deployment Operation and Maintenance** > **MAE-Deployment Guidelines** > **Enhanced Feature Management** > **Feature Operation and Maintenance**.
- Single/Batch configuration

  This function can be activated for a single base station or a batch of base stations on the MAE-Deployment. For detailed operations, see Feature Configuration Using the MAE-Deployment.

  **NOTE**

  Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

## 5.4.2 Activation Verification

### Monitoring Counters

Counters related to NSA DC can be subscribed on the MAE-Access in one-click mode.

- Check whether NSA PCC anchoring has taken effect and calculate the handover success rate using the counters listed below.

  This function has taken effect if all the counters below produce non-zero values. If any counter produces a zero value, check whether the **NSA_PCC_ANCHORING_SWITCH** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter is selected, **PccFreqCfg**.***PccDlEarfcn*** and **PccFreqCfg**.***NsaPccAnchoringPriority*** are set to non-zero values, and the UE capability (the DC combination in the *UE-MRDC-Capability* IE of the UECapabilityInformation message) supports NSA DC with the target frequency.

| Counter ID | Counter Name | NE |
| --- | --- | --- |
| 1526749449 | **L.NsaDC.PCCAnchor.HHO.PrepAttOut** | eNodeB |
| 1526749450 | **L.NsaDC.PCCAnchor.HHO.ExecAttOut** | eNodeB |
| 1526749451 | **L.NsaDC.PCCAnchor.HHO.ExecSuccOut** | eNodeB |

- Check whether non-data-volume-based NSA PCC anchoring has taken effect.

  This function has taken effect if the **L.NsaDC.PCCAnchor.HHO.ExecAttOut** counter value increases.

- Check whether NSA PCC anchoring based on LTE uplink coverage has taken effect.

  This function has taken effect if the **L.HHO.InterFreq.ULquality.PrepAttOut** counter value increases after LTE spectrum coordination is disabled and the **NSA_DC_FLEXIBLE_PCC_ANCHOR_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter is selected.

- Check whether blind PSCell addition has taken effect.

  Check the value of the **L.NsaDc.SgNB.Add.Blind.Att** counter, which indicates the total number of blind SgNB addition attempts for LTE-NR NSA DC UEs that treat the local cell as their PCell. If the value of the counter is not 0, this function has taken effect.

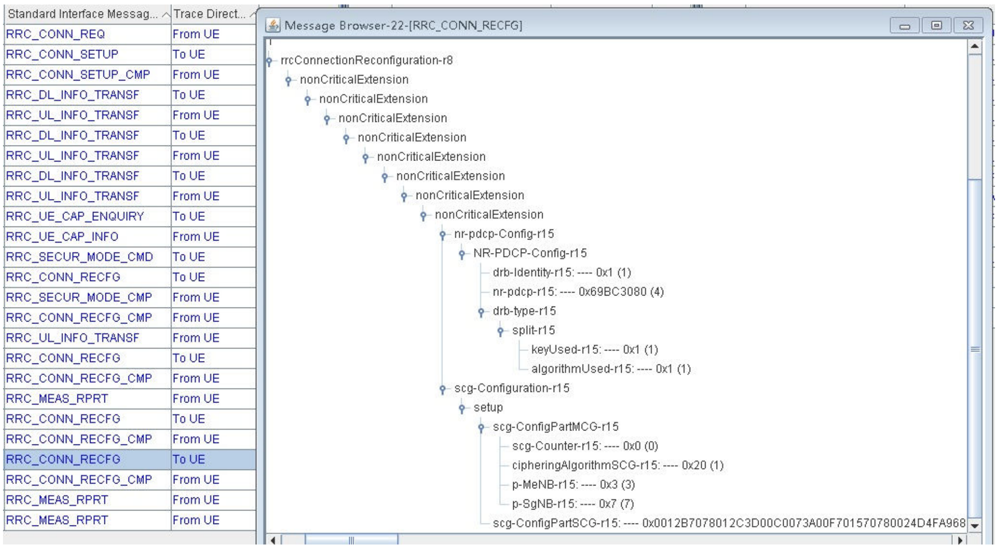
  Check the value of the **L.NsaDc.SgNB.Add.Blind.Succ** counter, which indicates the total number of successful blind SgNB additions for LTE-NR NSA DC UEs that treat the local cell as their PCell. If the value of the counter is not 0, this function has taken effect.

## Message Tracing

1. Log in to the MAE-Access. Choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**. The **Signaling Trace Management** window is displayed.
2. Trace UE random access.

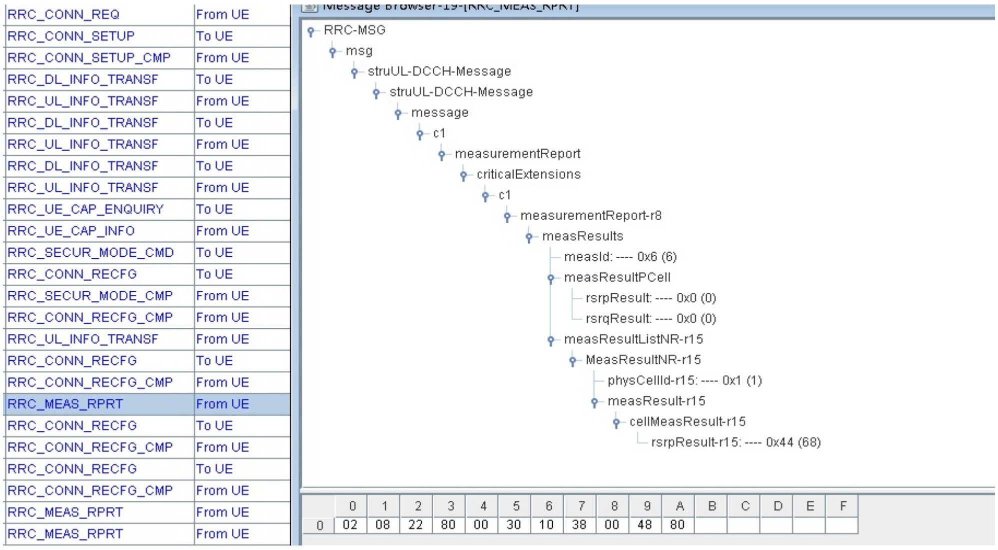
   Choose **Trace Type** > **LTE** > **Application Layer** > **Uu Interface Trace**. You can observe that the UE sends an RRC_CONN_SETUP_CMP message to the eNodeB to initiate an LTE access procedure.
3. (Optional) The eNodeB delivers an NR measurement configuration message to the UE.

   Choose **Trace Type** > **LTE** > **Application Layer** > **Uu Interface Trace**. You can observe that the RRC_CONN_RECFG message sent by the eNodeB to the UE includes an *EventB1* IE.




4. (Optional) The UE reports measurement results.

   Choose **Trace Type** > **LTE** > **Application Layer** > **Uu Interface Trace**. You can observe that the RRC_MEAS_RPRT message sent by the UE contains IEs related to the measured PCIs and signal strength of NR cells.




**NOTE**

- If blind PSCell configuration (also known as blind PSCell addition) is enabled, the eNodeB does not deliver measurement configurations for neighboring NR frequencies after entering the blind configuration procedure. Therefore, measurement configurations and measurement reports involved in **3** and **4** cannot be traced over the Uu interface.
- If blind PSCell configuration (also known as blind PSCell addition) is performed and the UE attempts to access a target cell without NR coverage, the access will fail. In this case, the UE sends an SCG Failure Information message to the eNodeB. The value of the **L.NsaDc.ScgFailure** counter increases. After blind PSCell addition is enabled, the SgNB addition success rate (**L.NsaDc.SgNB.Add.Succ** / **L.NsaDc.SgNB.Add.Att**) decreases. It is not recommended that blind PSCell addition be enabled if the SgNB addition success rate before function activation is less than 98%.

5. (Optional) Check whether SIB2 includes an *upperLayerIndication* IE.

   Choose **Trace Type** > **LTE** > **Application Layer** > **Uu Interface Trace**. If SIB2 sent by the base station includes an *upperLayerIndication* IE, the function has taken effect.

6. Check whether NSA PCC anchoring based on NR coverage has taken effect.

   Choose **Trace Type** > **LTE** > **Application Layer** > **Uu Interface Trace**. Check whether the base station sends a UE an RRC_CONN_RECFG message with IEs *EventB1* and *EventA5*. If so, this function has taken effect.

7. Check whether NSA PCC anchoring optimization for UEs in idle mode has taken effect.

   Choose **Trace Type** > **LTE** > **Application Layer** > **Uu Interface Trace**. Check whether the base station sends a UE an RRC_CONN_REL message with a *FreqPriorityEUTRA* IE that carries a non-anchor frequency. If so, this function has taken effect.

## 5.4.3 Network Monitoring

None

# 6 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

**NOTE**

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

**FAQ 1: How do I find the parameters related to a certain feature from parameter reference?**

**Step 1** Open the EXCEL file of parameter reference.

**Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.

**Step 3** Click **OK**. All parameters related to the feature are displayed.

**----End**

**FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?**

**Step 1** Open the EXCEL file of the used reserved parameter list.

**Step 2** On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.

**----End**

# 7 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

**NOTE**

You can find the EXCEL files of performance counter reference for the software version used on the live network from the product documentation delivered with that version.

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

**FAQ: How do I find the counters related to a certain feature from performance counter reference?**

**Step 1** Open the EXCEL file of performance counter reference.

**Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.

**Step 3** Click **OK**. All counters related to the feature are displayed.

**----End**

# 8 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

# 9 Reference Documents


- 3GPP TS 38.101: "NR; User Equipment (UE) radio transmission and reception"
- 3GPP TS 37.340: "E-UTRA and NR; Multi-connectivity; Stage-2"
- 3GPP TS 36.331: "E-UTRA; Radio Resource Control (RRC) Protocol specification"
- 3GPP TS 38.306: "NR; User Equipment (UE) radio access capabilities"
- 3GPP TS 38.213: "NR; Physical layer procedures for control"
- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- *Power Control* in 5G RAN feature documentation
- *Ultimate Cloud X Experience* in 5G RAN feature documentation
- *Massive MIMO Enhancements (FDD)* in eRAN feature documentation
- *CA Experience Improvement* in 5G RAN feature documentation
- *Multi-carrier Unified Scheduling* in eRAN feature documentation
- *UL CoMP* in eRAN feature documentation
- *Mobility Management in Connected Mode* in eRAN feature documentation
- *eMBMS* in eRAN feature documentation
- *VoLTE* in eRAN feature documentation
- *Video Experience Optimization* in eRAN feature documentation
- *Multi-band Optimal Carrier Selection* in eRAN feature documentation
- *RAN Sharing* in eRAN feature documentation
- *DRX* in 5G RAN feature documentation
- *High Speed Mobility* in eRAN feature documentation
- *Physical Channel Resource Management* in eRAN feature documentation
- *ANR Management* in eRAN feature documentation
- *LTE Spectrum Coordination* in eRAN feature documentation
- *Virtual 4T4R (FDD)* in eRAN feature documentation
- *Short TTI (FDD)* in eRAN feature documentation
- *MIMO* in eRAN feature documentation
- *Carrier Aggregation* in eRAN feature documentation
- *Idle Mode Management* in eRAN feature documentation

- *Downlink Scheduling* in eRAN feature documentation
- *Breathing Pilot* in eRAN feature documentation
- *Air Interface Latency Optimization* in eRAN feature documentation
- *WBB* in eRAN feature documentation
- *LTE and NR Zero Bufferzone* in eRAN feature documentation
- *UL and DL Decoupling* in 5G RAN feature documentation
- *NSA Mobility Management*
- *EPC-based NSA Basics*
- *SRAN Networking and Evolution Overview*
- *LTE FDD and NR Dynamic Spectrum Sharing*
- *LTE FDD and NR Hybrid Dynamic Spectrum Sharing*
- *EPC-based NSA Experience Improvement*
- *License Control Item Lists (FDD)* in eRAN feature documentation
- *License Control Item Lists (CIoT)* in eRAN feature documentation
- *License Control Item Lists (TDD)* in eRAN feature documentation
- *License Control Item Lists* in 5G RAN feature documentation